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Prevalence of anabolic steroid users seeking support from physicians and the type of support sought: a systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-056445
Article Type:	Original research
Date Submitted by the Author:	16-Aug-2021
Complete List of Authors:	Amaral, Julio Mario; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions Kimergård, Andreas; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions Deluca, Paolo; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions
Keywords:	Substance misuse < PSYCHIATRY, EPIDEMIOLOGY, PRIMARY CARE

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Prevalence of anabolic steroid users seeking support from physicians and the type of support sought: a systematic review and meta-analysis

Short title: Anabolic steroid users seeking support from physicians

Word count: 3,706

PROSPERO registration number: CDR42020177919

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Prevalence of anabolic steroid users seeking support from physicians and the type of support sought: a systematic review and meta-analysis

Abstract

Objectives: The main objective is to estimate the prevalence of androgenic-anabolic steroids' (AAS) users seeking support from physicians. Secondary objectives are to describe the type of support sought from physicians and to describe prevalence rates of AAS users seeking support from physicians in different periods of time, locations and subpopulations of AAS users. This study's protocol is registered in PROSPERO: CRD42020177919.

Design: Systematic review and meta-analysis.

Data sources: 25 studies published between 1988 and 2019 on MEDLINE, PsycINFO, Web of Science and SciELO.

Eligibility criteria: Peer-reviewed studies with AAS users that report the proportion of those seeking support from physicians; studies published at any time, in any language. The quality of studies was evaluated using the Mixed Methods Appraisal Tool.

Results: The overall prevalence of AAS users seeking support from physicians estimated by this study is 35.7%. Higher prevalence rates were observed among clients in needle and syringe programmes (48.3%), while lower rates were seen among adolescent AAS users (15.0%). A greater proportion of AAS users sought physicians for consultations (50.0%) rather than AAS-related information (20.2%). Higher prevalence rates of AAS users seeking support from physicians were found in studies published in Australia (56% to 77%).

Conclusion: A relevant proportion of AAS users seek support from physicians, with a wide range of prevalence rates seen among different subpopulations of AAS users. Results must be considered in the light of a limited sample of heterogenic studies.

Acknowledgement: The first author was funded by King's College London International Scholarship.

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Strengths and limitations of this study

- It is the first meta-analysis of studies investigating the prevalence of anabolic steroid users seeking support from physicians.
- Adds to the knowledge of help-seeking behaviours of people using anabolic steroids in different countries and among subpopulations of anabolic steroid users.
- Advances our understanding of the kind of support sought by people using anabolic steroids from physicians, as well as other sources of information such as the internet, friends and coaches.
- Findings are based on a limited sample of highly heterogenic studies.

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Prevalence of anabolic steroid users seeking support from physicians and the type of support sought: a systematic review and meta-analysis

1. Introduction

People using anabolic-androgenic steroids (AAS) may adopt strategies, such as medical consultations, exams to monitor their health and use of harm reduction services to prevent or treat adverse effects of AAS [8, 25, 26, 29, 52, 55, and 62]. Among these strategies, increased attention has been given to the support provided to AAS users by physicians [43, 31, 22, and 56], including consultations, information about AAS, and diagnosis [11 and 24].

Despite a relevant prevalence of recreational use of AAS worldwide [47 and 48] some physicians admit lack of training and experience in recognising and treating adverse effects of AAS [16, 17, 22, and 59]. Health conditions potentially associated with the use of AAS – such as dyslipidaemia, atherosclerotic disease, cardiomyopathy, testicular atrophy, gynecomastia, menstrual irregularities, depression, and anxiety – vary in prevalence and severity [44]. While only a small number of AAS users seem to seek medical treatment for these conditions [19], some studies have found that few physicians ask patients presenting these problems about their current or previous use of AAS [4, 26, and 43]. Some people who have experienced adverse health conditions associated with the use of AAS prefer not to inform their physicians about their use of AAS due to the stigma associated with AAS use, lack of trust in doctors' knowledge about AAS or concerns about AAS use being registered in their medical records [18, 22, 43, and 61]. These barriers in engagement and communication contribute to a gap in knowledge about AAS users' help-seeking behaviours, and the prevalence of AAS users seeking medical support remains largely unknown. Analysing this phenomena can add to the knowledge of how the prevalence of AAS users seeking support from physicians has changed through time, in different regions and among different subpopulations of AAS users.

The aim of this study is to estimate the prevalence of AAS users seeking support from medical professionals by conducting a systematic review and meta-analysis of peer-reviewed studies with people using AAS.

2. Methods

This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [35], and its protocol is registered in PROSPERO (registration number CRD42020177919). As a review of the literature, this study is exempted from ethical clearance by King's College London research ethics office.

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2.1 Search strategy and screening

A search strategy (Figure 1) was designed to retrieve studies containing the terms used to describe the use of AAS and other performance and image-enhancing drugs (PIEDs). Words such as 'survey', 'interview' and 'help-seeking behaviour' were not included in the search algorithm to avoid overlooking relevant studies without these terms in their titles or abstracts. There were no restrictions on date, location and language of studies. Searches were performed in June 2020 on MEDLINE, PsycINFO, Web of Science and SciELO.

Figure 1: Search strategy.

After identification of studies, eligibility criteria (Table 1) were applied in the screening of titles, abstracts and full-texts (Figure 2). Selection of studies was conducted by JA and submitted to AK and PD for assessment. The Rayyan QCRI platform [38] was used for managing searches, eliminating duplicates and screening. A secondary, 'manual' search for eligible studies was performed in the reference lists of included studies.

Figure 2: Search flow.

Table 1: Inclusion and exclusion criteria

Inclusion criteria	**Exclusion criteria**
Quantitative and qualitative studies with people using AAS for the purposes of image and performance enhancement.	Studies with patients using AAS for treatment of medical conditions. Studies with animal subjects and in vitro analysis of AAS effects.
Studies informing the proportion of AAS users seeking physicians to receive information and prevent and treat AAS-related health conditions.	Studies not including the proportion of AAS users in the sample who seek support from physicians. Studies where the contact with the physician cannot be understood as a health strategy adopted by the patient, such as involuntary admissions, post-mortem analysis and cohorts of patients selected for studies of specific health conditions.
Peer-reviewed studies. Studies published at any time. Studies published in any language, as long as it is possible to retrieve relevant data from the	'Grey literature' (non-peer-reviewed studies and reports). Case studies and interviews with a single AAS user.

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authors or articles (English summaries and abstracts).

2.2 Data synthesis

Data from selected studies was summarized by JA and reviewed by AK and PD. The following items were described (Table 2): First author, publication year, location of study, subpopulation of AAS users, sample size and gender of participants, the proportion of AAS users who sought support from physicians, and other sources of AAS information.

2.3 Quality assessment

Quality assessment of the 25 eligible studies was conducted by JA and reviewed by AK and PD. Quality and internal validity of studies were evaluated using the Mixed Methods Appraisal Tool (MMAT) for quantitative descriptive studies [42]. Quality analysis included two screening criteria: (a) 'Are there clear research questions?' and (b) 'Do the collected data allow to address the research questions?'. This was followed by five quality-assessment criteria: (i) 'Is the sampling strategy relevant to address the research question?', (ii) 'Is the sample representative of the target population?', (iii) 'Are the measurements appropriate?', (iv) 'Is the risk of nonresponse bias low?' and (v) 'Is the statistical analysis appropriate to answer the research question?'.

2.4 Meta-analysis

A meta-analysis, based on methods adopted in a similar study by Sagoe et al. [47], was performed to estimate the prevalence of AAS users seeking support from physicians. When a study informed more than one prevalence rate of AAS users seeking support from physicians, the higher prevalence rate of male AAS users seeking medical support was considered as the study's effect size. A random effects model – instead of fixed effects – was chosen because we could not assume to find similar dispersions of prevalence rates across studies [1]. Cochran's Q and I^2 were used to assess the variation in prevalence rates between studies. Q represents the weighted sum of squared deviations of prevalence rates of a group of studies [21]. Statistically significant Q values indicate that the dispersion of prevalence rates across studies is larger than the dispersion within subjects of a study due to sampling error [57]. Due to the small number of studies analysed, Q values were considered statistically significant if $p < 0.01$ [15]. Based on the Q value, the I^2 index describes the percentage of variation of prevalence rates across a group of studies that is due to heterogeneity (e.g. different sample sizes, populations or methods of accessing AAS users' health behaviours) rather than sampling error [20 and 21].

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After the assessment of the overall prevalence of AAS users seeking support from physicians, prevalence rates among subgroups of studies were compared. Studies were grouped as follows: publication year (before and after the year 2000), subpopulation of AAS users (gym users / recreational athletes, adolescents, bodybuilders/powerlifters, clients in needle and syringe exchange programmes (NSP), non-specified AAS users) and type of support sought (information, consultation). Q statistics for subgroup analysis represent the weighted comparison of prevalence rates. Partial prevalence rates were calculated for other sources of support chosen by respondents of selected studies: The internet, friends and coaches. Data unsuitable for subgroup meta-analysis due to a small number of studies in subgroups or insufficient information were summarised in narrative synthesis.

A meta-regression analysis was performed to observe variables that could account for the variance in the overall prevalence rate. For each moderator variable, the category with the highest number of studies was used as a contrast. Meta-analysis and meta-regression were conducted on Comprehensive Meta-Analysis, version 3.0 [7].

2.5 Patient and public involvement

It was not appropriate to involve patients or the public in the design, conduct, reporting or dissemination plans of this study.

3. Results

Table 2 shows the data synthesis of 25 included studies, published between 1988 and 2019. A total number of AAS users (nAAS) = 7,120 took part in the studies: 6,280 (88.2%) males, 116 (1.6%) females and 724 (10.2%) whose gender was not reported. The highest prevalence of AAS users seeking support from physicians was reported in Australia (77%, nAAS = 60) [33], and the lowest in South Africa (1.7%, nAAS = 512) [53]. Populations of each study and the type of support sought from physicians are presented in Table 2 using the terms employed by the studies' authors (verbatim).

Table 2: Data synthesis

First author	Publication year[1]	Location	Subpopulation, sample size[2] and gender	% of AAS users who sought support from physicians[4]	Other sources of AAS information	Risk of bias[5]
Yesalis [60]	1988	US	competing powerlifters N = 45 M **nAAS = 15 M**	**53%** consider a physician, pharmacist or medical literature as primary source of information on health effects of AAS	other athletes or coach (47%)	small sample size

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Johnson [27]	1989	US	11th grade high school students N = 853 M **nAAS = 95 M**	**29%** sought information about AAS from a physician or health professional	peers (51%), television (41%), muscle magazine (36%), newspaper (20%)	-
Tanner [51]	1995 (1992-93)	US	adolescents N = 3438 M, 3492 F **nAA S= 139 M, 45 F**	**10%** seek physicians as sources of information about AAS	television (92%), friends (67%), muscle magazines (27%)	-
Korkia [32]	1997 (1992)	UK	gym users N = **nAAS= 97 M, 3 F**	**35%** had regular medical checks for AAS use	n/a	-
Augé [5]	1999 (1995-96)	US	professional bodybuilders N= **nAAS= 14 M, 3 F**	Respondents who sought a physician contact during the study: 35.7% males, 100% females for routine medical examinations; 0% males and 33.3% females for treatment of medical conditions; **42.9%** males and 0% females, due to trauma	n/a	small sample size
Perry [41]	2005 (2002)	US	weightlifters and bodybuilders N = **nAAS = 207**[3]	seeking health care providers as sources of steroid information: **24.1%** often or seldom, 67.6% never	internet (91.8% often or seldom, 1.4% never), other users (99.2% often or seldom, 7.2% never), magazines (70.1% often or seldom, 16.9% never)	-
Parkinson [39]	2006 (2004)	Trans-region	AAS users N = **nAAS = 494 M, 6 F**	**37%** discussed AAS use with a physician	n/a	-
Striegel [50]	2006 (2001)	Germany	members of fitness centres N = 390 M, 231 F **nAAS = 84**[3]	**56%** had close contact with physicians	training colleagues (54.2%), pertinent literature (54.2%)	35.5% response rate
Cohen [12]	2007	US	AAS users N = **nAAS = 1955 M**	**66%** seek or informed willingness to seek medical supervision	n/a	-
Al-Falasi [2]	2008 (2006)	UAE	gym users N = 143 M, 20F **nAAS = 34 M**	**38%** seek or plan to consult a medical provider about AAS. 13% seek health care professionals as sources of knowledge about AAS	friends (50%), media (35%), internet (10%), fitness magazines (9%)	44% response rate
Larance [33]	2008 (2005)	Australia	PIED users N = **nAAS = 60 M**	**77%** had regular medical check-ups with a GP. 22% sought information about PIEDs from a doctor	internet sites (62%), friends (55%), contacts at the gym (18%), NSP (7%)	small sample size, number of non-respondents not informed
Posiadała [45]	2010 (2007)	Poland	men practising recreational sports N=50 M, **nAAS = 18 M**	**11%** have visited a doctor	n/a	small sample size
Gradidge [14]	2011	South Africa	adolescents N=100 M, **nAAS = 96 M**	**10%** seek physicians as sources of information about AAS	internet (74%), magazines (72%), friend (66%), coach (40%),	-

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					personal trainer (33%), parent (33%), brochures (31%), newspapers (29%), pharmacist (24%), television and school (23%), sibling (19%), books (16%), biokineticist (16%), journals (9%)	
Ip [25]	2011 (2009)	Trans-region	people regularly participating in strength-training exercise N=1277 M **nAAS = 506 M**	**56.1%** have routine health visits while using AAS	n/a	-
Santos [49]	2011 (2008)	Brazil	bodybuilders N = 123 M **nAAS = 41 M**	**10.5%** learn about AAS from health professionals	members from the fitness club (35.0%), TV (19.3%), instructors (12.2%), magazines (11.4%), newspapers (4.3%), others (7.0%)	number of non-respondents not informed
Hope [23]	2013 (2010-11)	UK	clients of NSP and outreach syringe services N = **nAAS = 395 M**	**45%** had seen a GP in the last year about their health	16% (A&E/walk-in), 17% (Sexual Health Clinic)	-
Mooney [36]	2016 (2015-16)	UK	gym users N = 177 M, 200 F **nAAS = 23[3]**	**5.1%** seek medical opinion about IPED	internet sites (33.7%)	number of incomplete questionnaires not informed
Rowe [46]	2016	Australia	PIED injectors at NSP N = **nAAS = 605 M**	**51.6%** visited a doctor within 1 month. 10.7% visited a doctor over 1 year ago. 33.9% seek doctors for injecting advice	other sources of injecting advice: mates (42.9%), internet (40.1%), NSP staff (21.2%), nurses (15.7%), personal trainers (12.5%), suppliers (12.4%), other sources (4.6%)	number of non-respondents not informed
Westerman [58]	2016 (2015)	Trans-region	AAS users N = **nAAS = 231 M**	**69.2%** regularly see a physician	n/a	unspecified location
Zahnow [62]	2017 (2015)	Trans-region	AAS users N = **nAAS = 253 M, 59 F**	**21.8%** visited a doctor in the last year	n/a	number of incomplete questionnaires not informed
Althobiti [3]	2018 (2016)	Saudi Arabia	gym users N = 4860 M **nAAS = 474 M**	**38%** seek or plan to seek medical provider as source of information about AAS	friends (41.5%), gym trainer (23.7%), social media (34.8%)	number of non-respondents not informed

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Hill [22]	2019 (2016)	UK	members of bodybuilding groups in social media N = **nAAS = 216**[3]	**50%** seek a private specialist for interpretation of health checks to monitor health while using AAS; 22% seek a NHS doctor	interpretation of health checks: friends (6%), other sources (22%)	sample's representativeness not informed
Jacka [26]	2019 (2014-15)	Australia	PIED users N = **nAAS = 246 M**	**56%** sought health professional as source of AAS information	internet and media (87%), social network (65%)	number of non-respondents and incomplete questionnaires not informed
Pereira [40]	2019 (2016-17)	Brazil	gym users N=559 M, 160 F **nAAS = 194**[3]	**63.9%** have medical follow-ups. 60.3% seek doctors as information sources about AAS.	friends (36.8%), coaches (36.1%), internet (18.6%), nutritionist (21.1%), other (4.6%)	number of non-respondents not informed
Uddin [53]	2019	Pakistan	gym users N = 841 M **nAAS = 512 M**	**1.7 %** seek physicians as sources of information about AAS	friends/gym partners (49.7%), the media (30.2%), muscle magazines (9.8%), coach/trainer (8.4%)	-

(1) Year of data collection in parenthesis, if available. **n/a**: Data not available. **(2)** Populations informed using terms adopted in each study, verbatim. **N**: total sample size. **nAAS**: number of AAS users in sample. **M**: males. **F**: females. **(3)** Gender not informed. **PIED**: performance and image-enhancing drugs. **NSP**: needle and syringe exchange programme. **(4)** Description of physicians' support informed using terms adopted in each study, verbatim. **GP**: General Practitioner. **NHS**: UK's National Health Service. **(5)** Risks of bias reported for unfulfilled MMAT criteria – see Results for details.

3.1 Quality assessment and risk of bias

Assessment of representativeness was based on how the authors of each article described the representativeness of their study's sample. Twelve studies presented low or not-informed sample representativeness [2, 22, 26, 32, 33, 36, 41, 45, 49, 58, 60, and 62]. Two studies had a response rate lower than 50% [2 and 50] and eight did not inform the number of non-respondents [3, 26, 33, 36, 40, and 49]. Further biases include unspecified location of online surveys' respondents [25, 39, and 58] and unspecified gender of AAS users [22, 36, 40, 41, 50, and 53].

3.2 Overall prevalence of AAS users seeking support from physicians

The prevalence rate of AAS users seeking support from physicians obtained from 25 studies was 35.7% (Table 3). Heterogeneity analysis suggested that the dispersion of results is not due to

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random error. A full list of results of prevalence rates' analysis and subgroup comparisons can be found in the supplementary files.

Table 3: Overall prevalence

	N	p%	95%CI	Q	df(Q)	*p* – value	I^2
Overall	25	35.7	28.6 – 43.5	794.3	24	< 0.001	97.0

N = number of studies; p% = prevalence; Q = heterogeneity statistic; df(Q) = Q's degrees of freedom;
I^2 = heterogeneity index

3.3 Time of publication

Selected studies published between 1988 and 2019 were grouped as published before (1988 to 1999) and after (2005 to 2019) the year 2000. No studies published between 2000 and 2004 were selected. The prevalence rate of AAS users seeking support from physicians obtained before the year 2000 was 29.0% and 37.6% after the year 2000 (Table 4). Heterogeneity analysis suggested that when grouped by year of publication the dispersion of results seen in these groups is not due to random error. Subgroup comparisons did not observe a statistically significant difference between these prevalence rates (Q = 86.0, p = 0.354).

Table 4: Prevalence rates by time of publication

	N	p%	95%CI	Q	df(Q)	*p* - value	I^2
Before 2000	5	29.0	16.6 – 45.5	34.3	4	< 0.001	88.2
After 2000	20	37.6	29.8 – 46.0	680.9	19	< 0.001	97.2

N = number of studies; p% = prevalence; Q = heterogeneity statistic; df(Q) = Q's degrees of freedom;
I^2 = heterogeneity index

3.4 Subpopulations of AAS users

The highest prevalence rates of AAS users seeking support from physicians were observed among NSP clients: 48.5%, followed by bodybuilders: 32.3%, gym users: 29.8% and adolescents: 15.0%. Studies with a non-specified population of AAS users shown a prevalence rate of 54.2% (Table 5). Heterogeneity analysis of the two studies with a sample of NSP clients showed that the dispersion of results in this subgroup cannot be attributed only to differences between the studies, but also to random error. Subgroup comparisons show a statistically significant difference between prevalence rates of NSP clients and adolescents (Q = 11.4, p = 0.001).

Table 5: Prevalence rates by subpopulation

	N	p%	95%CI	Q	df(Q)	*p*-value	I^2
NSP clients	2	48.3	23.9 – 73.6	4.0	1	0.044	75.3
Bodybuilders	5	32.3	18.2 – 50.6	41.5	4	<0.001	90.4
Gym users	9	29.8	18.4 – 44.4	212.4	8	<0.001	96.2
Adolescents	3	15.0	6.5 – 31.0	19.4	2	<0.001	89.7
Non-specified	6	54.2	37.8 – 69.6	292.2	5	<0.001	98.3

N = number of studies; p% = prevalence; Q = heterogeneity statistic; df(Q) = Q's degrees of freedom;

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I^2 = heterogeneity index; * *p*-value <0.001

3.5 Type of support sought from physicians

Highest prevalence rates of AAS users seeking support from physicians were observed among AAS users seeking physicians for medical consultations (50.0%) compared to information about AAS (20.2%) (Table 6). Subgroup comparison shown a statistically significant difference between these prevalence rates (Q = 17.6, p < 0.001).

Table 6: Prevalence rates by kind of support sought

	N	p%	95%CI	Q	df(Q)	*p*-value	I^2
Consultation	14	50.0	41.9 – 58.0	291.4	13	<0.001	95.5
Information	11	20.2	12.7 – 30.6	240.8	10	<0.001	95.9

N = number of studies; p% = prevalence; Q = heterogeneity statistic; df(Q) = Q's degrees of freedom;
I^2 = heterogeneity index; * *p*-value <0.001

3.6 Other sources of AAS-related information

In nine studies comparing physicians with the internet as sources of AAS-related information (Table 7), the prevalence rate of AAS users seeking the internet was 52%, and seeking from physicians was 31.4%. The difference was not considered statistically significant (Q = 3.7, p = 0.056).

Table 7: AAS users seeking information from the internet

	N	p%	95%CI	Q	df(Q)	*p*-value	I^2
Internet	9	52.3	33.9 – 70.1	350.9	8	<0.001	97.7
Physicians	9	31.4	22.2 – 42.3	148.9	8	<0.001	94.6

N = number of studies; p% = prevalence; Q = heterogeneity statistic; df(Q) = Q's degrees of freedom;
I^2 = heterogeneity index

Higher prevalence rates of AAS users seeking the internet ($p\%_{Int}$) compared to physicians ($p\%_{Phy}$) for AAS-related information were seen among bodybuilders in the US ($p\%_{Int}$ = 91.8, $p\%_{Phy}$ = 34.1%) [41]; PIED users in Australia ($p\%_{Int}$ = 62.0, $p\%_{Phy}$ = 22.0 [33], and $p\%_{Int}$ = 87, $p\%_{Phy}$ = 56 [26]); adolescents in South Africa ($p\%_{Int}$ = 74.0, $p\%_{Phy}$ = 10.0) [14]; and NSP clients in the UK ($p\%_{Int}$ =33.7, $p\%_{Phy}$ = 5.1) [36]. Small differences in prevalence rates of AAS seeking the internet compared to physicians as sources of AAS-related information were seen among gym users in the United Arab Emirates ($p\%_{Int}$ = 10.0, $p\%_{Phy}$ = 13.0) [2] and Saudi Arabia ($p\%_{Int}$ = 34.8, $p\%_{Phy}$ = 38.0) [3], and among NSP clients in Australia ($p\%_{Int}$ = 40.1, $p\%_{Phy}$ = 51.6) [46]. A higher prevalence of AAS users seeking physicians for information compared to from the internet was seen among gym users in Brazil ($p\%_{Int}$ = 18.6, $p\%_{Phy}$ = 60.3) [40].

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In twelve studies comparing physicians with friends as sources of AAS information (Table 8), the prevalence rate of AAS users seeking friends was 42.8% and it was 28.0% for seeking physicians. The difference was not considered statistically significant (Q = 4.8 p = 0.029).

Table 8: AAS users seeking information from friends

	N	p%	95%CI	Q	df(Q)	*p*-value	I^2
Friends	12	42.8	35.6 – 50.4	116.5	11	<0.001	90.6
Physicians	12	35.5	19.0 – 39.1	252.6	11	<0.001	95.6

N = number of studies; p% = prevalence; Q = heterogeneity statistic; df(Q) = Q's degrees of freedom; I^2 = heterogeneity index

Higher prevalence rates of AAS users seeking friends as sources of information were seen among adolescents in the US ($p\%_{Fri}$ = 51.0, $p\%_{Phy}$ = 29.0) [27]; gym users in the United Arab Emirates ($p\%_{Fri}$ = 50.0, $p\%_{Phy}$ = 13.0) [2]; PIED users in Australia ($p\%_{Fri}$ = 55.0, $p\%_{Phy}$ = 22.0) [33]; adolescents in South Africa ($p\%_{Fri}$ = 66.0, $p\%_{Phy}$ = 10.0) [14]; bodybuilders in Brazil ($p\%_{Fri}$ = 35.0, $p\%_{Phy}$ = 10.5) [49]; and gym users in Pakistan ($p\%_{Fri}$ = 49.7, $p\%_{Phy}$ = 1.7) [53]. Small differences in prevalence rates of AAS seeking friends compared to physicians as sources of information were seen among powerlifters in the US ($p\%_{Fri}$ = 47.0, $p\%_{Phy}$ = 53.0) [60]; gym users in Germany ($p\%_{Fri}$ = 54.2, $p\%_{Phy}$ = 56.0) [50]; NSP clients in Australia ($p\%_{Fri}$ = 42.9, $p\%_{Phy}$ = 51.6) [46]; and gym users in Saudi Arabia ($p\%_{Fri}$ = 41.5, $p\%_{Phy}$ = 38.0) [3]. A higher prevalence of AAS users seeking physicians for information compared to from friends was seen among members of online bodybuilding groups in the UK ($p\%_{Fri}$ = 6.0, $p\%_{Phy}$ = 50.0) [22] and gym users in Brazil ($p\%_{Fri}$ = 36.8, $p\%_{Phy}$ = 60.3) [40].

In five studies comparing physicians with coaches as sources of AAS information (Table 9), the prevalence rate of AAS users seeking information from coaches was 21.3% and 23.8% from physicians. The difference was not considered statistically significant (Q = 0.1, p = 0.065).

Table 9: AAS users seeking information from coaches

	N	p%	95%CI	Q	df(Q)	*p*-value	I^2
Coaches	5	21.3	12.2 – 34.6	114.9	4	<0.001	96.5
Physicians	5	23.8	11.6 – 42.6	200.3	4	<0.001	98.0

N = number of studies; p% = prevalence; Q = heterogeneity statistic; df(Q) = Q's degrees of freedom; I^2 = heterogeneity index

Higher prevalence rates of AAS users seeking coaches as sources of information were found among adolescents in South Africa ($p\%_{Coa}$ = 40.0, $p\%_{Phy}$ = 10.0) [14]. Small differences in prevalence rates were observed among gym users in Saudi Arabia ($p\%_{Coa}$ = 23.7, $p\%_{Phy}$ = 38.0) [3] and Pakistan ($p\%_{Coa}$ = 8.4, $p\%_{Phy}$ = 1.7) [53]. A higher prevalence of AAS users seeking physicians for information compared to from coaches was seen among NSP clients in Australia ($p\%_{Coa}$ = 12.5, $p\%_{Phy}$ = 51.6) [46] and gym users in Brazil ($p\%_{Coa}$ = 36.1, $p\%_{Phy}$ = 60.3) [40].

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3.7 Location of studies and gender of AAS users

The small number of studies from countries other than the US, Australia and the UK as well as with female AAS users precluded a subgroup meta-analysis of these data.

The majority of studies were published in the United States (n =6) with an aggregated population of 2,473 AAS users. Prevalence rates of those seeking support from physicians in the US ranged from 10% to 29% among adolescents [27 and 51], 24.1% to 53% among powerlifters and bodybuilders [5, 41, and 60] and 66% among a non-specified population of AAS users [12]. Four studies were conducted in the UK, with an aggregated population of 734 AAS users. Prevalence rates of AAS users seeking support from physicians in the UK ranged from 5.1% to 35% among gym users [32 and 36], 45% among NSP clients [23] and 50% among members of bodybuilding groups in social media [22]. The highest overall prevalence rate of AAS users seeking support from physicians was seen in studies published in Australia (n = 3), with an aggregated population of 911 AAS users: It ranged from 56.0% to 77.0% among non-specified PIED users [26 and 33], and 51.6% among NSP clients [46]. Two Brazilian studies, with an aggregated population of 235 AAS users, reported the following prevalence rates of those seeking support from physicians: 10.5% among bodybuilders [49] and 63.9% among gym users [40]. In four trans-regional studies, with an aggregated population of 1,549 AAS users, prevalence rates of those seeking support from physicians ranged from 21.8% to 69.2% [36, 25, 58, and 62]. Finally, prevalence rates of AAS users seeking support from physicians were represented by a single study from the following countries: Germany, 56% (84 gym users) [50]; Pakistan, 1.7% (512 gym users) [53]; Poland, 11% (18 recreational athletes; 45); Saudi Arabia, 38% (474 gym users) [3]; South Africa, 10% (96 gym users) [14]; and United Arab Emirates, 38% (34 gym users) [2].

Five studies informed the number of female AAS users included in the study: 45 adolescents and 3 bodybuilders in the US [5 and 51]; 3 gym users in the UK [32]; and 65 non-specified AAS users from trans-regional studies [39 and 62] Differences in physician-seeking behaviour between male and female AAS users were described by two studies. Among a cohort of 17 bodybuilders (3 females) in the US, 100% of the females sought a physician for routine medical examinations (against 35.7% of males); 33.3% of the females sought a physicians for treatment of medical conditions (against 0% of males); and 0% of females sought a physician due to trauma (42.9% males) [5]. A trans-regional study with a non-specified population of 312 AAS users (59 females) observed that males were less likely to seek support from physicians than females (odds ratio = 0.09, $p < 0.01$) [62].

3.8 Meta-regression

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A meta-regression was conducted using four different models to assess the influence of each group of co-variants over the dispersion of prevalence rates (Table 10). The influence of each group of co-variants over the overall prevalence is expressed by the R^2 index: 8% for publication year, 0% for region and subpopulations, and 27% for the type of support sought. The overall prevalence rate of AAS seeking support from physicians was significantly influenced by prevalence rates reported after the year 2000; prevalence rates from Asia; prevalence rates of gym users; and the prevalence rate of AAS users seeking information.

Table 10: Meta-regression

	Coefficient	SE	*p*-value	R^2
Publication year				0.08
After 2000*	-0.5071	0.1778	0.0044⁺	
Before 2000	-0.3896	0.4036	0.3377	
Region				0.00
Europe*	-0.6156	0.3861	0.1109	
Africa	-1.5362	0.9891	0.1204	
Asia	-3.4078	0.9899	0.0006⁺	
Middle East	0.1301	0.7343	0.8594	
North America	-0.0776	0.5331	0.8843	
Oceania	1.0904	0.6329	0.0849	
South America	-0.0056	0.7572	0.9942	
Trans-Region	0.4277	0.5762	0.4579	
Subpopulations				0.00
Gym users*	-0.8208	0.2881	0.0044⁺	
Adolescents	-0.9135	0.5580	0.1016	
Bodybuilders	0.0821	0.4837	0.8652	
NSP clients	0.7535	0.6268	0.2293	
Non-specified	0.9862	0.4348	0.0233	
Type of support				0.27
Consultation*	-0.0151	0.1891	0.9362	
Information	-0.3184	0.2931	<0.001⁺	

* reference category ⁺p < 0.01

4. Discussion

Data from this meta-analysis of data extracted from 25 studies suggests that the overall prevalence of AAS users seeking support from physicians is 35.7%. According to heterogeneity analysis, the dispersion in prevalence rates among the 25 eligible studies is due to differences between studies (e.g. time of publication, location, sampling strategy, etc.) and not random error. Results must be considered in the light of a small sample of highly heterogenic studies. Subgroup analysis suggested that the proportion of AAS users seeking support from physicians before (29.0%) and after the year 2000 (37.6%) is not significantly different. The analysis suggests that the overall proportion of AAS users seeking support from physicians has remained somehow stable through time, despite the availability of new sources of information such as the

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internet and social media. This comparison, however, involved a small number of studies (n = 5) published before the year 2000 against 25 studies published after 2000. Likewise, the sum of studies published before 2000 had an aggregated population of 411 AAS users, against an aggregated population of 6,709 AAS users in the sum of studies published after 2000.

The majority of AAS users sought physicians for consultations instead of information about AAS, possibly reflecting physicians' role in diagnosing and treating adverse effects of AAS [13]. The overall prevalence of AAS users seeking physicians for information was not considered significantly different from the prevalence of AAS users seeking other sources of information. However, only a minority (n = 3) of studies [22, 40, and 46] described higher prevalence of AAS users seeking physicians for information, while 9 studies [2, 14, 26, 27, 33, 36, 41, 49, and 53] described higher prevalence of AAS users seeking the internet, friends or coaches for AAS-related information. This difference is more aligned with findings of previous studies [9, 10, 18, 30, 34, 52, and 54], suggesting that the absence of a statistically significant difference between the prevalence of AAS users seeking physicians or other sources of information in this meta-analysis was limited by the number and heterogeneity of selected studies, leading to a wide and overlapping dispersion of prevalence rates.

Among different subpopulations of AAS users seeking support from physicians, higher prevalence rates were seen among NSP clients (48.3%), possibly suggesting a positive association between the use of harm reduction services and engagement with other health services [6 and 29]. Findings also suggest that about one third of bodybuilders (32.3%) and gym goers (29.8%) who use AAS seek support from physicians, while only 15% of adolescent AAS users seem to adopt this behaviour. This is in keeping with previous studies reporting lower levels of engagement with physicians among young AAS users [28 and 37].

Ranges of prevalence rates of AAS users seeking support from physicians varied widely in most countries. Among studies published in the US, prevalence rates of AAS users seeking support from physicians ranged from 10% [51] to 66% [12]. Similarly wide dispersions were observed in the UK – 5.1% [36] to 50% [22] – and Brazil – 10.5% [49] to 63.9% [40]. The smallest dispersion of prevalence was seen among Australian studies: 51.6% [46], 56% [26] and 77% [33]. Several countries had one single paper examined by this review, compromising the generalisation of results by region – as exemplified by the extremely low prevalence of AAS users seeking support from physicians in the only paper from Asia (1.7%) [53]. Likewise, the small number of studies reporting prevalence rates of female AAS users seeking support from physicians precluded further analysis of this behaviour among women.

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4.1 Limitations

Limitations of this meta-analysis include the heterogeneity of methods utilised by selected studies to access AAS users' physician-seeking behaviour, including online surveys with a non-specified population of AAS users, a longitudinal observation of a cohort of bodybuilders and questionnaires filled by high-school students. Different types of support sought from physicians, such as information and regular medical checks, were also examined together in the estimate of the overall prevalence. Furthermore, a significant variety of prevalence rates was seen among different subpopulations of AAS users, suggesting that the overall prevalence of AAS users seeking support from physicians might not be generalisable across these groups – and that different subpopulations of AAS users might adopt very distinct health strategies. Likewise, available data compromised regional generalisations of physician-seeking behaviour, namely among female AAS users.

Several factors could have influenced the prevalence rates of AAS users seeking support from physicians seen in this study. These include – but are not limited to – AAS users' perception of a need to seek medical support, local provisions of health services, access to doctors and other sources of support, cultural and legal frameworks regarding the use of AAS. Therefore, a higher number of high-quality studies and in-depth analysis are necessary to understand the prevalence rates of different populations of AAS users seeking support from physicians described in this meta-analysis.

5. Conclusion

This meta-analysis was the first to systematically investigate prevalence rates of AAS users seeking support from physicians worldwide. It examines studies published since 1995, describing physician-seeking behaviours of several subpopulations of AAS users in a diversity of settings and locations. The findings suggest that about one third of AAS users seek support from physicians.

Findings from this study suggests that seeking support from physicians is a behaviour adopted by a significant proportion AAS users but possibly dismissed by an even greater number of them. It highlights the importance of understanding the help-seeking behaviours of different populations of AAS users and improve the access of this population to medical support.

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Acknowledgements

The authors thank the participants and authors of the selected studies.

Contributors

JA conceptualised the study and performed the systematic review with support from AK and PD. JA led the meta-analysis which was discussed with AK and PD. All authors contributed to the preparation of the paper and approved the final version. Writing was led by JA.

Funding

The first author was funded by King's College London International Scholarship (RDPL3PRPCA - 2019-20).

Competing interests

None

Patient and public involvement

It was not appropriate to involve patients or the public in the design, conduct, reporting or dissemination plans of this study.

Ethics approval

As a review of the literature, this study was exempt from ethical approval by King's College London Research Ethics Office.

Provenance and peer reviewed

Not commissioned; externally peer reviewed.

Data sharing statement

Data from meta-analysis and subgroup comparisons is available in the supplementary files.

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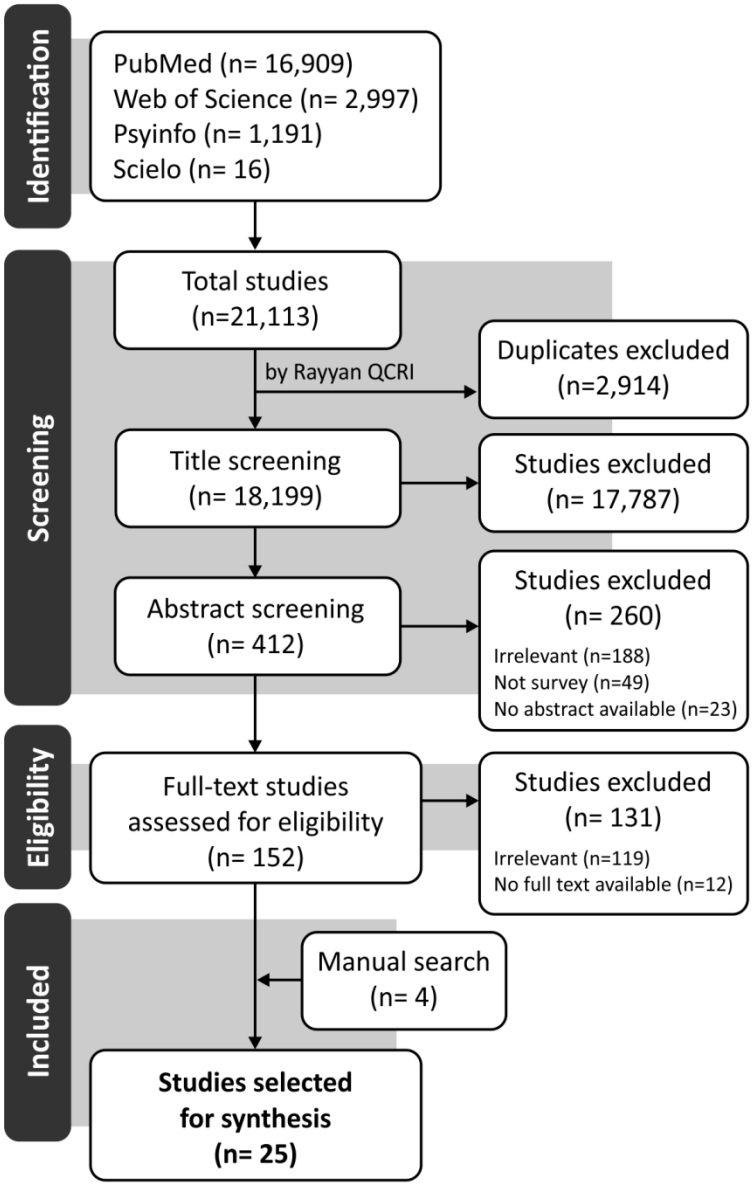
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Search Strategy: ((A1 OR A2) NOT (A3))			
field(s)	**booleans**	**search terms**	
MESH terms		agents, anabolic	A1
	OR	doping in sport	
	OR	drugs, performance enhancing	
	OR	testosterone congeners	
Title OR Abstract		anabolic steroid*	A2
	OR	anabolic androgenic	
	OR	anabolic agent*	
	OR	anabolic drug*	
	OR	androgen* abuse	
	OR	AAS	
	OR	performance enhanc*	
	OR	image enhanc*	
	OR	synthetic androgen*	
	OR	synthetic testosterone	
	OR	testosterone ester*	
	OR	doping	
	OR	designer steroid*	
	OR	"non prescript*" steroid*	
	OR	"non medic*" steroid*	
	OR	"non medic*" androgen*	
	OR	"non prescript*" androgen*	
Title OR Abstract		animal*	A3
	OR	bovine	
	OR	in vitro	
	OR	mice*	
	OR	pig	
	OR	pigs	
	OR	rat	
	OR	rats	
	OR	spectrometry	
	OR	nano*	
	OR	conductor*	
	OR	electro*	
	OR	optic*	
	OR	hydrogen	
	OR	superconductor*	
	OR	semiconductor	
	OR	catalyst*	
	OR	thermoelectric	
	OR	ions	

140x220mm (300 x 300 DPI)



104x160mm (300 x 300 DPI)

Study name	Events	Sample size	Event rate	Logit event rate	Std Error
Yesalis 1988	8	15	0.533	0.134	0.518
Johnson 1989	28	95	0.295	-0.872	0.550
Tanner 1995	18	184	0.098	-2.222	0.248
Korkia 1997	39	110	0.355	-0.599	0.199
Augé 1999	7	20	0.350	-0.619	0.469
Perry 2005	50	207	0.242	-1.144	0.162
Parkinson 2006	185	500	0.370	-0.532	0.093
Striegel 2006	47	84	0.560	0.239	0.220
Cohen 2007	1290	1955	0.660	0.663	0.048
Al-Falasi 2008	13	34	0.382	-0.480	0.353
Larance 2008	46	60	0.767	1.190	0.305
Posiadala 2010	2	18	0.111	-2.079	0.750
Gradidge 2011	10	96	0.104	-2.152	0.334
Ip 2011	284	506	0.561	0.246	0.090
Santos 2011	4	41	0.098	-2.225	0.526
Hope 2013	178	395	0.451	-0.198	0.101
Mooney 2016	1	23	0.043	-3.091	1.022
Rowe 2016	312	605	0.516	0.063	0.081
Westerman 2016	160	231	0.693	0.812	0.143
Zahnow 2017	68	312	0.218	-1.278	0.137
Althobiti 2018	180	474	0.380	-0.491	0.095
Hill 2019	108	216	0.500	0.000	0.136
Jacka 2019	138	246	0.561	0.245	0.128
Pereira 2019	124	194	0.639	0.572	0.149
Uddin 2019	9	512	0.018	-4.023	0.336

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Year of publication	Q	df(Q)	p-value
1988-1999 - 2005-2019	0.86	1	0.354

Region	Q	df(Q)	p-value
Africa - Asia	15.587	1	0.000
Africa - Europe	21.526	1	0.000
Africa - M.East	23.018	1	0.000
Africa - N.America	5.039	1	0.025
Africa - Oceania	**40.826**	**1**	**0.000**
Africa - S.America	0.91	1	0.340
Africa - T.Regional	14.52	1	0.000
Asia - Europe	88.504	1	0.000
Asia - M.East	102.799	1	0.000
Asia - N.America	26.027	1	0.000
Asia - Oceania	**121.987**	**1**	**0.000**
Asia - S.America	5.089	1	0.024
Asia - T.Regional	55.078	1	0.000
Europe - M.East	0.421	1	0.516
Europe - N.America	0.322	1	0.570
Europe - Oceania	**6.309**	**1**	**0.012**
Europe - S.America	0.094	1	0.759
Europe - T.Regional	0.132	1	0.716
M.East - N.America	0.119	1	0.731
M.East - Oceania	**14.08**	**1**	**0.000**
M.East - S.America	0.043	1	0.835
M.East - T.Regional	0.562	1	0.454
N.America - Oceania	3.195	1	0.074
N.America - S.America	0.004	1	0.950
N.America - T.Regional	0.528	1	0.467
Oceania - S.America	0.684	1	0.408
Oceania - T.Regional	1.659	1	0.198
S.America - T.Regional	0.167	1	0.683

Type of support	Q	df(Q)	p-value
information - consultatio	17.557	1	0.000

Population	Q	df(Q)	p-value
Gym - adolescents	2.310	1	0.129
Gym - bodybuilders	0.056	1	0.814
Gym - NSP	5.127	1	0.022
Gym - n/s	4.784	1	0.029
Adolescents - bodybuilde	2.618	1	0.106
Adolescents - NSP	11.444	1	0.001
Adolescents - n/s	10.555	1	0.001
Bodybuilders - NSP	2.738	1	0.098
Bodybuilders - n/s	3.087	1	0.079
NSP - n/s	0.393	1	0.531

Gym = gym users and recreational athletes n/s = not specified
NSP = clients of needle and syringe preogrammes

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Physicians vs Internet	nAAS	% Phy	**n Phy**	% Int	**n Int**
Perry 2005	207	24.1	**50**	91.8	**190**
Al-Falasi 2008	34	13	**4**	10	**3**
Larance 2008	60	22	**13**	62	**37**
Gradidge 2011	96	10	**10**	74	**71**
Mooney 2016	23	5.1	**1**	33.7	**8**
Rowe 2016	605	51.6	**312**	40.1	**243**
Althobiti 2018	474	38	**180**	34.8	**165**
Jacka 2019	246	56	**138**	87	**214**
Pereira 2019	194	60.3	**117**	18.6	**36**

Physicians vs Friends	nAAS	% Phy	**n Phy**	% Friends	**n Friends**
Yesalis 1988	15	53	**8**	47	**7**
Johnson 1989	95	29	**28**	51	**48**
Striegel 2006	84	56	**47**	54.2	**46**
Al-Falasi 2008	34	13	**4**	50	**17**
Larance 2008	60	22	**13**	55	**33**
Gradidge 2011	96	10	**10**	66	**63**
Santos 2011	41	10.5	**4**	35	**14**
Rowe 2016	605	51.6	**312**	42.9	**260**
Althobiti 2018	474	38	**180**	41.5	**197**
Hill 2019	216	50	**108**	6	**13**
Pereira 2019	194	60.3	**117**	36.8	**71**
Uddin 2019	512	1.7	**9**	49.7	**254**

Physicians vs Coaches	nAAS	% Phy	**n Phy**	% Coach	**n Coach**
Gradidge 2011	96	10	**10**	40	**38**
Rowe 2016	605	51.6	**312**	12.5	**76**
Althobiti 2018	474	38	**180**	23.7	**112**
Pereira 2019	194	60.3	**117**	36.1	**70**
Uddin 2019	512	1.7	**9**	8.4	**43**

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PRISMA 2020 Main Checklist

Topic	No.	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Section 1, Lines 126-128
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section 1, Lines 120-123
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section 1, Lines 123-125, 126-128
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 2, Line 157 (Table 1)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 2, Line 141
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 2, Line 144 (Figure 1)
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2, Lines 146-150
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 2, Lines 159-163
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 2, Lines 159-163
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 2, Lines 159-163

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Topic	No.	Item	Location where item is reported
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2, Lines 165-174
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Section 2, 176-200
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	Section 2, 198-200
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 2, 176-200
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 2, 176-200
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 2, 176-204
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Section 2, Lines 182-190
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Section 2, 201-204
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Section 2, Lines 201-204
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Section 2, 201-204
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 2, Line 154 (Figure 2)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	n/a
Study characteristics	17	Cite each included study and present its characteristics.	Section 3, Line 217 (Table 2)
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section 3, Lines 226-233
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Section 3, Line 217 (Table 2)
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section 3, Lines 226-233

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Topic	No.	Item	Location where item is reported
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Section 3, Lines 235-390
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Section 3, Lines 240-375
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Section 3, Lines 365-372
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Section 3, Lines 240-375
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Section 3, Lines 240-375
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section 4, Lines 377-419
	23b	Discuss any limitations of the evidence included in the review.	Section 4, Lines 420-438
	23c	Discuss any limitations of the review processes used.	Section 4, Lines 420-438
	23d	Discuss implications of the results for practice, policy, and future research.	Section 5, Lines 446-449
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Section 2, Lines 130-132
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Section 2, Lines 130-132
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	n/a
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Section 6, Lines 461-453
Competing interests	26	Declare any competing interests of review authors.	Section 6, Line 453
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Section 3, Lines 238-239

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PRIMSA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesize results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. MetaArXiv. 2020, September 14. DOI: 10.31222/osf.io/v7gm2. For more information, visit: www.prisma-statement.org

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BMJ Open

Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-056445.R1
Article Type:	Original research
Date Submitted by the Author:	07-Mar-2022
Complete List of Authors:	Amaral, Julio Mario; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions Kimergård, Andreas; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions Deluca, Paolo; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions
Primary Subject Heading:	Medical management
Secondary Subject Heading:	Addiction, General practice / Family practice, Health services research
Keywords:	Substance misuse < PSYCHIATRY, PRIMARY CARE, GENERAL MEDICINE (see Internal Medicine)

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Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Short title: Anabolic steroid users seeking support from physicians

Word count: 4,333

PROSPERO registration number: CDR42020177919

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Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Abstract

Objectives: To estimate the overall prevalence of androgenic-anabolic steroids (AAS) users seeking support from physicians. Secondary objectives are to compare this prevalence in different locations and among subpopulations of AAS users, discussing some of the factors that could have influenced the engagement of AAS users with physicians.

Design: Systematic review and meta-analysis.

Data sources: MEDLINE, PsycINFO, Web of Science and SciELO were searched in January 2022.

Eligibility criteria: Quantitative and qualitative studies reporting the number of AAS users who sought support from physicians, with no restrictions of language or time of publication.

Data extraction and synthesis: Two independent reviewers extracted data and assessed the quality of studies, including publication bias. A random-effects meta-analysis was performed to estimate the overall prevalence of AAS users seeking support from physicians, followed by pooled prevalence rates by studies' location and the subpopulation of AAS users.

Results: We identified 36 studies published between 1988 and 2021, involving 10,101 AAS users. The estimated overall prevalence of AAS users seeking support from physicians is 37.12% (95% CI 29.71 to 44.52). Higher prevalence rates were observed in studies from Australia (67.27%; 95%CI 42.29 to 87.25) and among clients of the needle and syringe exchange programme (54.13%; 95% CI 36.41 to 71.84). The lowest prevalence was observed among adolescent AAS users (17.27%; 95% CI 4.80 to 29.74).

Conclusion: Our findings suggest that about one-third of AAS users seek support from physicians, with remarkable differences between locations and subpopulations of AAS users. Further studies should investigate the factors influencing the engagement of AAS users with physicians.

Registration: PROSPERO - CRD42020177919.

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Strengths and limitations of this study

- This is the first meta-analysis of studies investigating the prevalence of AAS users seeking support from physicians.
- It corroborates the existence of a wide range of prevalence rates of AAS users seeking support from physicians, depending on the studies' location and subpopulation of AAS users.
- It adds to the knowledge of help-seeking behaviours of people using AAS and contributes to the research of factors influencing the access of AAS users to health services.
- Limitations include significant risks of bias from selected studies and a scarcity of data from several countries.

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Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Introduction

Anabolic-androgenic steroids (AAS) are synthetic androgens with several potential effects. AAS can be used to treat conditions such as male hypogonadism, pathological loss of muscle mass and anaemia [1], with findings suggesting their efficacy in the treatment of depression [2] and conditions associated with type 2 diabetes [3]. Beneficial effects of AAS include the increasing of muscle mass, feelings of wellbeing and boosted energy, enhancement of body image, and improvement of athletic and occupational performance [4,5]. However, the use of AAS can increase the risk of several adverse health conditions such as acne, testicular atrophy, gynecomastia, clitoromegaly, hypomania, anxiety, dyslipidaemia and high haematocrit – therefore increasing the risk of myocardial infarction and stroke [6]. Despite the risks, many AAS users refrain from seeking physicians for AAS-related information or to treat health conditions potentially associated with the use of AAS [7]. Among factors possibly influencing the prevalence of AAS seeking support from physicians are the legal status of AAS, AAS users' engagement with health services and their perceptions of the services provided by physicians to people using AAS [8]. In countries where the possession of AAS without a medical prescription is illegal, it is reasonable to expect that some AAS users will refrain from disclosing the use of AAS to a physician [9]. The legal status of AAS can also influence the service provided to AAS users by physicians, as doctors are usually not allowed to prescribe AAS for the purposes of enhancement or to regulate hormonal levels after the use of AAS, a practice also known as post-cycle therapy (PCT) [10]. Whilst physicians are trained to treat the use of other illegal substances such as heroin and cocaine, many of them admit a lack of training and experience in recognising and treating adverse effects of AAS [11–14]. Another reason given by some AAS users to refrain from seeking medical support is the stigma and judgmental attitudes experienced in their contact with health professionals [13,15]. Due to these factors – and possibly others, such as the provision of health services in their locality, financial limitations, etc. – some AAS users rely on self-conducted research to manage their use and adverse effects of AAS and/or seek the support of informal sources such as friends and online forums [16,17]. Finally, some AAS users reported not seeking physicians simply because they did not feel the need to do so, due to an absence of adverse effects or to a perception that these effects can be managed without the help of a medical professional [13].

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In addition to the legal status of AAS use and the access to health services in different countries, different help-seeking behaviours are seen across subpopulations of AAS users [18]. Younger AAS users, for instance, seem to be less likely to engage with health services [19]. Some strength athletes tend to rely on other athletes perceived as knowledgeable about AAS [20–22], and AAS users who are clients of the needle and syringe exchange programme (NSP) seem to be more likely to interact with health services [23,24]. However, to this date, no systematic comparison of the prevalence of AAS users seeking the support of physicians has been produced.

The objective of this study is to estimate the overall prevalence of AAS users seeking support from physicians by conducting a systematic review and meta-analysis of surveys and interviews with AAS users. The secondary objectives are to compare the prevalence of AAS users seeking support from physicians in different locations and among subpopulations of AAS users.

Methods

This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [25], and its protocol is registered in PROSPERO (registration number CRD42020177919). As a review of the literature, this study is exempted from ethical clearance by King's College London research ethics office.

A search strategy was designed to retrieve studies describing surveys and interviews with people using AAS. Searches were performed on MEDLINE, PsycINFO, Web of Science and SciELO in January 2022 with no restrictions on the date, location and language of studies. The search algorithm adapted to each database can be found in the supplementary material. A supplementary search was performed on the reference lists of eligible studies. Two independent researchers (JA and AK) performed the screening and data extraction. Disagreements were resolved by discussion with the third researcher (PD).

Results of the searches were exported to Rayyan QCRI [26] for screening and removing of duplicates. Titles, abstracts and the full text of studies were screened for eligibility against our inclusion and exclusion criteria (Table 1). A spreadsheet was utilised to summarise descriptive data of selected studies, i.e. location of study, subpopulation of AAS users, the number of participants, number and gender of AAS users, and the number of AAS users who sought support from physicians. Two independent reviewers (JA and AK) performed the data extraction and assessed the risk of bias. Disagreements were resolved by the third reviewer (PD).

Table 1: Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria

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Quantitative and qualitative studies with people using AAS for the purposes of image and performance enhancement.	Studies with patients using AAS for treatment of medical conditions. Studies with prisoners, who have limited access to external health services. Studies with animal subjects and in vitro analysis of AAS effects.
Studies informing the number of AAS users seeking physicians to receive information and prevent and treat AAS-related health conditions. Studies where the source of support or information can be understood as being a physician (e.g. doctor, medical doctor, health professional).	Studies not informing the number of AAS users in the sample who seek support from physicians. Studies where the contact with the physician cannot be understood as participant's choice, such as involuntary admissions, post-mortem analysis and cohorts of patients selected for studies of specific health conditions.
Peer-reviewed studies. Studies published at any time. Studies published in any language, as long as it is possible to retrieve relevant data from the authors or articles.	'Grey literature' (non-peer-reviewed studies and reports). Case studies and interviews with a single AAS user.

The risk of bias was assessed in two stages. Initially, the quality and internal validity of studies were evaluated using the Mixed Methods Appraisal Tool (MMAT)[27]. The MMAT is comprised of five quality-assessment criteria: (i) 'Is the sampling strategy relevant to address the research question?'; (ii) 'Is the sample representative of the target population?'; (iii) 'Are the measurements appropriate?'; (iv) 'Is the risk of nonresponse bias low?'; and (v) 'Is the statistical analysis appropriate to answer the research question?'. In the second stage, studies were assessed for risk of publication bias.

The data synthesis for meta-analysis was performed extracting the number of AAS users who reported seeking any kind of support from physicians in the selected studies. For the purpose of effect size calculations, the number of AAS users in each study was used as the number of participants of interest, and the number of those who informed seeking a physician was used as the number of cases in each study. When a study informed more than one number or percentage of AAS users seeking support from physicians, the higher value of male AAS users seeking medical support was considered, as only a few studies included female participants.

A meta-analysis was performed to estimate the overall prevalence of AAS users seeking support from physicians. A random-effects model was chosen to better incorporate the dispersions of prevalence rates across studies and the different approaches to the research question [28]. Heterogeneity was measured using the I^2 index, which describes the percentage of variation of prevalence rates across a group of studies that is due to differences between studies (e.g. different sample sizes, populations or methods) [29,30]. A Baujat plot was produced to identify

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studies that could influence the overall result [31]. The secondary outcomes were measured with the prevalence rates of studies grouped by location and subpopulation of AAS users, and the type of support sought from physicians was discussed in a narrative synthesis.

Univariate and multivariate meta-regression were performed to measure the impact of the moderators 'location of study' and 'subpopulation of AAS users' on the variance of the overall prevalence rate. For each moderator variable, the category with the highest number of studies was used as a contrast. Meta-analysis and meta-regression were conducted in R [32] using the metafor package [33]. A full description of the codes can be found in the supplementary material.

Patient and public involvement

It was not appropriate to involve patients or the public in the design, conduct, reporting or dissemination plans of this study.

3. Results

The searches identified 11,906 studies. After the removal of duplicates, 9,876 studies were screened by title and abstract. Among 285 full-text studies were assessed for eligibility, 31 were included in the review. A supplementary search on the reference list of included papers and previous reviews led to the inclusion of another five studies. A total of 36 studies were included in the review, as shown in flowchart of Figure 1.

Figure 1: Flowchart of the inclusion of studies in the review

Table 2 shows the summary characteristics of the studies included in this review. The studies were published between 1988 and 2021, with a total number of AAS users (nAAS) = 10,101, being 9,278 (91.85%) males, 179 (1.77%) females and 644 (6.38%) whose sex was not reported. Eight (22.22%) of the selected studies were located in the USA, six (16.67%) in continental Europe, five (13.89%) in the UK, four (1.11%) in Australia, three (8.33%) in Brazil, five (13.89%) in Africa, Asia or the Middle East and five (13.89%) studies were trans-regional. The selected studies presented a wide range of nAAS (median = 97.50; range = 4 to 2,385) and number of AAS users seeking support from physicians (nPhy; median = 31.50; range = 1 to 1,290). Regarding the subpopulation of AAS users, the most common were studies with nonspecific AAS users such as gym users, recreational athletes or people using AAS (21; 58.33%), followed by those with strength athletes such as bodybuilders, powerlifters or weightlifters (9; 25.00%), studies with adolescents (4; 11.11%) and studies with clients of the NSP (2; 5.56%).

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The majority of studies with adolescents and strength athletes were located in the USA, followed by a single study from different countries. The two studies with NSP clients were located in either the UK or Australia. All the five trans-regional regional studies were conducted with a nonspecific subpopulation of AAS users.

Table 2: Summary characteristics of selected studies

Authors, year	Location	Subpopulation	n	nAAS	nPhys n (%)
Yesalis et al., 1988 [34]	US	strenght athletes	45	15 M	8 (53.33)
Johnson et al., 1989 [35]	US	adolescents	853	95 M	28 (29.47)
Kisling et al., 1989 [36]	Denmark	nonspecific	157	85 M	21 (24.71)
Lindstrom et al., 1990 [37]	Sweden	strenght athletes	138	138 M	12 (8.70)
Terney et al., 1990 [38]	US	adolescents	2113	94 (67 M/27 F)	5 (5.32)
Tanner et al., 1995 [39]	US	adolescents	6930	184 (139 M/45 F)	33 (17.93)
Korkia et al., 1997 [40]	UK	nonspecific	1667	110 (97 M/13 F)	39 (35.45)
Bolding et al., 1999 [41]	UK	nonspecific	1004	81 M	25 (30.86)
Augé et al., 1999 [42]	US	strenght athletes	17	17 (14 M/3 F)	8 (47.05)
Peters et al., 1999 [43]	Australia	nonspecific	100	100 (94 M/6 F)	42 (42.00)
Perry et al., 2005 [44]	US	strenght athletes	207	207*	46 (22.22)
Parkinson et al., 2005 [45]	Trans-region	nonspecific	500	500 (494 M/6 F)	185 (37.00)
Pope et al., 2004 [7]	US	strenght athletes	80	43 M	16 (37.21)
Striegel et al., 2006 [46]	Germany	nonspecific	621	84 (75 M/9 F)	47 (55.95)
Cohen et al., 2007 [47]	US	nonspecific	1955	1955 M	1290 (65.98)
Al-Falasi et al., 2008 [48]	UAE	nonspecific	154	34 M	4 (11.76)
Larance et al., 2008 [49]	Australia	nonspecific	60	60 M	46 (76.66)
Posiadała et al., 2010 [50]	Poland	nonspecific	50	18 M	2 (11.11)
Gradidge et al., 2011 [51]	South Africa	adolescents	100	4 M	1 (25.00)
Ip et al., 2011[52]	Trans-region	nonspecific	1277	506 M	387 (76.48)
Santos et al., 2011 [53]	Brazil	strenght athletes	123	41 M	4 (9.76)
Hope et al., 2013 [54]	UK	NSP clients	395	395 M	178 (45.06)
Raschka et al., 2013 [55]	Germany	nonspecific	484	79 (62 M/ 17 F)	30 (37.97)
Rowe et al., 2016 [56]	Australia	NSP clients	605	605 M	382 (63.14)
Westerman et al., 2016	Trans-region	nonspecific	231	231 M	153 (66.23)
Mooney et al., 2017 [57]	UK	nonspecific	377	26*	1 (3.85)
Zahnow et al., 2017 [19]	Trans-region	nonspecific	195	195*	68 (34.87)
Althobiti et al., 2018 [58]	Saudi Arabia	nonspecific	4860	476 M	181 (38.0)
Hill et al., 2019 [13]	UK	strenght athletes	350	216*	91 (42.0)
Jacka et al., 2019 [59]	Australia	nonspecific	267	267 M	237 (88.76)
Macedo et al., 2019 [60]	Brazil	nonspecific	40	25 M	9 (36.00)
Pany et al., 2019 [61]	India	strenght athletes	74	74 M	24 (32.43)
Pereira et al., 2019 [62]	Brazil	nonspecific	719	194 (149 M/45 F)	117 (60.31)
Uddin et al., 2019 [63]	Pakistan	nonspecific	841	512 M	9 (1.76)
Bonnecaze et al., 2020 [64]	Trans-region	nonspecific	2385	2385 M	1047 (43.90)
Jokipalo et al., 2021 [65]	Finland	strenght athletes	50	50 (42 M/8 F)	15 (30.0)

AAS: androgenic-anabolic steroids. nAAS: number of AAS users in each study. nPhys: number of AAS users who informed seeking support from physicians. M = males. F = females. NSP: Needle and syringe exchange programme. * Sex of AAS users not informed.

Risk of bias

According to the MMAT quality criteria, only seven (19.44%) studies did not present a risk of bias. The most common risks were non-representative sample sizes of AAS users and inappropriate measurements of the number of AAS seeking support from physicians. One of the reasons for the high frequency of inappropriate measurements is due to the fact that, in some studies, the number of AAS seeking support from physicians was not clearly stated, and had to be extrapolated from the study's total sample size, including users and non-users of AAS (see full assessment in the supplementary material, eTable 1).

There was no evidence of publication bias, as indicated by visual inspection of the funnel plot (Figure 2), Egger's test for asymmetry (bias coefficient = 0.937, p = 0.349, 95% CI 0.49 to 0.45), and Rank correlation test (τ = -0.086, p = 0.4731).

Figure 2: Funnel plot of studies included in the analysis with pseudo 95% confidence intervals

Meta-analysis of the overall prevalence of AAS users seeking support from physicians

The overall prevalence of AAS users seeking support from physicians obtained from 36 studies is 37.12% (SE = 0.38, p < 0.001, 95% CI 27.71 to 44.52; I^2 = 98.03). The smallest prevalence rate (1.76%, 95% CI 0.61 to 2.91; nAAS = 512) was observed in a study with gym users in Pakistan [63]. The highest prevalence rate (88.76%, 95% CI 77.46 to 100.06; nAAS = 267) was reported in a study with nonspecific AAS users in Australia [59]. Figure 3 shows a forest plot of prevalence rates, ordered by the effect sizes of all studies.

Figure 3: Forest plot of the pooled prevalence of AAS users seeking support from physicians

Meta-analysis of prevalence rates of studies grouped by location

When grouped by the location of studies, the highest prevalence of AAS users seeking support from physicians was seen among studies taking place in Australia (67.27%, 95% CI 47.29 to 87.25; I^2 = 91.6%), followed by trans-regional studies (51.48%, 95% CI 35.26 to 67.71; I^2 = 97.4%). The lowest prevalence of AAS users seeking support from physicians was seen in studies located in

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Africa, Asia or Middle East (21.02%, 95% CI 5.26 to 36.79, I^2 = 95.7%). A forest plot of the prevalence rates of studies grouped by location is shown in Figure 4.

Figure 4: Forest plot of prevalence rates of studies, grouped by location

Meta-analysis of prevalence rates of studies grouped by subpopulation of AAS users

The highest prevalence of AAS users seeking support from physicians was seen among studies with NSP clients (54.13%, 95% CI 36.41 to 71.84; I^2 = 93.3%), followed by studies with nonspecific AAS users (41.67%, 95% CI 31.23 to 52.12; I^2 = 98.6%) and studies with strength athletes (27.83%; 95% CI 17.97 to 37.69; I^2 = 86.3%). The lowest prevalence of AAS users seeking support from physicians was seen in studies with adolescents (17.27%, 95% CI 4.80 to 29.74; I^2 = 87.0%). A forest plot of the prevalence rates of studies grouped by subpopulation of AAS users is shown in Figure 5.

Figure 5: Forest plot of prevalence rates of studies, grouped by subpopulation of AAS users

Meta-regression exploring the variance in the prevalence of AAS users seeking support from physicians

The results of the meta-regression analyses are shown in Table 3. Univariable analyses showed that the prevalence of AAS users seeking support from physicians was significantly higher among studies located in Australia (b = 0.35, 95% CI 0.11 to 0.58, p = 0.005) and the univariable model for the location of studies was considered statistically significant (χ^2 (6) = 17.92, p = 0.007). An overall multivariable model with all the variables was considered statistically significant (χ^2 (9) = 22.85, p = 0.007) and reduced the I^2 from 98.0% to 94.8%. In the multivariable analysis, only the subpopulation of adolescents (b = -0.28, 95% CI -0.56 to -0.01, p = 0.048) was considered a significant predictor of the prevalence of AAS users seeking support from physicians.

Table 3: Univariable and multivariable predictors of the prevalence of AAS users seeking support from physicians (n = 36)

		Univariable			Multivariable		
Variable	**No**	**Regression coefficient (95% CI)**	**SE**	**p value**	**Regression coefficient (95% CI)**	**SE**	**p value**
Location							
USA	8	1	-	-	1	-	-
UK	5	-0.01 (-0.23 to 0.21)	0.11	0.908	-0.16 (-0.42 to 0.10)	0.13	0.218
Continental Europe	6	-0.05 (-0.26 to 0.16)	0.11	0.629	-0.18 (-0.42 to 0.06)	0.12	0.144
Australia	4	0.35 (0.11 to 0.58)	0.12	0.005**	0.17 (-0.12 to 0.47)	0.15	0.247

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Brazil	3	0.02 (-0.24 to 0.29)	0.14	0.860	-0.10 (-0.39 to 0.19)	0.15	0.490
Africa, Asia or Middle East	5	-0.12 (-0.34 to 0.11)	0.12	0.316	-0.23 (-0.48 to 0.02)	0.13	0.068
Trans-regional	5	0.19 (-0.03 to 0.40)	0.11	0.089	0.03 (-0.24 to 0.30)	0.14	0.826
χ^2 (6) = 17.92, p = 0.007							
Subpopulation							
Nonspecific AAS users	21	1	-	-	1	-	-
Adolescents	4	-0.23 (-0.47 to 0.01)	0.12	0.055	-0.28 (-0.56 to -0.01)	0.14	0.048*
Strenght athletes	9	-0.12 (-0.29 to 0.05)	0.09	0.160	-0.09 (-0.27 to 0.09)	0.09	0.316
NSP clients	2	0.13 (-0.18 to 0.43)	0.15	0.416	0.05 (-0.24 to 0.34)	0.15	0.731
χ^2 (3) = 6.20, p = 0.102					χ^2 (9) = 22.85, p = 0.007		

*p < 0.01, *p < 0.05. SE: standard error

4. Discussion

We conducted a systematic review and meta-analysis and pooled data from 36 studies to estimate that the overall prevalence of AAS users seeking support from physicians is 37.12%. Higher prevalence rates of AAS users seeking support from physicians were seen among studies located in Australia (67.27%) and studies with NSP clients (54.13%). Lower prevalence rates were seen among studies located in Africa, Asia or the Middle East (21.02%) and studies with adolescents (17.27%). We discuss some factors that could have influenced the prevalence of AAS users seeking support from physicians in the studies. One factor possibly influencing the totality of studies is the fact that men, who represent the majority of AAS users [66], are generally less likely to seek medical support [67] – a tendency corroborated by the only two studies comparing the prevalence of male and female AAS users seeking support from physicians [19,42]. Other potential factors include the legal status of AAS use [9], the engagement of AAS users with health services [8], AAS users' perceptions of the service provided by physicians [7,24,68], and the stigma experienced by AAS users Yu 2015, Dunn 2016, Hill 2019].

Our findings suggest that the legal status of AAS use is not always associated with the prevalence of AAS users seeking support from physicians. For example, studies located in Australia – where the possession of AAS is illegal [69] – showed the highest pooled prevalence of AAS users seeking support from physicians (51.13%). Besides, the estimated prevalence of AAS users seeking support from physicians in the USA (32.91%) – where the possession of AAS is considered a federal crime [70]– was similar to the estimated prevalence in the UK (31.43%) and Brazil (35.23%), where the use and possession of AAS are not illegal [71,72]. Likewise, a low prevalence of AAS users seeking support from physicians was seen in studies from countries in Africa, Asia or the Middle East (21.02%), where anecdotal reports suggest loose enforcement of the prohibition of AAS use [73–76] or, in the case of India, there are no laws regulating the use and commerce of AAS [77]. Two studies from German showed prevalence rates of AAS users seeking

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support from physicians of 37.97% [55] and 55.95% [46], while a single study from Poland reported a prevalence of 11.11% [50], despite the use of AAS not being illegal in both countries [78,79]. Nevertheless, the legal status of AAS use could have influenced the small prevalence of AAS users seeking support from physicians (8.70%) seen in a single study from Denmark, where the use of AAS is not only illegal but where gym users can be subject to urinalysis to screen for the use of AAS [80]. Other factors might have influenced the prevalence of AAS users seeking support from physicians across the locations analysed by this study. In Australia, for example, there is an effort to educate physicians about the management of the non-prescribed use of AAS [81] which could reflect a willingness to discuss AAS use with the medical community.

The engagement of AAS users with health services could have influenced the comparatively high prevalence (54.13%) of AAS users seeking support from physicians among clients of NSP services that provide information about injection practices, adverse effects of AAS, and probably advising their clients to seek a physician if necessary [54,56]. Among NSP clients, those seeking support from physicians were more likely to have diagnostic screening for health conditions potentially associated with the use of AAS – despite some NSP clients considering physicians a less reliable source of information about AAS than NSP workers [56]. Despite the growing numbers of AAS users seeking the NSP, only a minority of primary NSP units provide advice about AAS [82], as the majority of NSP clients access the service in retail pharmacies providing the exchange of injectable material [83].

The lower pooled prevalence of AAS users seeking support from physicians was seen among adolescents (17.27%). Although low rates of engagement with health services are seen among adolescents in general [84], some factors could have influenced an even lower prevalence of adolescent AAS users seeking support from physicians. Firstly, the prevalence of some health conditions potentially associated with the use of AAS – such as cardiovascular disease [85] – is lower among adolescents. Secondly, it is possible that the illegality of AAS use has deterred adolescents from seeking physicians more than other subpopulations of AAS users. As highlighted by Terney et al. [38], physicians were allowed to prescribe AAS for enhancement purposes in the USA until 1988 and were frequently reported as a source of AAS to adolescents [38,39,86]. As near all of the selected studies with adolescents were located in the USA, it is possible that the criminalization of AAS use has driven adolescent AAS users further away from seeking the support of physicians.

Among strength athletes who use AAS, we estimated a prevalence of seeking support from physicians of 27.83%. Strength athletes who use AAS have been described as having a perception that the use of AAS can be safely managed, namely with the support of other AAS users who share their objectives, training routines and lifestyles [20,22]. The self-research and trial-and-

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error experiences with AAS, combined with aesthetical and performance goals frequently considered exaggerated by people outside of the community of strength athletes [87] probably contribute to the perception that physicians are less knowledgeable about AAS than some members of this subpopulation. Furthermore, it is possible that, because of their muscular appearance, some strength athletes who use AAS are even more subject to stigma than other AAS users [88].

Results of the multivariate meta-regression analysis showed that the prevalence rate of studies with adolescents was the only variable with a statistically significant impact on the overall prevalence of AAS users seeking support from physicians. Although current data suggest that adolescents represent a minority of the total population of AAS users [5], it is important to consider the under-representation of this population of AAS users in studies with nonspecific AAS users, trans-regional studies, and all the locations apart from the USA. Since the totality of studies with nonspecific AAS users and trans-regional studies in this review were performed with adults aged 16 or older, it is possible that the absence of adolescent AAS users in those studies represented a bias in their prevalence of AAS users seeking support from physicians.

Regarding the kind of support sought by AAS users from physicians among the 36 selected studies, 16 (44.44%) reported that AAS users sought physicians as a source of information about the use and adverse effects of AAS. The remaining studies described AAS users' contact with physicians in many different ways, such as having close contact with physicians [46], seeking a doctor for interpretation of health checks [13], and disclosing the use of AAS to a physician [7,64]. We considered that further exploration of the types of support sought by AAS users would lie beyond the scope of this study, as they have been recently investigated by other reviews [17,89,90].

Limitations of this study.

Among the limitations of this study, we highlight that the prevalence of AAS users seeking support from physicians varied widely across studies, and this variation was only minimally explained by the meta-regression and the comparisons between subgroups of studies. Selected studies investigated the help-seeking behaviours of AAS users in many different ways – e.g. interviews, anonymous surveys, longitudinal studies – and were synthesised as a single variable for comparison. This review did not distinguish data from male and female AAS users, as the sex of participants or differences in help-seeking behaviour between male and female participants were not reported by the majority of studies. The review only included studies from a few countries and, among those, many contributed with a single study unable to represent the local population of AAS users. As discussed in this review, the engagement of AAS users with

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physicians can be influenced by several factors, including attitudes that can vary widely between locations and subpopulations of AAS users. These limitations can compromise the generalisation of our results, and further studies are necessary to better understand the help-seeking behaviours of AAS users, namely among understudied locations and subpopulations of AAS users.

Implications for practice and policy

Available data suggest that factors such as the criminalisation of AAS use, the scarcity of physicians' knowledge about AAS and stigma against AAS users are barriers to the access of some AAS users to physicians. The results of this review can indicate locations and subpopulations of AAS users with higher engagement with physicians, so successful strategies can be replicated. Likewise, our results indicate the existence of under-studied and possibly under-supported populations of AAS users.

5. Conclusion

This meta-analysis was the first to systematically investigate the prevalence of AAS users seeking support from physicians. Our findings suggest that the overall prevalence of AAS users seeking support from physicians is 37.12%, with considerable variation across locations and among subpopulations of AAS users. This study highlights the importance of understanding the help-seeking behaviours of AAS users and improving their access to physicians.

Acknowledgements

We thank the participants and authors of the selected studies, the developers of R and the mefator package, and the reviewers of this journal for their valuable comments.

Contributors

JA planned the study and conducted the meta-analysis. JA and AK conducted the selection of studies and the assessment of the risk of bias. AK and PD contributed to the interpretation of data. JA, AK and PD reviewed and approved the final draft.

Funding

The first author was funded by King's College London International Scholarship (RDPL3PRPCA - 2019-20).

Competing interests

None

Ethics approval

As a review of the literature, this study was exempt from ethical approval by King's College London Research Ethics Office.

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Provenance and peer reviewed

Not commissioned; externally peer reviewed.

Data availability statement

No additional data are available.

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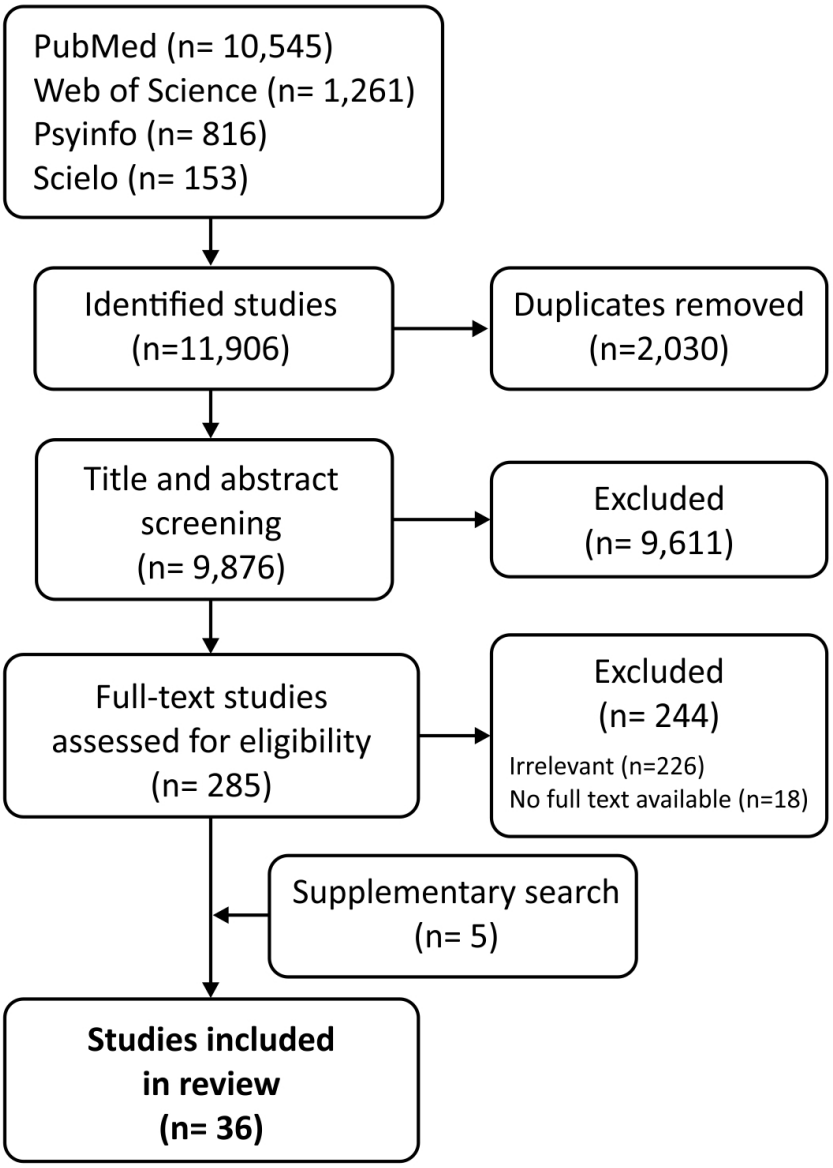


Figure 1: Flowchart of the inclusion of studies in the review

92x128mm (300 x 300 DPI)

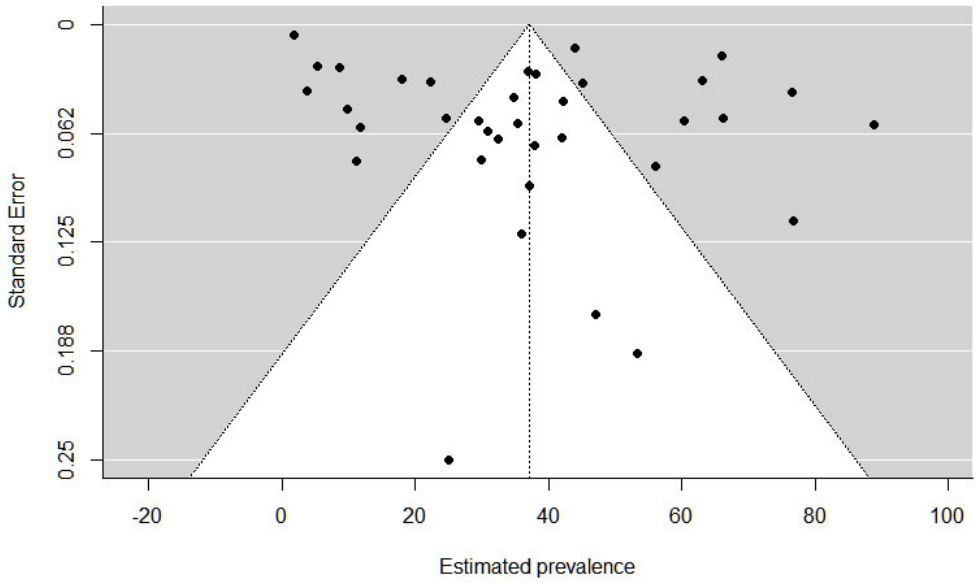


Figure 2: Funnel plot of studies included in the analysis with pseudo 95% confidence intervals

162x98mm (120 x 120 DPI)

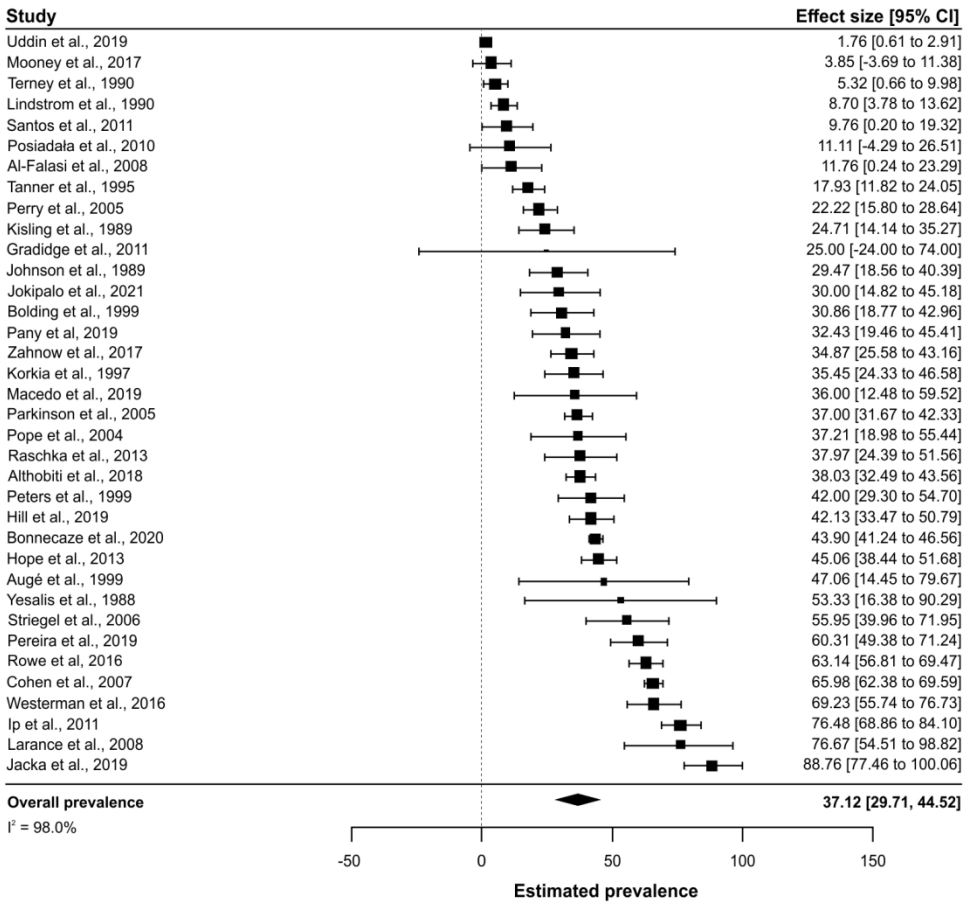


Figure 3: Forest plot of the pooled prevalence of AAS users seeking support from physicians

138x126mm (300 x 300 DPI)

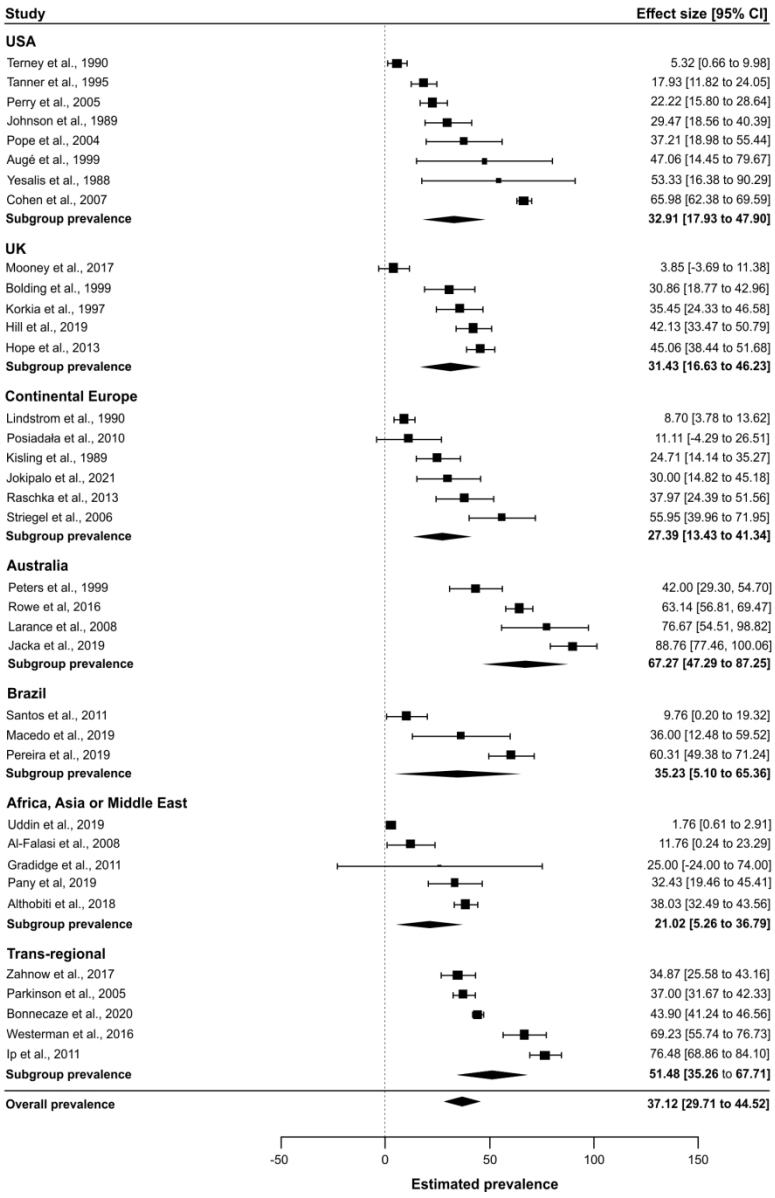


Figure 4: Forest plot of prevalence rates of studies, grouped by location

138x207mm (300 x 300 DPI)

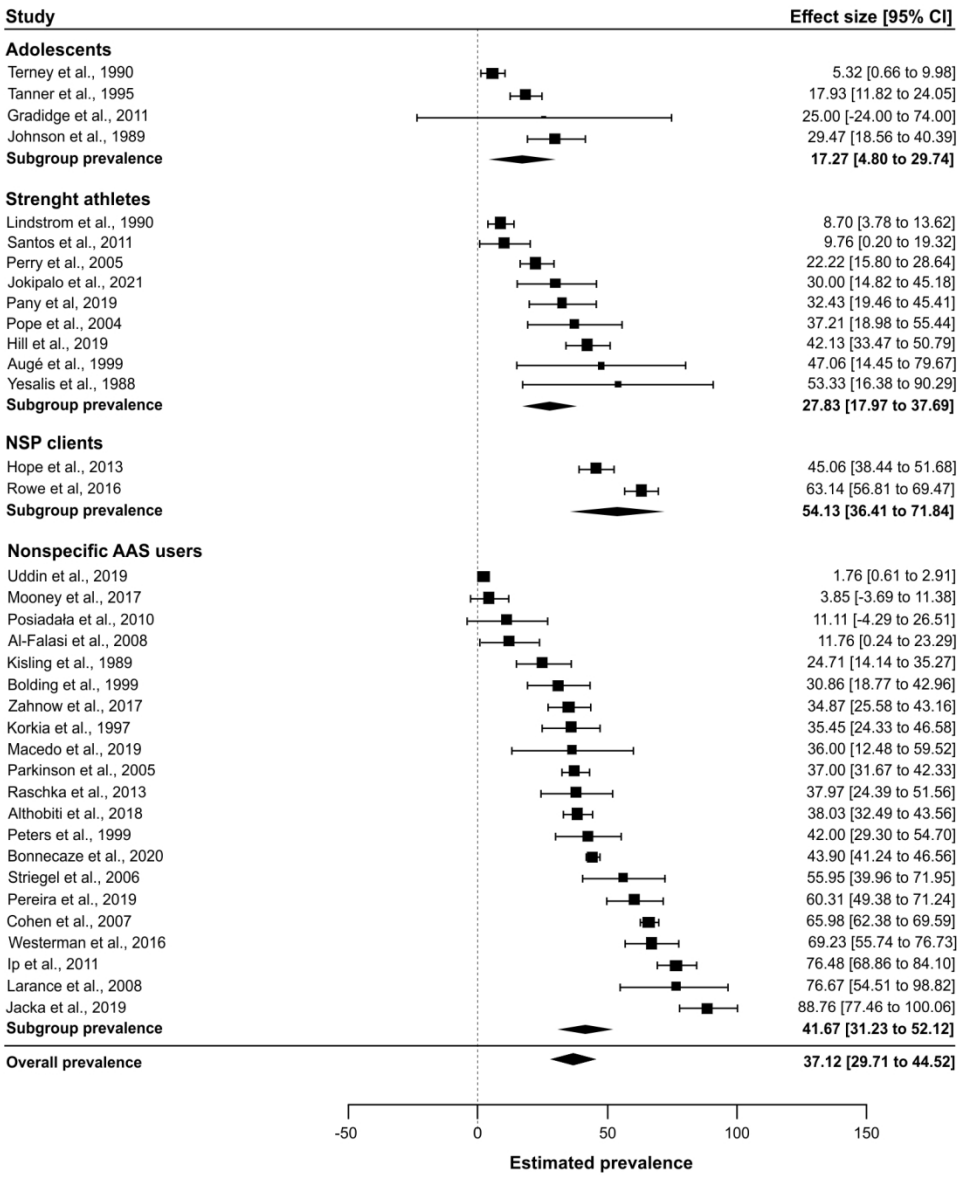


Figure 5: Forest plot of prevalence rates of studies, grouped by subpopulation of AAS users

138x166mm (300 x 300 DPI)

Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Amaral JMX, Kimergård A, Deluca P

SUPPLEMENTARY MATERIAL

Search algorithms

PUBMED (results: 10,545)

((agents, anabolic[MeSH Terms]) OR (doping in sport[MeSH Terms]) OR (drugs, performance enhancing[MeSH Terms])) OR ((anabolic steroid*[Title/Abstract]) OR (anabolic androgenic[Title/Abstract]) OR (anabolic agent*[Title/Abstract]) OR (anabolic drug*[Title/Abstract]) OR (androgen* abuse[Title/Abstract]) OR (performance enhanc*[Title/Abstract]) OR (image enhanc*[Title/Abstract]) OR (synthetic androgen*[Title/Abstract]) OR (synthetic testosterone[Title/Abstract]) OR (testosterone ester*[Title/Abstract]) OR (designer steroid*[Title/Abstract]) OR (doping[Title/Abstract]) OR ("non prescript*" steroid*[Title/Abstract]) OR ("non medic*" steroid*[Title/Abstract]) OR ("non medic*" androgen*[Title/Abstract]) OR ("non prescript*" androgen*[Title/Abstract])) NOT ((review[Title/Abstract]) OR (animal*[Title/Abstract]) OR (bovine[Title/Abstract]) OR (in vitro[Title/Abstract]) OR (mice*[Title/Abstract]) OR (pig[Title/Abstract]) OR (pigs[Title/Abstract]) OR (rat[Title/Abstract]) OR (rats[Title/Abstract]) OR (mouse[Title/Abstract]) OR (spectrometry[Title/Abstract]) OR (nano*[Title/Abstract]) OR (conductor*[Title/Abstract]) OR (semiconductor[Title/Abstract]) OR (electro*[Title/Abstract]) OR (optic*[Title/Abstract]) OR (hydrogen[Title/Abstract]) OR (superconductor*[Title/Abstract]) OR (catalyst*[Title/Abstract]) OR (thermoelectric[Title/Abstract]) OR (ions[Title/Abstract])) AND (humans[Filter])

Web of Science (results: 1,261)

((TS=("doping in sport" OR "performance enhancing drug*"OR "anabolic steroid*" OR "anabolic androgenic*" OR "anabolic agent*" OR "anabolic drug*")) NOT (TS=(review OR animal* OR bovine OR “in vitro” OR mice* OR pig OR pigs OR rat OR rats OR mouse OR spectrometry OR nano* OR conductor* OR semiconductor OR electro* OR optic* OR hydrogen OR superconductor* OR catalyst* OR thermoelectric OR ions))) Filters: Articles, Sport Sciences, Medicine (all), Psychology (all), Toxicology

APA PsycNet (results: 816)

((anabolic or doping or performance enhancing) not (review or animal or bovine or vitro or mice* or pig* or rat* or mouse or spectrometry or nano* or conductor* or semiconductor or electro* or optic* or hydrogen or superconductor* or catalyst* or thermoelectric or ions)).ti - limit to human

Scielo (results: 153)

(ti:((ti:((ti:(anabolic)) OR (ti:(doping)) OR (ti:(performance enhancing)))) AND NOT (ti:(review)))) AND NOT (ti:((ti:(animal)) AND NOT (ti:(bovine)) AND NOT (ti:(vitro)) AND NOT (ti:(mice*)) AND NOT (ti:(pig*)) AND NOT (ti:(rat*)) AND NOT (ti:(mouse)) AND NOT (ti:(spectrometry)) AND NOT (ti:(nano*)) AND NOT (ti:(conductor*)) AND NOT (ti:(semiconductor)) AND NOT (ti:(electro*)) AND NOT (ti:(optic*)) AND NOT (ti:(hydrogen)) AND NOT (ti:(superconductor*)) AND NOT (ti:(catalyst*)) AND NOT (ti:(thermoelectric)) AND NOT (ti:(ions))))

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eTable 1: Risk of bias

	Risk of bias: MMAT criteria*					
Authors, year	**1**	**2**	**3**	**4**	**5**	**Observations**
Yesalis et al., 1988	Y	N	?	Y	Y	Small sample of AAS users. Source of information described as 'Physician/pharmacist/medical literature'.
Johnson et al., 1989	Y	Y	Y	Y	Y	n/a
Kisling et al., 1989	Y	Y	?	Y	Y	Nonresponse rate not informed.
Lindstrom et al., 1990	Y	Y	Y	Y	N	The number of AAS users seeking a physician must be inferred from the text.
Terney et al., 1990	Y	Y	N	Y	Y	The number of AAS users seeking support from physicians was extrapolated from the total number of participants.
Peters et al., 1993	Y	Y	Y	?	Y	Nonresponse rate not informed.
Tanner et al., 1995	Y	Y	N	Y	Y	The number of AAS users seeking support from physicians was extrapolated from the total number of participants.
Korkia et al., 1997	Y	Y	Y	N	Y	Nonresponse rate = 41%.
Bolding et al., 1999	Y	Y	?	Y	Y	Support described as 'regular health checks to screen for steroid-related health problems'.
Auge et al., 1999	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Perry et al., 2005	Y	Y	N	Y	Y	The number of AAS users seeking a physician must be inferred from a graph and was extrapolated from the total number of participants.
Parkinson et al., 2005	Y	Y	Y	?	Y	Nonresponse rate not informed.
Pope et al., 2004	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Striegel et al., 2006	Y	Y	Y	?	Y	Nonresponse rate not informed.
Cohen et al., 2007	Y	Y	?	?	Y	Nonresponse rate not informed. Support described as 'willing to seek medical supervision'.
Al-Falasi et al., 2008	Y	N	N	N	Y	Nonresponse rate = 56%. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Larance et al., 2008	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Posiadala et al., 2010	Y	N	Y	N	?	Small sample of AAS users. Nonresponse rate = 28.57%.
Gradidge et al., 2011	Y	Y	N	Y	Y	Small sample of AAS users. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Ip et al., 2011	Y	Y	Y	Y	Y	n/a
Santos et al., 2011	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Hope et al., 2013	Y	Y	Y	Y	Y	n/a
Raschka et al., 2013	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Mooney et al., 2016	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Rowe et al., 2016	Y	Y	Y	Y	Y	n/a
Westerman et al., 2016	Y	Y	Y	Y	Y	n/a
Zahnow et al., 2017	Y	Y	Y	?	Y	Nonresponse rate not informed.
Althobiti et al., 2018	Y	Y	N	?	Y	Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Hill et al., 2019	Y	Y	Y	Y	Y	n/a
Jacka et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.

Macedo et al., 2019	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Pany et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.
Pereira et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.
Uddin et al., 2019	Y	Y	N	Y	Y	The number of AAS users seeking a physician must be inferred from the text.
Bonnecaze et al., 2020	Y	Y	Y	Y	Y	n/a
Jokipalo et al., 2021	Y	Y	Y	Y	Y	n/a

* MMAT's quality-assessment criteria:
1 - Is the sampling strategy relevant to address the research question?
2 - Is the sample representative of the target population?
3 - Are the measurements appropriate?
4 - Is the risk of nonresponse bias low?
5 - 'Is the statistical analysis appropriate to answer the research question?'
Y = yes. N = no. ? = not clear.

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R codes for the meta-analysis, forest plots and meta-regression
Results of the Baujat plot

The table below contains the data of the file 'metaAAS.csv'. The column 'ti' contains the number of androgenic-anabolic steroids (AAS) users in each study, and the column 'xi', the number of AAS users who sought support from physicians.

study id	studies	ti	xi	locat	pop
1	Yesalis et al.,1988	15	8	0	1
2	Johnson et al.,1989	95	28	0	0
3	Kisling et al.,1989	85	21	2	3
4	Lindstrom et al.,1990	138	12	2	1
5	Terney et al.,1990	94	5	0	0
6	Peters et al.,1993	100	42	3	3
7	Tanner et al.,1995	184	33	0	0
8	Korkia et al.,1997	110	39	1	3
9	Bolding et al.,1999	81	25	1	3
10	Auge et al.,1999	17	8	0	1
11	Perry et al.,2005	207	46	0	1
12	Parkinson et al.,2005	500	185	6	3
13	Pope et al.,2004	43	16	0	1
14	Striegel et al.,2006	84	47	2	3
15	Cohen et al.,2007	1955	1290	0	3
16	Al-Falasi et al.,2008	34	4	5	3
17	Larance et al.,2008	60	46	3	3
18	Posiadala et al.,2010	18	2	2	3
19	Gradidge et al.,2011	4	1	5	0
20	Ip et al.,2011	506	387	6	3
21	Santos et al.,2011	41	4	4	1
22	Hope et al.,2013	395	178	1	2
23	Raschka et al.,2013	79	30	2	3
24	Mooney et al.,2016	26	1	1	3
25	Rowe et al.,2016	605	382	3	2
26	Westerman et al.,2016	231	153	6	3
27	Zahnow et al.,2017	195	68	6	3
28	Althobiti et al.,2018	476	181	5	3
29	Hill et al.,2019	216	91	1	1
30	Jacka et al.,2019	267	237	3	3
31	Macedo et al.,2019	25	9	4	3
32	Pany et al.,2019	74	24	5	1
33	Pereira et al.,2019	194	117	4	3
34	Uddin et al.,2019	512	9	5	3
35	Bonnecaze et al.,2020	2385	1047	6	3
36	Jokipalo et al.,2021	50	15	2	1

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R codes

```
## Loading packages previously installed
library("robumeta")
library("metafor")
library("dplyr")
library("forestplot")
library("checkmate")
library("nloptr")
library("meta")
library("CompQuadForm")

## Meta-analysis of all studies
metaAAS <- read.csv("metaAAS.csv")
View(metaAAS)
metaAAS <- escalc(measure="IR", xi=xi, ti=ti, data=metaAAS, slab=paste (studies))
res <- rma(yi, vi, data=metaAAS)
res
confint(res)

## Baujat plot
b_res <- rma(xi, ti, data=metaAAS, slab=study.id)
baujat(b_res)
inf <- influence(res)
print(inf)
plot(inf)


### funnel plot
funnel(res,atransf=mytransf,xlab = "Estimated prevalence")
regtest(res)
ranktest(res)
warnings()


### Cross-tabulation of the number of studies – location vs subpopulation
addmargins(table(metaAAS$locat, metaAAS$pop))


### Setup functions for the forest plot
mlabfun <- function(text, res) {
 list(bquote(paste(.(text),
            " (Q = ", .(formatC(res$QE, digits=2, format="f")),
            ", df = ", .(res$k - res$p),
            ", p ", .(metafor:::.pval(res$QEp, digits=2, showeq=TRUE, sep=" ")), "; ",
            I^2, " = ", .(formatC(res$I2, digits=1, format="f")), "%, ",
            tau^2, " = ", .(formatC(res$tau2, digits=2, format="f")), ")")))}
mytransf<-function(x)
 (x)*100


## Forest plot of all studies
forest (res, transf=mytransf, cex=.7,header=c("Authors and Year", "Effect size [95%
CI]"),xlab="Prevalence rate",
      mlab=mlabfun("RE Model for All Studies", res), order="obs", refline=NA)
```

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```
## Forest plot - subgroup LOCATION
forest (res, transf=mytransf, cex=.6, refline=NA,top=0,
      order=metaAAS$locat,
      rows=c(65:58,53:49,44:39,34:31,26:24,19:15,10:6),
      mlab=mlabfun("RE Model for All Studies", res),
      header=c("Authors and year", "Effect size [95% CI]"),
      xlab="Estimated prevalence")


### switch to bold italic font
par(font=4)

### add text for the subgroups
text(-130, c(54,45,35,27,20,11), pos=4,
   c("UK","Continental Europe","Australia","Brazil",
     "Africa,Asia and Middle East","Trans-regional"))

### fit random-effects model in the seven subgroups
res.0 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="0"), data=metaAAS)
res.1 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="1"), data=metaAAS)
res.2 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="2"), data=metaAAS)
res.3 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="3"), data=metaAAS)
res.4 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="4"), data=metaAAS)
res.5 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="5"), data=metaAAS)
res.6 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="6"), data=metaAAS)

### add summary polygons for the seven subgroups
addpoly(cex=.5,res.0, row=56.5, mlab=mlabfun("RE Model for Subgroup", res.0),transf=mytransf)
addpoly(cex=.5,res.1, row=47.5, mlab=mlabfun("RE Model for Subgroup", res.1),transf=mytransf)
addpoly(cex=.5,res.2, row=37.5, mlab=mlabfun("RE Model for Subgroup", res.2),transf=mytransf)
addpoly(cex=.5,res.3, row=29.5, mlab=mlabfun("RE Model for Subgroup", res.3),transf=mytransf)
addpoly(cex=.5,res.4, row=22.5, mlab=mlabfun("RE Model for Subgroup", res.4),transf=mytransf)
addpoly(cex=.5,res.5, row=13.5, mlab=mlabfun("RE Model for Subgroup", res.5),transf=mytransf)
addpoly(cex=.5,res.6, row=4.5, mlab=mlabfun("RE Model for Subgroup", res.6),transf=mytransf)


### fit meta-regression model to test for subgroup differences
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(locat), data=metaAAS)
resMeta
confint(resMeta)

### add text for the test of subgroup differences
text(-130, -3, pos=4, cex=0.6,
   bquote(paste("Test of moderator (location): ",Q[M], " = ",
            .(formatC(resMeta$QM, digits=2, format="f")),
            ", df = ", .(resMeta$p - 1),
                        ", p = ", .(formatC(resMeta$QMp, digits=2, format="f")))))
```

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```
##Forest plot -subgroup SUBPOPULATION
forest (res, transf=mytransf, cex=.6, refline=NA,top=0,
      order=metaAAS$pop,
      rows=c(53:50,45:37,32:31,26:6),
      mlab=mlabfun("RE Model for All Studies", res),
      header=c("Authors and year", "Effect size [95% CI]"),
      xlab="Estimated prevalence")

### fit random-effects model in the four subgroups
res.0 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="0"), data=metaAAS)
res.1 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="1"), data=metaAAS)
res.2 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="2"), data=metaAAS)
res.3 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="3"), data=metaAAS)

### add summary polygons for the four subgroups
addpoly(cex=.5,res.0, row=48, mlab=mlabfun("RE Model for Subgroup", res.0),transf=mytransf)
addpoly(cex=.5,res.1, row=35, mlab=mlabfun("RE Model for Subgroup", res.1),transf=mytransf)
addpoly(cex=.5,res.2, row=29, mlab=mlabfun("RE Model for Subgroup", res.2),transf=mytransf)
addpoly(cex=.5,res.3, row=4, mlab=mlabfun("RE Model for Subgroup", res.3),transf=mytransf)


### fit meta-regression model to test for subgroup differences
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(pop), data=metaAAS)
resMeta
confint(resMeta)

### add text for the test of subgroup differences
text(-130, -3, pos=4, cex=0.6,
   bquote(paste("Test of moderator (sub-population): ",Q[M], " = ",
            .(formatC(resMeta$QM, digits=2, format="f")),
            ", df = ", .(resMeta$p - 1),
            ", p = ", .(formatC(resMeta$QMp, digits=2, format="f")))))


### Univariate meta-regression model. Moderator: LOCATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(locat), ref="0"),  data=metaAAS)
resMeta
confint(resMeta,transf=mytransf)


### Univariate meta-regression model. Moderator: SUBPOPULATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(pop), ref="3"), data=metaAAS)
resMeta
confint(resMeta)

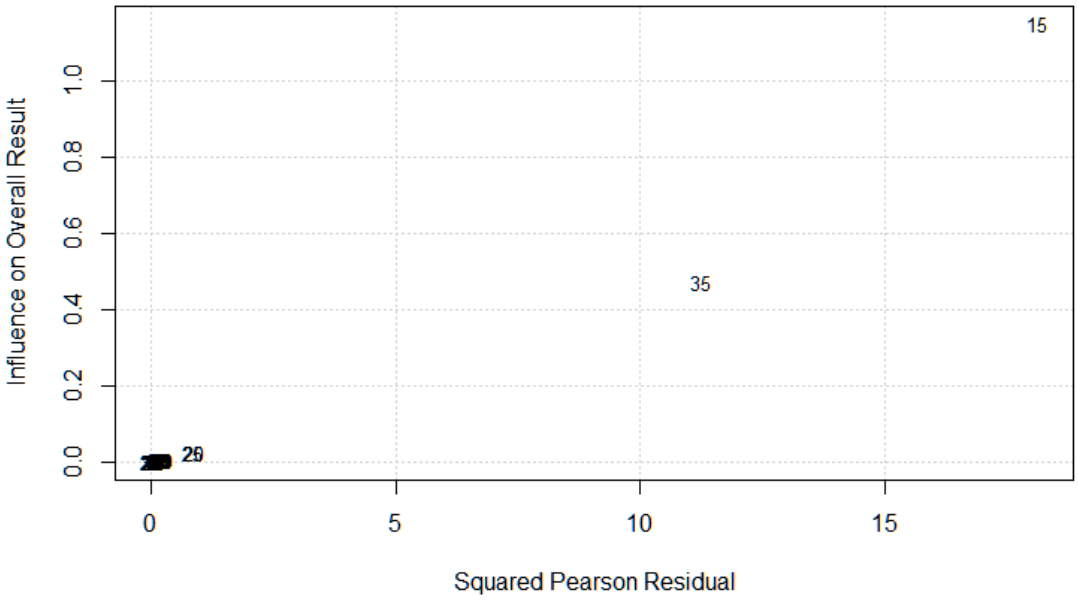


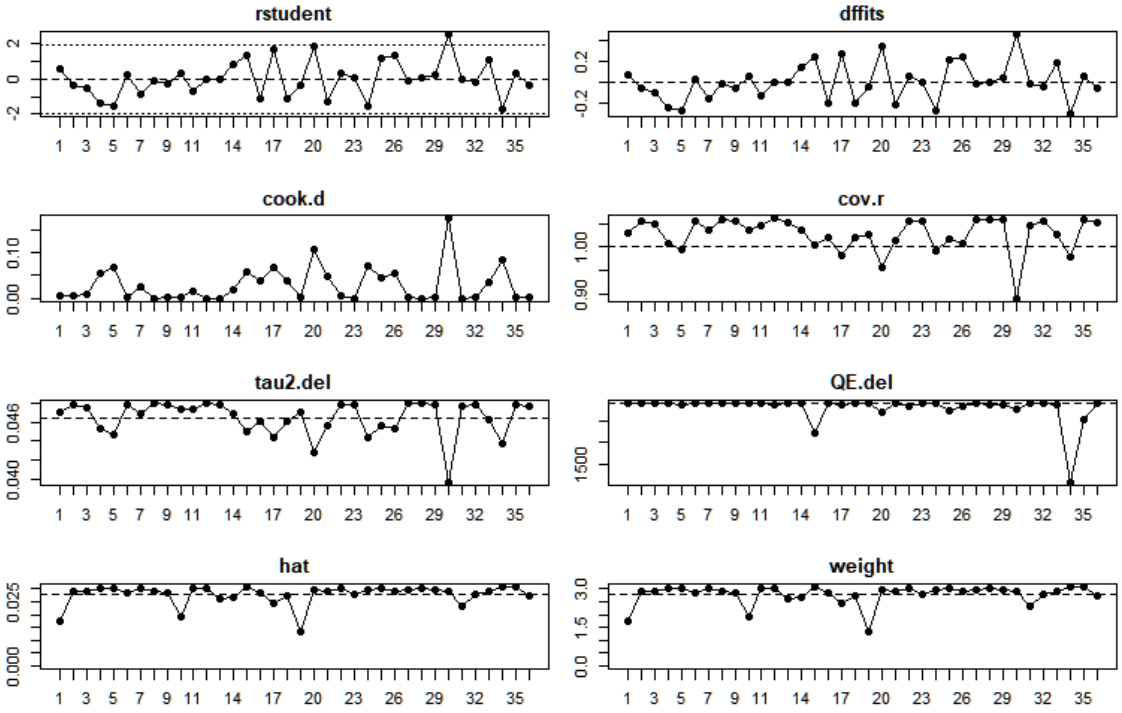
### Multivariate meta-regression model. Moderators: LOCATION + SUBPOPULATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(locat)+relevel(factor(pop), ref="3"),
data=metaAAS)
resMeta
confint(resMeta)
```

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Results of the Baujat plot

	rstudent	dffits	cook.d	cov.r	tau2.del	QE.del	hat	weight	dfbs	inf
Yesalis et al.,1988	0.5691	0.0751	0.0057	1.0296	0.0470	2901.5926	0.0174	1.7416	0.0749	
Johnson et al.,1989	-0.3444	-0.0615	0.0039	1.0562	0.0477	2901.1899	0.0288	2.8822	-0.0615	
Kisling et al.,1989	-0.5615	-0.0986	0.0099	1.0507	0.0475	2903.7126	0.0289	2.8937	-0.0986	
Lindstrom et al.,1990	-1.3464	-0.2365	0.0546	1.0080	0.0453	2888.7020	0.0303	3.0336	-0.2363	
Terney et al.,1990	-1.5188	-0.2662	0.0681	0.9939	0.0446	2872.4020	0.0304	3.0377	-0.2658	
Peters et al.,1993	0.2168	0.0346	0.0012	1.0567	0.0478	2891.9587	0.0282	2.8198	0.0346	
Tanner et al.,1995	-0.8914	-0.1576	0.0250	1.0380	0.0468	2904.9166	0.0301	3.0115	-0.1576	
Korkia et al.,1997	-0.0751	-0.0154	0.0002	1.0593	0.0479	2896.1842	0.0288	2.8752	-0.0154	
Bolding et al.,1999	-0.2797	-0.0501	0.0026	1.0566	0.0478	2901.0530	0.0284	2.8416	-0.0501	
Auge et al.,1999	0.3664	0.0502	0.0025	1.0367	0.0473	2902.0604	0.0193	1.9265	0.0501	
Perry et al.,2005	-0.6882	-0.1224	0.0152	1.0477	0.0473	2903.7785	0.0301	3.0053	-0.1225	
Parkinson et al.,2005	-0.0059	-0.0036	0.0000	1.0626	0.0480	2858.4148	0.0303	3.0265	-0.0036	
Pope et al.,2004	0.0035	-0.0017	0.0000	1.0536	0.0478	2900.9937	0.0259	2.5919	-0.0017	
Striegel et al.,2006	0.8253	0.1366	0.0188	1.0357	0.0468	2884.0498	0.0269	2.6891	0.1365	
Cohen et al.,2007	1.3765	0.2473	0.0594	1.0032	0.0451	2203.3151	0.0305	3.0526	0.2470	
Al-Falasi et al.,2008	-1.1563	-0.1977	0.0388	1.0208	0.0460	2903.5873	0.0286	2.8616	-0.1977	
Larance et al.,2008	1.6754	0.2655	0.0680	0.9820	0.0444	2878.6129	0.0241	2.4114	0.2664	
Posiadala et al.,2010	-1.1536	-0.1919	0.0366	1.0200	0.0461	2904.0446	0.0271	2.7140	-0.1920	
Gradidge et al.,2011	-0.3687	-0.0438	0.0019	1.0253	0.0470	2904.9093	0.0131	1.3108	-0.0436	
Ip et al.,2011	1.9001	0.3414	0.1077	0.9560	0.0428	2681.1281	0.0298	2.9778	0.3407	
Santos et al.,2011	-1.2671	-0.2186	0.0470	1.0136	0.0456	2901.6068	0.0292	2.9249	-0.2186	
Hope et al.,2013	0.3645	0.0618	0.0039	1.0575	0.0477	2842.9150	0.0300	3.0010	0.0619	
Raschka et al.,2013	0.0375	0.0040	0.0000	1.0575	0.0479	2897.1920	0.0279	2.7864	0.0040	
Mooney et al.,2016	-1.5763	-0.2731	0.0714	0.9891	0.0444	2889.9271	0.0298	2.9798	-0.2728	
Rowe et al.,2016	1.2229	0.2170	0.0463	1.0150	0.0457	2712.0790	0.0301	3.0072	0.2168	
Westerman et al.,2016	1.3474	0.2350	0.0539	1.0053	0.0452	2825.5656	0.0290	2.8960	0.2349	
Zahnow et al.,2017	-0.1027	-0.0204	0.0004	1.0610	0.0479	2890.1445	0.0296	2.9607	-0.0204	
Althobiti et al.,2018	0.0413	0.0047	0.0000	1.0624	0.0480	2856.9754	0.0302	3.0228	0.0047	
Hill et al.,2019	0.2275	0.0373	0.0014	1.0592	0.0479	2876.4907	0.0295	2.9508	0.0373	
Jacka et al.,2019	2.5277	0.4496	0.1749	0.8926	0.0397	2756.3671	0.0287	2.8694	0.4492	
Macedo et al.,2019	-0.0457	-0.0092	0.0001	1.0485	0.0476	2902.8855	0.0235	2.3471	-0.0091	
Pany et al.,2019	-0.2084	-0.0378	0.0015	1.0570	0.0478	2900.6346	0.0281	2.8096	-0.0378	
Pereira et al.,2019	1.0596	0.1829	0.0333	1.0252	0.0462	2848.8713	0.0288	2.8819	0.1829	
Uddin et al.,2019	-1.7165	-0.3019	0.0858	0.9758	0.0437	1099.7502	0.0307	3.0725	-0.3012	
Bonnecaze et al.,2020	0.3139	0.0533	0.0029	1.0598	0.0478	2520.6043	0.0306	3.0626	0.0534	
Jokipalo et al.,2021	-0.3118	-0.0543	0.0030	1.0537	0.0477	2902.8279	0.0272	2.7230	-0.0543	





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PRISMA 2020 Main Checklist

Topic	No.	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Section 1, Lines 126-128
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section 1, Lines 120-123
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section 1, Lines 123-125, 126-128
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 2, Line 157 (Table 1)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 2, Line 141
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 2, Line 144 (Figure 1)
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2, Lines 146-150
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 2, Lines 159-163
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 2, Lines 159-163
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 2, Lines 159-163

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Topic	No.	Item	Location where item is reported
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2, Lines 165-174
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Section 2, 176-200
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	Section 2, 198-200
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 2, 176-200
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 2, 176-200
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 2, 176-204
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Section 2, Lines 182-190
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Section 2, 201-204
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Section 2, Lines 201-204
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Section 2, 201-204
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 2, Line 154 (Figure 2)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	n/a
Study characteristics	17	Cite each included study and present its characteristics.	Section 3, Line 217 (Table 2)
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section 3, Lines 226-233
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Section 3, Line 217 (Table 2)
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section 3, Lines 226-233

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Topic	No.	Item	Location where item is reported
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Section 3, Lines 235-390
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Section 3, Lines 240-375
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Section 3, Lines 365-372
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Section 3, Lines 240-375
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Section 3, Lines 240-375
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section 4, Lines 377-419
	23b	Discuss any limitations of the evidence included in the review.	Section 4, Lines 420-438
	23c	Discuss any limitations of the review processes used.	Section 4, Lines 420-438
	23d	Discuss implications of the results for practice, policy, and future research.	Section 5, Lines 446-449
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Section 2, Lines 130-132
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Section 2, Lines 130-132
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	n/a
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Section 6, Lines 461-453
Competing interests	26	Declare any competing interests of review authors.	Section 6, Line 453
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Section 3, Lines 238-239

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PRIMSA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesize results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. MetaArXiv. 2020, September 14. DOI: 10.31222/osf.io/v7gm2. For more information, visit: www.prisma-statement.org

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BMJ Open

Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-056445.R2
Article Type:	Original research
Date Submitted by the Author:	14-Apr-2022
Complete List of Authors:	Amaral, Julio Mario; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions Kimergård, Andreas; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions Deluca, Paolo; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions
Primary Subject Heading:	Medical management
Secondary Subject Heading:	Addiction, General practice / Family practice, Health services research
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Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Short title: Anabolic steroid users seeking support from physicians

Word count: 4,569

PROSPERO registration number: CDR42020177919

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Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Abstract

Objectives: To estimate the overall prevalence of androgenic-anabolic steroids (AAS) users seeking support from physicians. Secondary objectives are to compare this prevalence in different locations and among subpopulations of AAS users, discussing some of the factors that could have influenced the engagement of AAS users with physicians.

Design: Systematic review and meta-analysis.

Data sources: MEDLINE, PsycINFO, Web of Science and SciELO were searched in January 2022.

Eligibility criteria: Quantitative and qualitative studies reporting the number of AAS users who sought support from physicians, with no restrictions of language or time of publication.

Data extraction and synthesis: Two independent reviewers extracted data and assessed the quality of studies, including publication bias. A random-effects meta-analysis was performed to estimate the overall prevalence of AAS users seeking support from physicians, followed by pooled prevalence rates by studies' location and the subpopulation of AAS users.

Results: We identified 36 studies published between 1988 and 2021, involving 10,101 AAS users. The estimated overall prevalence of AAS users seeking support from physicians is 37.12% (95% CI 29.71 to 44.52). Higher prevalence rates were observed in studies from Australia (67.27%; 95%CI 42.29 to 87.25) and among clients of the needle and syringe exchange programme (54.13%; 95% CI 36.41 to 71.84). The lowest prevalence was observed among adolescent AAS users (17.27%; 95% CI 4.80 to 29.74).

Conclusion: Our findings suggest that about one-third of AAS users seek support from physicians, with remarkable differences between locations and subpopulations of AAS users. Further studies should investigate the factors influencing the engagement of AAS users with physicians.

Registration: PROSPERO - CRD42020177919.

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Strengths and limitations of this study

- This is the first meta-analysis of studies investigating the prevalence of AAS users seeking support from physicians.
- It analyses 36 studies published between 1988 and 202, involving 10,101 AAS users.
- The R codes used in the meta-analysis and meta-regression are available in the supplementary files, allowing complete and free reproducibility of our results.
- Our results are based on studies from a small number of countries, with a limited representativeness of the subpopulations of AAS users.

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Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Introduction

Anabolic-androgenic steroids (AAS) are synthetic androgens with several potential effects. AAS can be used to treat conditions such as male hypogonadism, pathological loss of muscle mass and anaemia [1], with findings suggesting their efficacy in the treatment of depression [2] and conditions associated with type 2 diabetes [3]. Beneficial effects of AAS include the increasing of muscle mass, feelings of wellbeing and boosted energy, enhancement of body image, and improvement of athletic and occupational performance [4,5]. However, the use of AAS can increase the risk of several adverse health conditions such as acne, testicular atrophy, gynecomastia, clitoromegaly, hypomania, anxiety, dyslipidaemia and high haematocrit – therefore increasing the risk of myocardial infarction and stroke [6]. Despite the risks, many AAS users refrain from seeking physicians for AAS-related information or to treat health conditions potentially associated with the use of AAS [7]. Among factors possibly influencing the prevalence of AAS seeking support from physicians are the legal status of AAS, AAS users' engagement with health services and their perceptions of the services provided by physicians to people using AAS [8]. In countries where the possession of AAS without a medical prescription is illegal, it is reasonable to expect that some AAS users will refrain from disclosing the use of AAS to a physician [9]. The legal status of AAS can also influence the service provided to AAS users by physicians, as doctors are usually not allowed to prescribe AAS for the purposes of enhancement or to regulate hormonal levels after the use of AAS, a practice also known as post-cycle therapy (PCT) [10]. Whilst physicians are trained to treat the use of other illegal substances such as heroin and cocaine, many of them admit a lack of training and experience in recognising and treating adverse effects of AAS [11–14]. Another reason given by some AAS users to refrain from seeking medical support is the stigma and judgmental attitudes experienced in their contact with health professionals [13,15]. Due to these factors – and possibly others, such as the provision of health services in their locality, financial limitations, etc. – some AAS users rely on self-conducted research to manage their use and adverse effects of AAS and/or seek the support of informal sources such as friends and online forums [16,17]. Finally, some AAS users reported not seeking physicians simply because they did not feel the need to do so, due to an absence of adverse effects or to a perception that these effects can be managed without the help of a medical professional [13].

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In addition to the legal status of AAS use and the access to health services in different countries, different help-seeking behaviours are seen across subpopulations of AAS users [18]. Younger AAS users, for instance, seem to be less likely to engage with health services [19]. Some strength athletes tend to rely on other athletes perceived as knowledgeable about AAS [20–22], and AAS users who are clients of the needle and syringe exchange programme (NSP) seem to be more likely to interact with health services [23,24]. However, to this date, no systematic comparison of the prevalence of AAS users seeking the support of physicians has been produced.

The objective of this study is to estimate the overall prevalence of AAS users seeking support from physicians by conducting a systematic review and meta-analysis of surveys and interviews with AAS users. The secondary objectives are to compare the prevalence of AAS users seeking support from physicians in different locations and among subpopulations of AAS users.

Methods

This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [25], and its protocol is registered in PROSPERO (registration number CRD42020177919). As a review of the literature, this study is exempted from ethical clearance by King's College London research ethics office.

A search strategy was designed to retrieve studies describing surveys and interviews with people using AAS. Searches were performed on MEDLINE, PsycINFO, Web of Science and SciELO in January 2022 with no restrictions on the date, location and language of studies. The search algorithm adapted to each database can be found in the supplementary material (eTable 1). A supplementary search was performed on the reference lists of eligible studies. Two independent researchers (JA and AK) performed the screening, data extraction and assessed the risk of bias. Disagreements were resolved by discussion with the third researcher (PD).

Results of the searches were exported to Rayyan QCRI [26] for screening and removing of duplicates. Titles, abstracts and the full text of studies were screened for eligibility against our inclusion and exclusion criteria (Table 1). A spreadsheet was utilised to summarise descriptive data of selected studies, i.e. location of study, subpopulation of AAS users, the number of participants, number and gender of AAS users, and the number of AAS users who sought support from physicians.

Table 1: Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
Quantitative and qualitative studies with people using AAS for the purposes of image and performance enhancement.	Studies with patients using AAS for treatment of medical conditions.

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	Studies with prisoners, who have limited access to external health services. Studies with animal subjects and in vitro analysis of AAS effects.
Studies informing the number of AAS users seeking physicians to receive information and prevent and treat AAS-related health conditions. Studies where the source of support or information can be understood as being a physician (e.g. doctor, medical doctor, health professional).	Studies not informing the number of AAS users in the sample who seek support from physicians. Studies where the contact with the physician cannot be understood as participant's choice, such as involuntary admissions, post-mortem analysis and cohorts of patients selected for studies of specific health conditions.
Peer-reviewed studies. Studies published at any time. Studies published in any language, as long as it is possible to retrieve relevant data from the authors or articles.	'Grey literature' (non-peer-reviewed studies and reports). Case studies and interviews with a single AAS user.

The risk of bias was assessed in two stages. Initially, the quality and internal validity of studies were evaluated using the Mixed Methods Appraisal Tool (MMAT)[27]. The MMAT is comprised of five quality-assessment criteria: (i) 'Is the sampling strategy relevant to address the research question?'; (ii) 'Is the sample representative of the target population?'; (iii) 'Are the measurements appropriate?'; (iv) 'Is the risk of nonresponse bias low?'; and (v) 'Is the statistical analysis appropriate to answer the research question?'. In the second stage, studies were assessed for risk of publication bias. The risk of publication bias was assessed by visual inspection of asymetry in a funnel plot [28], Egger's test for asymmetry [29], and a rank correlation test [30].

The data synthesis for meta-analysis was performed extracting the number of AAS users who reported seeking any kind of support from physicians in the selected studies. For the purpose of effect size calculations, the number of AAS users in each study was used as the number of participants of interest, and the number of those who informed seeking a physician was used as the number of cases in each study. When a study informed more than one number or percentage of AAS users seeking support from physicians, the higher value of male AAS users seeking medical support was considered, as only a few studies included female participants.

A meta-analysis was performed to estimate the overall prevalence of AAS users seeking support from physicians. A random-effects model was chosen to better incorporate the dispersions of prevalence rates across studies and the different approaches to the research question [31]. Heterogeneity was measured using the I^2 index, which describes the percentage of variation of prevalence rates across a group of studies that is due to differences between studies (e.g. different sample sizes, populations or methods) [32,33]. A Baujat plot was produced to identify

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studies that could influence the overall result [34]. The secondary outcomes were measured by the prevalence rates of studies grouped by location and subpopulation of AAS users. Univariate and multivariate meta-regression were performed to measure the impact of the moderators 'location of study' and 'subpopulation of AAS users' on the variance of the overall prevalence rate. For each moderator variable, the category with the highest number of studies was used as a comparator. Meta-analysis and meta-regression were conducted in R [35] using the metafor package [36]. A full description of the codes can be found in the supplementary material.

Patient and public involvement

It was not appropriate to involve patients or the public in the design, conduct, reporting or dissemination plans of this study.

3. Results

The searches identified 11,906 studies. After the removal of duplicates, 9,876 studies were screened by title and abstract. Among 285 full-text studies were assessed for eligibility, 31 were included in the review. A supplementary search on the reference list of included papers and previous reviews led to the inclusion of another five studies. A total of 36 studies were included in the review, as shown in flowchart of Figure 1.

Figure 1: Flowchart of the inclusion of studies in the review

Table 2 shows the summary characteristics of the studies included in this review. The studies were published between 1988 and 2021, with a total number of AAS users (nAAS) = 10,101, being 9,278 (91.85%) males, 179 (1.77%) females and 644 (6.38%) whose sex was not reported. Eight of the selected studies were located in the US (nAAS = 2,610; 25.84%), six in continental Europe (nAAS = 454; 4.49%), five in the UK (nAAS = 828; 8.20%), four in Australia (nAAS = 1,032; 10.22%), three in Brazil (nAAS = 260; 2.57%), five in Africa, Asia or the Middle East (nAAS = 1,110; 10.89%) and five studies were trans-regional (nAAS = 3,817; 37.79%). The selected studies presented a wide range of nAAS (median = 97.50; range = 4 to 2,385) and number of AAS users seeking support from physicians (nPhy; median = 31.50; range = 1 to 1,290). Regarding the subpopulation of AAS users, the most common were studies with nonspecific AAS users such as gym users and recreational athletes (21 studies, nAAS = 7,923; 78.44%), followed by studies with strength athletes such as bodybuilders, powerlifters or weightlifters (9 studies, nAAS = 801; 7.93%), studies with adolescents (4 studies, nAAS = 377; 3.73%) and studies with NSP clients (2 studies, nAAS = 1,000; 9.90%).

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A large proportion of studies with adolescents (three out of four) and strength athletes (four out of nine) were located in the US, followed by a single study from different countries. The two studies with NSP clients were located either in the UK or Australia. All the five trans-regional studies were conducted with a nonspecific subpopulation of AAS users.

Table 2: Summary characteristics of selected studies

Authors, year	Location	Subpopulation	n	nAAS	nPhys n (%)
Yesalis et al., 1988 [37]	US	strenght athletes	45	15 M	8 (53.33)
Johnson et al., 1989 [38]	US	adolescents	853	95 M	28 (29.47)
Kisling et al., 1989 [39]	Denmark	nonspecific	157	85 M	21 (24.71)
Lindstrom et al., 1990 [40]	Sweden	strenght athletes	138	138 M	12 (8.70)
Terney et al., 1990 [41]	US	adolescents	2113	94 (67 M/27 F)	5 (5.32)
Tanner et al., 1995 [42]	US	adolescents	6930	184 (139 M/45 F)	33 (17.93)
Korkia et al., 1997 [43]	UK	nonspecific	1667	110 (97 M/13 F)	39 (35.45)
Bolding et al., 1999 [44]	UK	nonspecific	1004	81 M	25 (30.86)
Augé et al., 1999 [45]	US	strenght athletes	17	17 (14 M/3 F)	8 (47.05)
Peters et al., 1999 [46]	Australia	nonspecific	100	100 (94 M/6 F)	42 (42.00)
Perry et al., 2005 [47]	US	strenght athletes	207	207*	46 (22.22)
Parkinson et al., 2005 [48]	Trans-region	nonspecific	500	500 (494 M/6 F)	185 (37.00)
Pope et al., 2004 [7]	US	strenght athletes	80	43 M	16 (37.21)
Striegel et al., 2006 [49]	Germany	nonspecific	621	84 (75 M/9 F)	47 (55.95)
Cohen et al., 2007 [50]	US	nonspecific	1955	1955 M	1290 (65.98)
Al-Falasi et al., 2008 [51]	UAE	nonspecific	154	34 M	4 (11.76)
Larance et al., 2008 [52]	Australia	nonspecific	60	60 M	46 (76.66)
Posiadała et al., 2010 [53]	Poland	nonspecific	50	18 M	2 (11.11)
Gradidge et al., 2011 [54]	South Africa	adolescents	100	4 M	1 (25.00)
Ip et al., 2011[55]	Trans-region	nonspecific	1277	506 M	387 (76.48)
Santos et al., 2011 [56]	Brazil	strenght athletes	123	41 M	4 (9.76)
Hope et al., 2013 [23]	UK	NSP clients	395	395 M	178 (45.06)
Raschka et al., 2013 [57]	Germany	nonspecific	484	79 (62 M/ 17 F)	30 (37.97)
Rowe et al., 2016 [58]	Australia	NSP clients	605	605 M	382 (63.14)
Westerman et al., 2016	Trans-region	nonspecific	231	231 M	153 (66.23)
Mooney et al., 2017 [59]	UK	nonspecific	377	26*	1 (3.85)
Zahnow et al., 2017 [19]	Trans-region	nonspecific	195	195*	68 (34.87)
Althobiti et al., 2018 [60]	Saudi Arabia	nonspecific	4860	476 M	181 (38.00)
Hill et al., 2019 [13]	UK	strenght athletes	350	216*	91 (42.00)
Jacka et al., 2019 [61]	Australia	nonspecific	267	267 M	237 (88.76)
Macedo et al., 2019 [62]	Brazil	nonspecific	40	25 M	9 (36.00)
Pany et al., 2019 [63]	India	strenght athletes	74	74 M	24 (32.43)
Pereira et al., 2019 [64]	Brazil	nonspecific	719	194 (149 M/45 F)	117 (60.31)
Uddin et al., 2019 [65]	Pakistan	nonspecific	841	512 M	9 (1.76)
Bonnecaze et al., 2020 [66]	Trans-region	nonspecific	2385	2385 M	1047 (43.90)
Jokipalo et al., 2021 [67]	Finland	strenght athletes	50	50 (42 M/8 F)	15 (30.0)

AAS: androgenic-anabolic steroids. nAAS: number of AAS users in each study. nPhys: number of AAS users who informed seeking support from physicians. M = males. F = females. NSP: Needle and syringe exchange programme. * Sex of AAS users not informed.

Risk of bias

According to the MMAT quality criteria, only seven (19.44%) studies did not present a risk of bias. The following number of studies presented a high risk of bias in each of the MMAT criteria: Inappropriate or unclear sampling strategy (n = 0); sample representativeness low or unclear (n = 10; 27.78%); inappropriate or unclear measurements (n = 14; 38.89%); high or unclear risk of nonresponse bias (n = 19; 52.77%); and inappropriate or unclear statistical analysis (n = 2; 5.56%). The main cause of inappropriate or unclear measurements in some studies was the fact that the number of AAS seeking support from physicians was not clearly stated, requiring an extrapolation from the total number of participants (see full assessment in the supplementary material, eTable 2).

There was no evidence of publication bias, as indicated by visual inspection of the funnel plot (Figure 2), Egger's test for asymmetry (bias coefficient = 0.937, p = 0.349, 95% CI 0.49 to 0.45), and a rank correlation test (τ = -0.086, p = 0.4731).

Figure 2: Funnel plot of studies included in the analysis with pseudo 95% confidence intervals

Meta-analysis of the overall prevalence of AAS users seeking support from physicians

The overall prevalence of AAS users seeking support from physicians obtained from 36 studies is 37.12% (95% CI 27.71 to 44.52). The smallest prevalence rate (1.76%; 95% CI 0.61 to 2.91; nAAS = 512) was observed in a study with gym users in Pakistan [65]. The highest prevalence rate (88.76%, 95% CI 77.46 to 100.06; nAAS = 267) was reported in a study with nonspecific AAS users in Australia [61]. Figure 3 shows a forest plot of prevalence rates, ordered by the effect sizes of all studies.

Figure 3: Forest plot of the pooled prevalence of AAS users seeking support from physicians

Meta-analysis of prevalence rates of studies grouped by location

When grouped by the location of studies, the highest prevalence of AAS users seeking support from physicians was seen among studies taking place in Australia (67.27%; 95% CI 47.29 to 87.25), followed by trans-regional studies (51.48%; 95% CI 35.26 to 67.71). The lowest

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prevalence of AAS users seeking support from physicians was seen in studies located in Africa, Asia or Middle East (21.02%; 95% CI 5.26 to 36.79). A forest plot of the prevalence rates of studies grouped by location is shown in Figure 4.

Figure 4: Forest plot of prevalence rates of studies, grouped by location

Meta-analysis of prevalence rates of studies grouped by subpopulation of AAS users

The highest prevalence of AAS users seeking support from physicians was seen among studies with NSP clients (54.13%; 95% CI 36.41 to 71.84), followed by studies with nonspecific AAS users (41.67%; 95% CI 31.23 to 52.12) and studies with strength athletes (27.83%; 95% CI 17.97 to 37.69). The lowest prevalence of AAS users seeking support from physicians was seen in studies with adolescents (17.27%; 95% CI 4.80 to 29.74). A forest plot of the prevalence rates of studies grouped by subpopulation of AAS users is shown in Figure 5.

Figure 5: Forest plot of prevalence rates of studies, grouped by subpopulation of AAS users

Meta-regression exploring the variance in the prevalence of AAS users seeking support from physicians

The results of the meta-regression analyses are shown in Table 3. Univariable analyses showed that the prevalence of AAS users seeking support from physicians was significantly higher among studies located in Australia (β = 0.35, 95% CI 0.11 to 0.58, p = 0.005) and the univariable model for the location of studies was considered statistically significant (χ^2 (6) = 17.92, p = 0.007). An overall multivariable model with all the variables was considered statistically significant (χ^2 (9) = 22.85, p = 0.007) and reduced the I^2 from 98.0% to 94.8%. In the multivariable analysis, only the subpopulation of adolescents (β = -0.28, 95% CI -0.56 to -0.01, p = 0.048) was considered a significant predictor of the prevalence of AAS users seeking support from physicians.

Table 3: Univariable and multivariable predictors of the prevalence of AAS users seeking support from physicians (n = 36)

		Univariable			Multivariable		
Variable	**No**	**Regression coefficient (95% CI)**	**SE**	**p value**	**Regression coefficient (95% CI)**	**SE**	**p value**
Location							
US	8	1	-	-	1	-	-
UK	5	-0.01 (-0.23 to 0.21)	0.11	0.908	-0.16 (-0.42 to 0.10)	0.13	0.218
Continental Europe	6	-0.05 (-0.26 to 0.16)	0.11	0.629	-0.18 (-0.42 to 0.06)	0.12	0.144
Australia	4	0.35 (0.11 to 0.58)	0.12	0.005**	0.17 (-0.12 to 0.47)	0.15	0.247

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Brazil	3	0.02 (-0.24 to 0.29)	0.14	0.860	-0.10 (-0.39 to 0.19)	0.15	0.490
Africa, Asia or Middle East	5	-0.12 (-0.34 to 0.11)	0.12	0.316	-0.23 (-0.48 to 0.02)	0.13	0.068
Trans-regional	5	0.19 (-0.03 to 0.40)	0.11	0.089	0.03 (-0.24 to 0.30)	0.14	0.826
χ^2 (6) = 17.92, p = 0.007							
Subpopulation							
Nonspecific AAS users	21	1	-	-	1	-	-
Adolescents	4	-0.23 (-0.47 to 0.01)	0.12	0.055	-0.28 (-0.56 to -0.01)	0.14	0.048*
Strenght athletes	9	-0.12 (-0.29 to 0.05)	0.09	0.160	-0.09 (-0.27 to 0.09)	0.09	0.316
NSP clients	2	0.13 (-0.18 to 0.43)	0.15	0.416	0.05 (-0.24 to 0.34)	0.15	0.731
χ^2 (3) = 6.20, p = 0.102					χ^2 (9) = 22.85, p = 0.007		

*p < 0.01, *p < 0.05. SE: standard error

4. Discussion

We conducted a systematic review and meta-analysis and pooled data from 36 studies to estimate that the overall prevalence of AAS users seeking support from physicians is 37.12%. Higher prevalence rates of AAS users seeking support from physicians were seen among studies located in Australia (67.27%) and studies with NSP clients (54.13%). Lower prevalence rates were seen among studies located in Africa, Asia or the Middle East (21.02%) and studies with adolescents (17.27%). We discuss some factors that could have influenced the prevalence of AAS users seeking support from physicians in the studies. One factor possibly influencing the totality of studies is the fact that men, who represent the majority of AAS users [68], are generally less likely to seek medical support [69] – a tendency corroborated by the only two studies comparing the prevalence of male and female AAS users seeking support from physicians [19,45]. Other potential factors include the legal status of AAS use [9], the engagement of AAS users with health services [8], AAS users' perceptions of the service provided by physicians [7,24,70], and the stigma experienced by AAS users [13,15,71].

Our findings suggest that the legal status of AAS use is not always associated with the prevalence of AAS users seeking support from physicians. For example, studies located in Australia – where the possession of AAS is illegal [72] – showed the highest pooled prevalence of AAS users seeking support from physicians (51.13%). Besides, the estimated prevalence of AAS users seeking support from physicians in the US (32.91%) – where the possession of AAS is considered a federal crime [73]– was similar to the estimated prevalence in the UK (31.43%) and Brazil (35.23%), where the use and possession of AAS are not illegal [74,75]. Likewise, a low prevalence of AAS users seeking support from physicians was seen in studies from countries in Africa, Asia or the Middle East (21.02%), where anecdotal reports suggest loose enforcement of the prohibition of AAS use [76–79] or, in the case of India, there are no laws regulating the use and commerce of AAS [80]. Two studies from German showed prevalence rates of AAS users seeking support from

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physicians of 37.97% [57] and 55.95% [49], while a single study from Poland reported a prevalence of 11.11% [53], despite the use of AAS not being illegal in both countries [81,82]. Nevertheless, the legal status of AAS use could have influenced the small prevalence of AAS users seeking support from physicians (8.70%) seen in a single study from Denmark, where the use of AAS is not only illegal but where gym users can be subject to urinalysis to screen for the use of AAS [83]. Other factors might have influenced the prevalence of AAS users seeking support from physicians across the locations analysed by this study. In Australia, for example, there is an effort to educate physicians about the management of the non-prescribed use of AAS [84] which could reflect a willingness to discuss AAS use with the medical community.

The engagement of AAS users with health services could have influenced the comparatively high prevalence (54.13%) of AAS users seeking support from physicians among clients of NSP services that provide information about injection practices, adverse effects of AAS, and probably advising their clients to seek a physician if necessary [23,58]. Among NSP clients, those seeking support from physicians were more likely to have diagnostic screening for health conditions potentially associated with the use of AAS – despite some NSP clients considering physicians a less reliable source of information about AAS than NSP workers [58]. Despite the growing numbers of AAS users seeking the NSP, only a minority of primary NSP units provide advice about AAS [85], as the majority of NSP clients access the service in retail pharmacies providing the exchange of injectable material [86].

The lower pooled prevalence of AAS users seeking support from physicians was seen among adolescents (17.27%). Although low rates of engagement with health services are seen among adolescents in general [87], some factors could have influenced an even lower prevalence of adolescent AAS users seeking support from physicians. Firstly, the prevalence of some health conditions potentially associated with the use of AAS – such as cardiovascular disease [88] – is lower among adolescents. Secondly, it is possible that the illegality of AAS use has deterred adolescents from seeking physicians more than other subpopulations of AAS users. As observed by Terney et al. [41], physicians were allowed to prescribe AAS for enhancement purposes in the US until 1988 and were frequently reported as a source of AAS to adolescents [41,42,89]. As near all of the selected studies with adolescents were located in the US, it is possible that the criminalization of AAS use has driven adolescent AAS users further away from seeking the support of physicians.

Among strength athletes who use AAS, we estimated a prevalence of seeking support from physicians of 27.83%. Strength athletes who use AAS have been described as having a perception that the use of AAS can be safely managed, namely with the support of other AAS users who share their objectives, training routines and lifestyles [20,22]. The self-research and trial-and-

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error experiences with AAS, combined with aesthetical and performance goals frequently considered exaggerated by people outside of the community of strength athletes [90] probably contribute to the perception that physicians are less knowledgeable about AAS than some members of this subpopulation. Furthermore, it is possible that, because of their muscular appearance, some strength athletes who use AAS are even more subject to stigma than other AAS users [91].

Results of the multivariate meta-regression analysis showed that the prevalence rate of studies with adolescents was the only variable with a statistically significant impact on the overall prevalence of AAS users seeking support from physicians. Although current data suggest that adolescents represent a minority of the total population of AAS users [5], it is important to consider the under-representation of this population of AAS users in studies with nonspecific AAS users, trans-regional studies, and all the locations apart from the US. Since the totality of studies with nonspecific AAS users and trans-regional studies in this review were performed with adults aged 16 or older, it is possible that the absence of adolescent AAS users in those studies represented a bias in their prevalence of AAS users seeking support from physicians.

Regarding the kind of support sought by AAS users from physicians among the 36 selected studies, 16 (44.44%) reported that AAS users sought physicians as a source of information about the use and adverse effects of AAS. The remaining studies described AAS users' contact with physicians in many different ways, such as having close contact with physicians [49], seeking a doctor for interpretation of health checks [13], and disclosing the use of AAS to a physician [7,66]. We considered that further exploration of the types of support sought by AAS users would lie beyond the scope of this study, as they have been recently investigated by other reviews [17,92,93].

Limitations of this study.

The prevalence of AAS users seeking support from physicians varied widely across the selected studies, and this variation was only minimally explained by the meta-regression and the comparisons between subgroups of studies. Furthermore, the selected studies investigated the help-seeking behaviours of AAS users in many different ways – e.g. interviews, anonymous surveys, longitudinal studies – and were synthesised as a single variable for comparison. The pooled prevalence in different locations was based on a limited sample of highly heterogenic studies. For instance, the estimated prevalence of AAS users seeking support from physicians from Australia was based on only four studies – three with a nonspecific population of AAS users and one with NSP clients – comprising 10.2% of the total number of AAS users from selected studies. This review did not distinguish data from male and female AAS users, as the sex of

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participants or differences in help-seeking behaviour between male and female participants were not reported by the majority of studies. The review only included studies from a few countries and, among those, many contributed with a single study unable to represent the local population of AAS users. As discussed in this review, the engagement of AAS users with physicians can be influenced by several factors, including attitudes that can vary widely between locations and subpopulations of AAS users. These limitations can compromise the generalisation of our results, and further studies are necessary to better understand the help-seeking behaviours of AAS users, namely among understudied locations and subpopulations of AAS users.

Implications for practice and policy

Available data suggest that factors such as the criminalisation of AAS use, the scarcity of physicians' knowledge about AAS and stigma against AAS users are barriers to the access of some AAS users to physicians. The results of this review can indicate locations and subpopulations of AAS users with higher engagement with physicians, so successful strategies can be replicated. Likewise, our results indicate the existence of under-studied and possibly under-supported populations of AAS users.

5. Conclusion

This meta-analysis was the first to systematically investigate the prevalence of AAS users seeking support from physicians. Our findings suggest that the overall prevalence of AAS users seeking support from physicians is 37.12%, with considerable variation across locations and among subpopulations of AAS users. This study highlights the importance of understanding the help-seeking behaviours of AAS users and improving their access to physicians.

Acknowledgements

We thank the participants and authors of the selected studies, the developers of R and the mefator package, and the reviewers of this journal for their valuable comments.

Contributors

JA planned the study and conducted the meta-analysis. JA and AK conducted the selection of studies and the assessment of the risk of bias. AK and PD contributed to the interpretation of data. JA, AK and PD reviewed and approved the final draft.

Funding

The first author was funded by King's College London International Scholarship (RDPL3PRPCA - 2019-20).

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Competing interests

None

Ethics approval

As a review of the literature, this study was exempt from ethical approval by King's College London Research Ethics Office.

Provenance and peer reviewed

Not commissioned; externally peer reviewed.

Data availability statement

No additional data are available.

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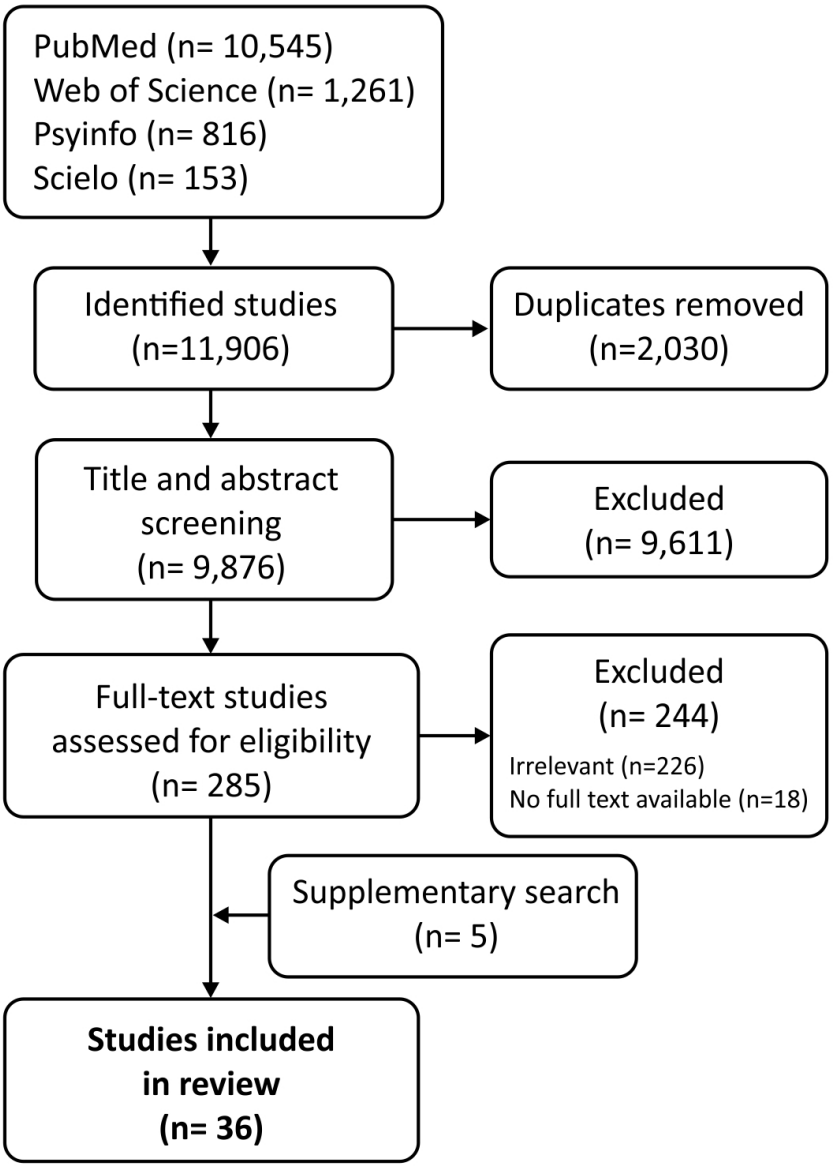


Figure 1: Flowchart of the inclusion of studies in the review

92x128mm (300 x 300 DPI)

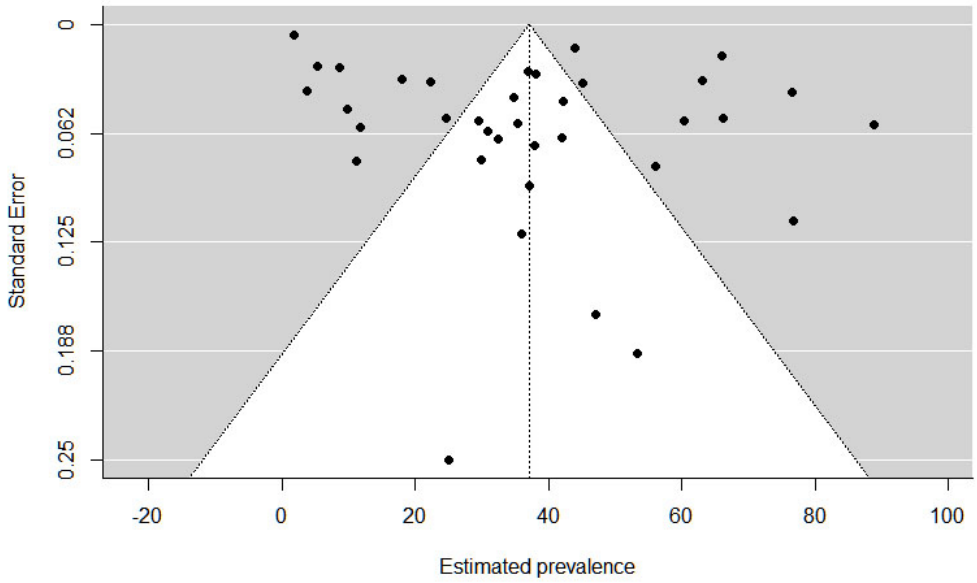


Figure 2: Funnel plot of studies included in the analysis with pseudo 95% confidence intervals

162x98mm (120 x 120 DPI)

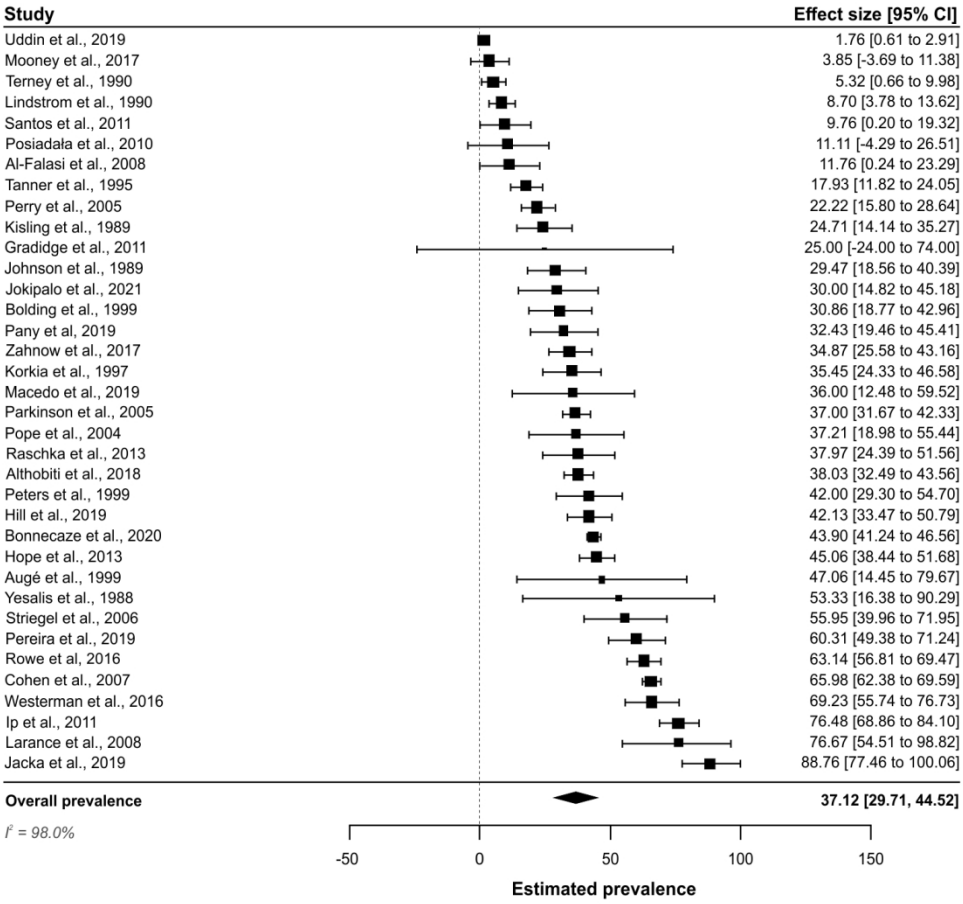


Figure 3: Forest plot of the pooled prevalence of AAS users seeking support from physicians

138x126mm (300 x 300 DPI)

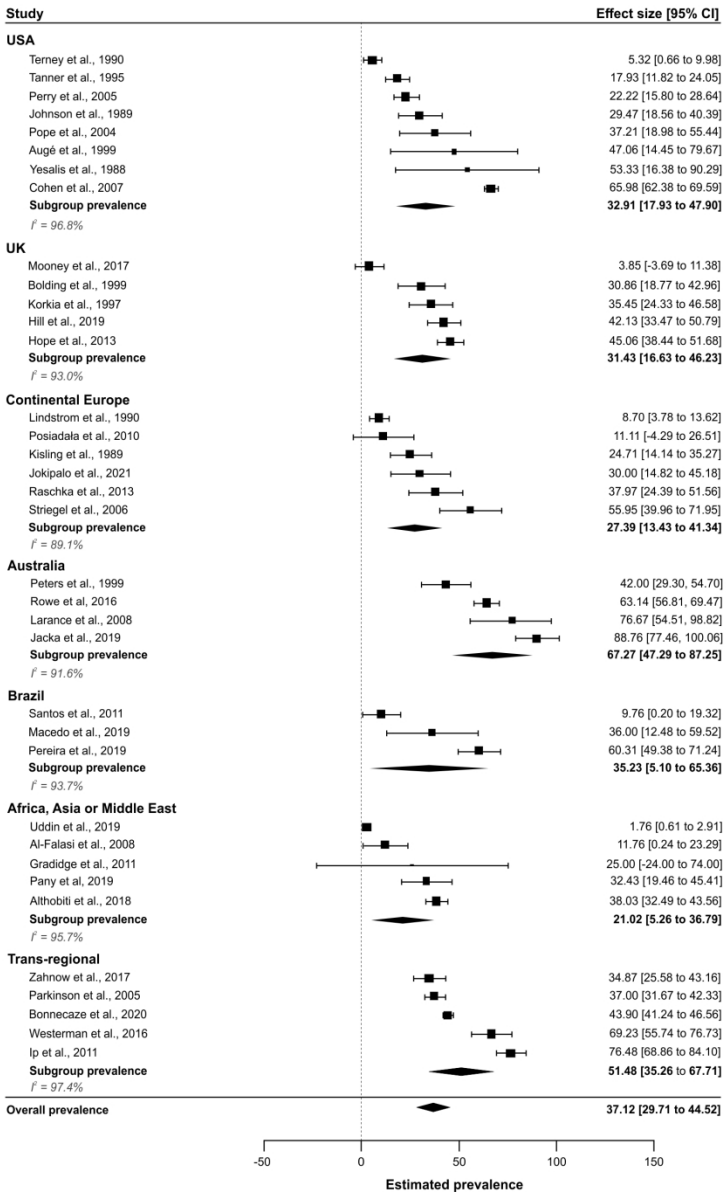


Figure 4: Forest plot of prevalence rates of studies, grouped by location

138x221mm (300 x 300 DPI)

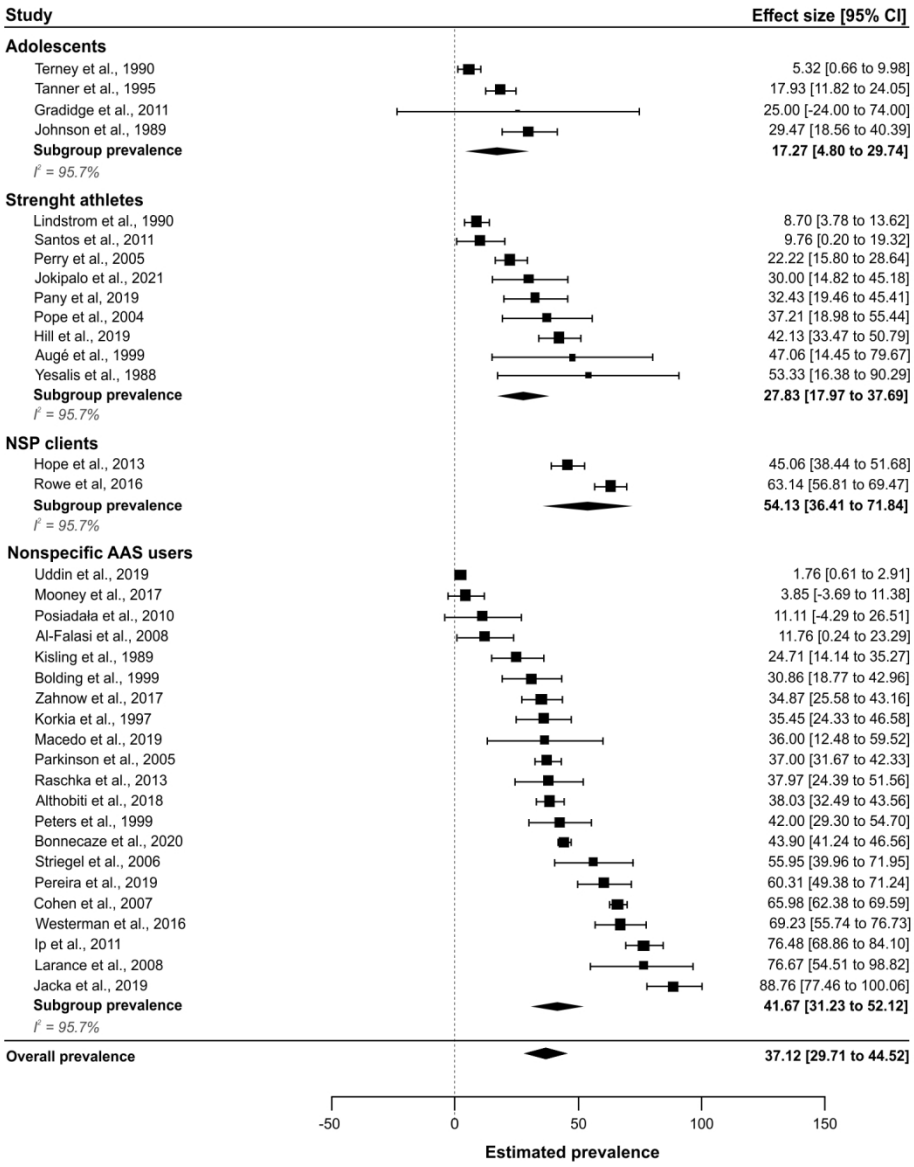


Figure 5: Forest plot of prevalence rates of studies, grouped by subpopulation of AAS users

138x174mm (300 x 300 DPI)

Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Amaral JMX, Kimergård A, Deluca P

SUPPLEMENTARY MATERIAL

eTable 1: Search algorithms

PUBMED (results: 10,545)
((agents, anabolic[MeSH Terms]) OR (doping in sport[MeSH Terms]) OR (drugs, performance enhancing[MeSH Terms])) OR ((anabolic steroid*[Title/Abstract]) OR (anabolic androgenic[Title/Abstract]) OR (anabolic agent*[Title/Abstract]) OR (anabolic drug*[Title/Abstract]) OR (androgen* abuse[Title/Abstract]) OR (performance enhanc*[Title/Abstract]) OR (image enhanc*[Title/Abstract]) OR (synthetic androgen*[Title/Abstract]) OR (synthetic testosterone[Title/Abstract]) OR (testosterone ester*[Title/Abstract]) OR (designer steroid*[Title/Abstract]) OR (doping[Title/Abstract]) OR ("non prescript*" steroid*[Title/Abstract]) OR ("non medic*" steroid*[Title/Abstract]) OR ("non medic*" androgen*[Title/Abstract]) OR ("non prescript*" androgen*[Title/Abstract])) NOT ((review[Title/Abstract]) OR (animal*[Title/Abstract]) OR (bovine[Title/Abstract]) OR (in vitro[Title/Abstract]) OR (mice*[Title/Abstract]) OR (pig[Title/Abstract]) OR (pigs[Title/Abstract]) OR (rat[Title/Abstract]) OR (rats[Title/Abstract]) OR (mouse[Title/Abstract]) OR (spectrometry[Title/Abstract]) OR (nano*[Title/Abstract]) OR (conductor*[Title/Abstract]) OR (semiconductor[Title/Abstract]) OR (electro*[Title/Abstract]) OR (optic*[Title/Abstract]) OR (hydrogen[Title/Abstract]) OR (superconductor*[Title/Abstract]) OR (catalyst*[Title/Abstract]) OR (thermoelectric[Title/Abstract]) OR (ions[Title/Abstract])) AND (humans[Filter])
Web of Science (results: 1,261)
((TS=("doping in sport" OR "performance enhancing drug*"OR "anabolic steroid*" OR "anabolic androgenic*" OR "anabolic agent*" OR "anabolic drug*")) NOT (TS=(review OR animal* OR bovine OR “in vitro” OR mice* OR pig OR pigs OR rat OR rats OR mouse OR spectrometry OR nano* OR conductor* OR semiconductor OR electro* OR optic* OR hydrogen OR superconductor* OR catalyst* OR thermoelectric OR ions))) Filters: Articles, Sport Sciences, Medicine (all), Psychology (all), Toxicology
APA PsycNet (results: 816) A
((anabolic or doping or performance enhancing) not (review or animal or bovine or vitro or mice* or pig* or rat* or mouse or spectrometry or nano* or conductor* or semiconductor or electro* or optic* or hydrogen or superconductor* or catalyst* or thermoelectric or ions)).ti - limit to human
Scielo (results: 153)
(ti:((ti:((ti:(anabolic)) OR (ti:(doping)) OR (ti:(performance enhancing)))) AND NOT (ti:(review)))) AND NOT (ti:((ti:(animal)) AND NOT (ti:(bovine)) AND NOT (ti:(vitro)) AND NOT (ti:(mice*)) AND NOT (ti:(pig*)) AND NOT (ti:(rat*)) AND NOT (ti:(mouse)) AND NOT (ti:(spectrometry)) AND NOT (ti:(nano*)) AND NOT (ti:(conductor*)) AND NOT (ti:(semiconductor)) AND NOT (ti:(electro*)) AND NOT (ti:(optic*)) AND NOT (ti:(hydrogen)) AND NOT (ti:(superconductor*)) AND NOT (ti:(catalyst*)) AND NOT (ti:(thermoelectric)) AND NOT (ti:(ions))))

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eTable 2: Risk of bias

	Risk of bias: MMAT criteria*					
Authors, year	**1**	**2**	**3**	**4**	**5**	**Observations**
Yesalis et al., 1988	Y	N	?	Y	Y	Small sample of AAS users. Source of information described as 'Physician/pharmacist/medical literature'.
Johnson et al., 1989	Y	Y	Y	Y	Y	n/a
Kisling et al., 1989	Y	Y	?	Y	Y	Nonresponse rate not informed.
Lindstrom et al., 1990	Y	Y	Y	Y	N	The number of AAS users seeking a physician must be inferred from the text.
Terney et al., 1990	Y	Y	N	Y	Y	The number of AAS users seeking support from physicians was extrapolated from the total number of participants.
Peters et al., 1993	Y	Y	Y	?	Y	Nonresponse rate not informed.
Tanner et al., 1995	Y	Y	N	Y	Y	The number of AAS users seeking support from physicians was extrapolated from the total number of participants.
Korkia et al., 1997	Y	Y	Y	N	Y	Nonresponse rate = 41%.
Bolding et al., 1999	Y	Y	?	Y	Y	Support described as 'regular health checks to screen for steroid-related health problems'.
Auge et al., 1999	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Perry et al., 2005	Y	Y	N	Y	Y	The number of AAS users seeking a physician must be inferred from a graph and was extrapolated from the total number of participants.
Parkinson et al., 2005	Y	Y	Y	?	Y	Nonresponse rate not informed.
Pope et al., 2004	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Striegel et al., 2006	Y	Y	Y	?	Y	Nonresponse rate not informed.
Cohen et al., 2007	Y	Y	?	?	Y	Nonresponse rate not informed. Support described as 'willing to seek medical supervision'.
Al-Falasi et al., 2008	Y	N	N	N	Y	Nonresponse rate = 56%. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Larance et al., 2008	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Posiadala et al., 2010	Y	N	Y	N	?	Small sample of AAS users. Nonresponse rate = 28.57%.
Gradidge et al., 2011	Y	Y	N	Y	Y	Small sample of AAS users. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Ip et al., 2011	Y	Y	Y	Y	Y	n/a
Santos et al., 2011	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Hope et al., 2013	Y	Y	Y	Y	Y	n/a
Raschka et al., 2013	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Mooney et al., 2016	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Rowe et al., 2016	Y	Y	Y	Y	Y	n/a
Westerman et al., 2016	Y	Y	Y	Y	Y	n/a
Zahnow et al., 2017	Y	Y	Y	?	Y	Nonresponse rate not informed.
Althobiti et al., 2018	Y	Y	N	?	Y	Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Hill et al., 2019	Y	Y	Y	Y	Y	n/a
Jacka et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.

Macedo et al., 2019	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Pany et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.
Pereira et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.
Uddin et al., 2019	Y	Y	N	Y	Y	The number of AAS users seeking a physician must be inferred from the text.
Bonnecaze et al., 2020	Y	Y	Y	Y	Y	n/a
Jokipalo et al., 2021	Y	Y	Y	Y	Y	n/a

* MMAT's quality-assessment criteria:
1 - Is the sampling strategy relevant to address the research question?
2 - Is the sample representative of the target population?
3 - Are the measurements appropriate?
4 - Is the risk of nonresponse bias low?
5 - 'Is the statistical analysis appropriate to answer the research question?'
Y = yes. N = no. ? = not clear.

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R codes for the meta-analysis, forest plots and meta-regression
Results of the Baujat plot

The table below contains the data of the file 'metaAAS.csv'. The column 'ti' contains the number of androgenic-anabolic steroids (AAS) users in each study, and the column 'xi', the number of AAS users who sought support from physicians.

study id	studies	ti	xi	locat	pop
1	Yesalis et al.,1988	15	8	0	1
2	Johnson et al.,1989	95	28	0	0
3	Kisling et al.,1989	85	21	2	3
4	Lindstrom et al.,1990	138	12	2	1
5	Terney et al.,1990	94	5	0	0
6	Peters et al.,1993	100	42	3	3
7	Tanner et al.,1995	184	33	0	0
8	Korkia et al.,1997	110	39	1	3
9	Bolding et al.,1999	81	25	1	3
10	Auge et al.,1999	17	8	0	1
11	Perry et al.,2005	207	46	0	1
12	Parkinson et al.,2005	500	185	6	3
13	Pope et al.,2004	43	16	0	1
14	Striegel et al.,2006	84	47	2	3
15	Cohen et al.,2007	1955	1290	0	3
16	Al-Falasi et al.,2008	34	4	5	3
17	Larance et al.,2008	60	46	3	3
18	Posiadala et al.,2010	18	2	2	3
19	Gradidge et al.,2011	4	1	5	0
20	Ip et al.,2011	506	387	6	3
21	Santos et al.,2011	41	4	4	1
22	Hope et al.,2013	395	178	1	2
23	Raschka et al.,2013	79	30	2	3
24	Mooney et al.,2016	26	1	1	3
25	Rowe et al.,2016	605	382	3	2
26	Westerman et al.,2016	231	153	6	3
27	Zahnow et al.,2017	195	68	6	3
28	Althobiti et al.,2018	476	181	5	3
29	Hill et al.,2019	216	91	1	1
30	Jacka et al.,2019	267	237	3	3
31	Macedo et al.,2019	25	9	4	3
32	Pany et al.,2019	74	24	5	1
33	Pereira et al.,2019	194	117	4	3
34	Uddin et al.,2019	512	9	5	3
35	Bonnecaze et al.,2020	2385	1047	6	3
36	Jokipalo et al.,2021	50	15	2	1

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R codes

```
## Loading packages previously installed
library("robumeta")
library("metafor")
library("dplyr")
library("forestplot")
library("checkmate")
library("nloptr")
library("meta")
library("CompQuadForm")

## Meta-analysis of all studies
metaAAS <- read.csv("metaAAS.csv")
View(metaAAS)
metaAAS <- escalc(measure="IR", xi=xi, ti=ti, data=metaAAS, slab=paste (studies))
res <- rma(yi, vi, data=metaAAS)
res
confint(res)

## Baujat plot
b_res <- rma(xi, ti, data=metaAAS, slab=study.id)
baujat(b_res)
inf <- influence(res)
print(inf)
plot(inf)


### funnel plot
funnel(res,atransf=mytransf,xlab = "Estimated prevalence")
regtest(res)
ranktest(res)
warnings()


### Cross-tabulation of the number of studies – location vs subpopulation
addmargins(table(metaAAS$locat, metaAAS$pop))


### Setup functions for the forest plot
mlabfun <- function(text, res) {
 list(bquote(paste(.(text),
            " (Q = ", .(formatC(res$QE, digits=2, format="f")),
            ", df = ", .(res$k - res$p),
            ", p ", .(metafor:::.pval(res$QEp, digits=2, showeq=TRUE, sep=" ")), "; ",
            I^2, " = ", .(formatC(res$I2, digits=1, format="f")), "%, ",
            tau^2, " = ", .(formatC(res$tau2, digits=2, format="f")), ")"))))}
mytransf<-function(x)
 (x)*100


## Forest plot of all studies
forest (res, transf=mytransf, cex=.7,header=c("Authors and Year", "Effect size [95%
CI]"),xlab="Prevalence rate",
      mlab=mlabfun("RE Model for All Studies", res), order="obs", refline=NA)
```

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```
## Forest plot - subgroup LOCATION
forest (res, transf=mytransf, cex=.6, refline=NA,top=0,
      order=metaAAS$locat,
      rows=c(65:58,53:49,44:39,34:31,26:24,19:15,10:6),
      mlab=mlabfun("RE Model for All Studies", res),
      header=c("Authors and year", "Effect size [95% CI]"),
      xlab="Estimated prevalence")


### switch to bold italic font
par(font=4)

### add text for the subgroups
text(-130, c(54,45,35,27,20,11), pos=4,
   c("UK","Continental Europe","Australia","Brazil",
     "Africa,Asia and Middle East","Trans-regional"))

### fit random-effects model in the seven subgroups
res.0 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="0"), data=metaAAS)
res.1 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="1"), data=metaAAS)
res.2 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="2"), data=metaAAS)
res.3 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="3"), data=metaAAS)
res.4 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="4"), data=metaAAS)
res.5 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="5"), data=metaAAS)
res.6 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="6"), data=metaAAS)

### add summary polygons for the seven subgroups
addpoly(cex=.5,res.0, row=56.5, mlab=mlabfun("RE Model for Subgroup", res.0),transf=mytransf)
addpoly(cex=.5,res.1, row=47.5, mlab=mlabfun("RE Model for Subgroup", res.1),transf=mytransf)
addpoly(cex=.5,res.2, row=37.5, mlab=mlabfun("RE Model for Subgroup", res.2),transf=mytransf)
addpoly(cex=.5,res.3, row=29.5, mlab=mlabfun("RE Model for Subgroup", res.3),transf=mytransf)
addpoly(cex=.5,res.4, row=22.5, mlab=mlabfun("RE Model for Subgroup", res.4),transf=mytransf)
addpoly(cex=.5,res.5, row=13.5, mlab=mlabfun("RE Model for Subgroup", res.5),transf=mytransf)
addpoly(cex=.5,res.6, row=4.5, mlab=mlabfun("RE Model for Subgroup", res.6),transf=mytransf)


### fit meta-regression model to test for subgroup differences
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(locat), data=metaAAS)
resMeta
confint(resMeta)

### add text for the test of subgroup differences
text(-130, -3, pos=4, cex=0.6,
   bquote(paste("Test of moderator (location): ",Q[M], " = ",
            .(formatC(resMeta$QM, digits=2, format="f")),
            ", df = ", .(resMeta$p - 1),
                        ", p = ", .(formatC(resMeta$QMp, digits=2, format="f")))))
```

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```
##Forest plot -subgroup SUBPOPULATION
forest (res, transf=mytransf, cex=.6, refline=NA,top=0,
      order=metaAAS$pop,
      rows=c(53:50,45:37,32:31,26:6),
      mlab=mlabfun("RE Model for All Studies", res),
      header=c("Authors and year", "Effect size [95% CI]"),
      xlab="Estimated prevalence")

### fit random-effects model in the four subgroups
res.0 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="0"), data=metaAAS)
res.1 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="1"), data=metaAAS)
res.2 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="2"), data=metaAAS)
res.3 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="3"), data=metaAAS)

### add summary polygons for the four subgroups
addpoly(cex=.5,res.0, row=48, mlab=mlabfun("RE Model for Subgroup", res.0),transf=mytransf)
addpoly(cex=.5,res.1, row=35, mlab=mlabfun("RE Model for Subgroup", res.1),transf=mytransf)
addpoly(cex=.5,res.2, row=29, mlab=mlabfun("RE Model for Subgroup", res.2),transf=mytransf)
addpoly(cex=.5,res.3, row=4, mlab=mlabfun("RE Model for Subgroup", res.3),transf=mytransf)


### fit meta-regression model to test for subgroup differences
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(pop), data=metaAAS)
resMeta
confint(resMeta)

### add text for the test of subgroup differences
text(-130, -3, pos=4, cex=0.6,
   bquote(paste("Test of moderator (sub-population): ",Q[M], " = ",
            .(formatC(resMeta$QM, digits=2, format="f")),
            ", df = ", .(resMeta$p - 1),
            ", p = ", .(formatC(resMeta$QMp, digits=2, format="f")))))


### Univariate meta-regression model. Moderator: LOCATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(locat), ref="0"),  data=metaAAS)
resMeta
confint(resMeta,transf=mytransf)


### Univariate meta-regression model. Moderator: SUBPOPULATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(pop), ref="3"), data=metaAAS)
resMeta
confint(resMeta)

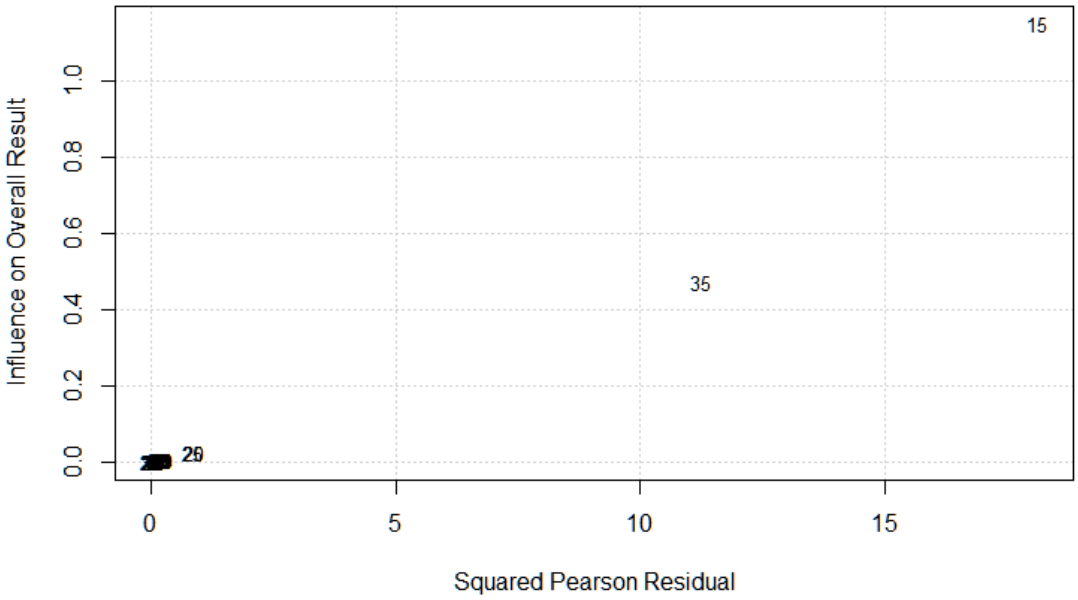


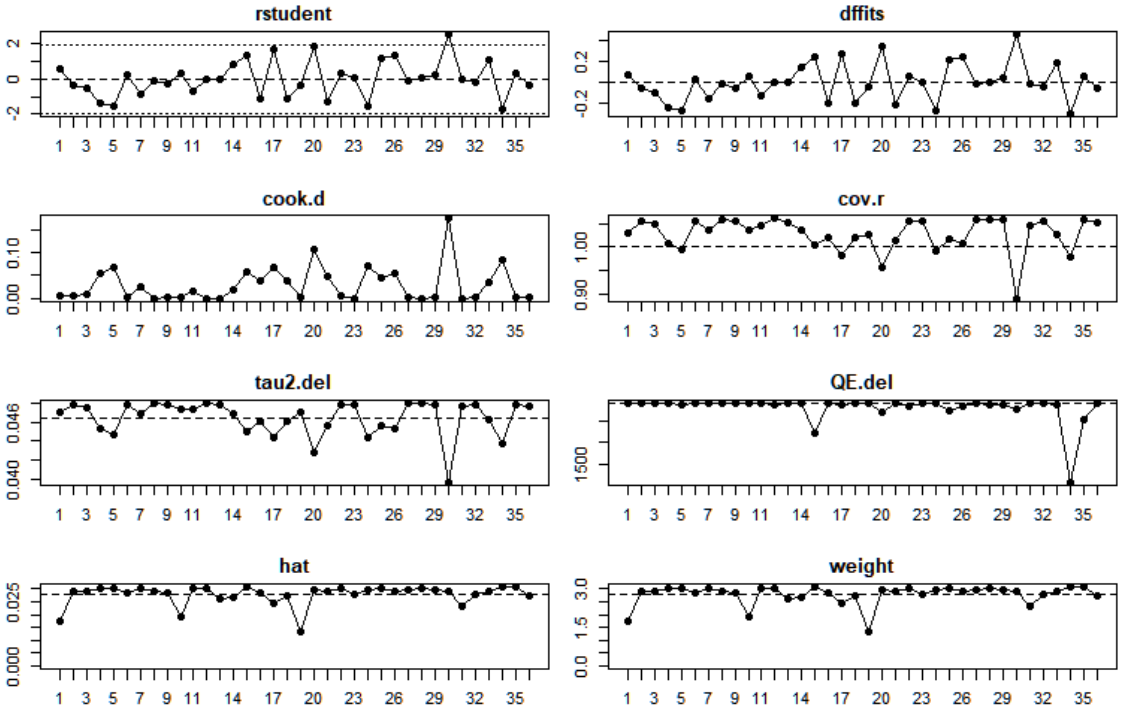
### Multivariate meta-regression model. Moderators: LOCATION + SUBPOPULATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(locat)+relevel(factor(pop), ref="3"),
data=metaAAS)
resMeta
confint(resMeta)
```

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Results of the Baujat plot

	rstudent	dffits	cook.d	cov.r	tau2.del	QE.del	hat	weight	dfbs	inf
Yesalis et al.,1988	0.5691	0.0751	0.0057	1.0296	0.0470	2901.5926	0.0174	1.7416	0.0749	
Johnson et al.,1989	-0.3444	-0.0615	0.0039	1.0562	0.0477	2901.1899	0.0288	2.8822	-0.0615	
Kisling et al.,1989	-0.5615	-0.0986	0.0099	1.0507	0.0475	2903.7126	0.0289	2.8937	-0.0986	
Lindstrom et al.,1990	-1.3464	-0.2365	0.0546	1.0080	0.0453	2888.7020	0.0303	3.0336	-0.2363	
Terney et al.,1990	-1.5188	-0.2662	0.0681	0.9939	0.0446	2872.4020	0.0304	3.0377	-0.2658	
Peters et al.,1993	0.2168	0.0346	0.0012	1.0567	0.0478	2891.9587	0.0282	2.8198	0.0346	
Tanner et al.,1995	-0.8914	-0.1576	0.0250	1.0380	0.0468	2904.9166	0.0301	3.0115	-0.1576	
Korkia et al.,1997	-0.0751	-0.0154	0.0002	1.0593	0.0479	2896.1842	0.0288	2.8752	-0.0154	
Bolding et al.,1999	-0.2797	-0.0501	0.0026	1.0566	0.0478	2901.0530	0.0284	2.8416	-0.0501	
Auge et al.,1999	0.3664	0.0502	0.0025	1.0367	0.0473	2902.0604	0.0193	1.9265	0.0501	
Perry et al.,2005	-0.6882	-0.1224	0.0152	1.0477	0.0473	2903.7785	0.0301	3.0053	-0.1225	
Parkinson et al.,2005	-0.0059	-0.0036	0.0000	1.0626	0.0480	2858.4148	0.0303	3.0265	-0.0036	
Pope et al.,2004	0.0035	-0.0017	0.0000	1.0536	0.0478	2900.9937	0.0259	2.5919	-0.0017	
Striegel et al.,2006	0.8253	0.1366	0.0188	1.0357	0.0468	2884.0498	0.0269	2.6891	0.1365	
Cohen et al.,2007	1.3765	0.2473	0.0594	1.0032	0.0451	2203.3151	0.0305	3.0526	0.2470	
Al-Falasi et al.,2008	-1.1563	-0.1977	0.0388	1.0208	0.0460	2903.5873	0.0286	2.8616	-0.1977	
Larance et al.,2008	1.6754	0.2655	0.0680	0.9820	0.0444	2878.6129	0.0241	2.4114	0.2664	
Posiadala et al.,2010	-1.1536	-0.1919	0.0366	1.0200	0.0461	2904.0446	0.0271	2.7140	-0.1920	
Gradidge et al.,2011	-0.3687	-0.0438	0.0019	1.0253	0.0470	2904.9093	0.0131	1.3108	-0.0436	
Ip et al.,2011	1.9001	0.3414	0.1077	0.9560	0.0428	2681.1281	0.0298	2.9778	0.3407	
Santos et al.,2011	-1.2671	-0.2186	0.0470	1.0136	0.0456	2901.6068	0.0292	2.9249	-0.2186	
Hope et al.,2013	0.3645	0.0618	0.0039	1.0575	0.0477	2842.9150	0.0300	3.0010	0.0619	
Raschka et al.,2013	0.0375	0.0040	0.0000	1.0575	0.0479	2897.1920	0.0279	2.7864	0.0040	
Mooney et al.,2016	-1.5763	-0.2731	0.0714	0.9891	0.0444	2889.9271	0.0298	2.9798	-0.2728	
Rowe et al.,2016	1.2229	0.2170	0.0463	1.0150	0.0457	2712.0790	0.0301	3.0072	0.2168	
Westerman et al.,2016	1.3474	0.2350	0.0539	1.0053	0.0452	2825.5656	0.0290	2.8960	0.2349	
Zahnow et al.,2017	-0.1027	-0.0204	0.0004	1.0610	0.0479	2890.1445	0.0296	2.9607	-0.0204	
Althobiti et al.,2018	0.0413	0.0047	0.0000	1.0624	0.0480	2856.9754	0.0302	3.0228	0.0047	
Hill et al.,2019	0.2275	0.0373	0.0014	1.0592	0.0479	2876.4907	0.0295	2.9508	0.0373	
Jacka et al.,2019	2.5277	0.4496	0.1749	0.8926	0.0397	2756.3671	0.0287	2.8694	0.4492	
Macedo et al.,2019	-0.0457	-0.0092	0.0001	1.0485	0.0476	2902.8855	0.0235	2.3471	-0.0091	
Pany et al.,2019	-0.2084	-0.0378	0.0015	1.0570	0.0478	2900.6346	0.0281	2.8096	-0.0378	
Pereira et al.,2019	1.0596	0.1829	0.0333	1.0252	0.0462	2848.8713	0.0288	2.8819	0.1829	
Uddin et al.,2019	-1.7165	-0.3019	0.0858	0.9758	0.0437	1099.7502	0.0307	3.0725	-0.3012	
Bonnecaze et al.,2020	0.3139	0.0533	0.0029	1.0598	0.0478	2520.6043	0.0306	3.0626	0.0534	
Jokipalo et al.,2021	-0.3118	-0.0543	0.0030	1.0537	0.0477	2902.8279	0.0272	2.7230	-0.0543	





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PRISMA 2020 Main Checklist

Topic	No.	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Section 1, Lines 126-128
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section 1, Lines 120-123
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section 1, Lines 123-125, 126-128
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 2, Line 157 (Table 1)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 2, Line 141
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 2, Line 144 (Figure 1)
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2, Lines 146-150
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 2, Lines 159-163
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 2, Lines 159-163
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 2, Lines 159-163

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Topic	No.	Item	Location where item is reported
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2, Lines 165-174
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Section 2, 176-200
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	Section 2, 198-200
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 2, 176-200
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 2, 176-200
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 2, 176-204
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Section 2, Lines 182-190
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Section 2, 201-204
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Section 2, Lines 201-204
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Section 2, 201-204
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 2, Line 154 (Figure 2)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	n/a
Study characteristics	17	Cite each included study and present its characteristics.	Section 3, Line 217 (Table 2)
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section 3, Lines 226-233
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Section 3, Line 217 (Table 2)
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section 3, Lines 226-233

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Topic	No.	Item	Location where item is reported
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Section 3, Lines 235-390
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Section 3, Lines 240-375
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Section 3, Lines 365-372
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Section 3, Lines 240-375
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Section 3, Lines 240-375
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section 4, Lines 377-419
	23b	Discuss any limitations of the evidence included in the review.	Section 4, Lines 420-438
	23c	Discuss any limitations of the review processes used.	Section 4, Lines 420-438
	23d	Discuss implications of the results for practice, policy, and future research.	Section 5, Lines 446-449
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Section 2, Lines 130-132
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Section 2, Lines 130-132
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	n/a
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Section 6, Lines 461-453
Competing interests	26	Declare any competing interests of review authors.	Section 6, Line 453
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Section 3, Lines 238-239

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PRIMSA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesize results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. MetaArXiv. 2020, September 14. DOI: 10.31222/osf.io/v7gm2. For more information, visit: www.prisma-statement.org

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BMJ Open

Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-056445.R3
Article Type:	Original research
Date Submitted by the Author:	30-May-2022
Complete List of Authors:	Amaral, Julio Mario; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions Kimergård, Andreas; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions Deluca, Paolo; King's College London Institute of Psychiatry Psychology and Neuroscience, Addictions
Primary Subject Heading:	Medical management
Secondary Subject Heading:	Addiction, General practice / Family practice, Health services research
Keywords:	Substance misuse < PSYCHIATRY, PRIMARY CARE, GENERAL MEDICINE (see Internal Medicine)

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Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

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Word count: 4,930

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Abstract

Objectives: To estimate the overall prevalence of androgenic-anabolic steroids (AAS) users seeking support from physicians. Secondary objectives are to compare this prevalence in different locations and among subpopulations of AAS users, and to discuss some of the factors that could have influenced the engagement of AAS users with physicians.

Design: Systematic review and meta-analysis.

Data sources: MEDLINE, PsycINFO, Web of Science and SciELO were searched in January 2022.

Eligibility criteria: Quantitative and qualitative studies reporting the number of AAS users who sought support from physicians, with no restrictions of language or time of publication.

Data extraction and synthesis: Two independent reviewers extracted data and assessed the quality of studies, including publication bias. A random-effects meta-analysis was performed to estimate the overall prevalence of AAS users seeking support from physicians, followed by pooled prevalence rates by studies' location and the subpopulation of AAS users.

Results: We identified 36 studies published between 1988 and 2021, involving 10,101 AAS users. The estimated overall prevalence of AAS users seeking support from physicians is 37.12% (95% CI 29.71 to 44.52). Higher prevalence rates were observed in studies from Australia (67.27%; 95%CI 42.29 to 87.25) and among clients of the needle and syringe exchange programme (54.13%; 95% CI 36.41 to 71.84). The lowest prevalence was observed among adolescent AAS users (17.27%; 95% CI 4.80 to 29.74).

Conclusion: Our findings suggest that about one-third of AAS users seek support from physicians, with remarkable differences between locations and subpopulations of AAS users. Further studies should investigate the factors influencing the engagement of AAS users with physicians.

Registration: PROSPERO, CRD42020177919.

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Strengths and limitations of this study

- This review analyses 36 studies published between 1988 and 2021, involving 10,101 AAS users.
- It compares the pooled prevalence rates of AAS seeking support from physicians between different locations and subpopulations of AAS users.
- The R codes used in the meta-analysis and meta-regression are available in the supplementary files, allowing quick reproducibility of the study.
- Our results are based on studies from a small number of countries, with a limited representativeness of the subpopulations of AAS users.

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Introduction

Anabolic-androgenic steroids (AAS) are synthetic androgens with several potential effects. In clinical settings, AAS can be used to treat conditions such as male hypogonadism, pathological loss of muscle mass and anaemia [1], with findings suggesting their efficacy in the treatment of depression [2] and conditions associated with type 2 diabetes [3]. Beneficial effects of AAS include the increasing of muscle mass, feelings of wellbeing and boosted energy, enhancement of body image, and improvement of athletic and occupational performance [4,5]. However, the use of AAS can increase the risk of several adverse health conditions such as acne, testicular atrophy, gynecomastia, clitoromegaly, hypomania, anxiety, dyslipidaemia and high haematocrit – therefore increasing the risk of myocardial infarction and stroke [6]. Despite the risks, many AAS users refrain from seeking physicians for AAS-related information or to treat health conditions potentially associated with the use of AAS [7]. Among factors possibly influencing the prevalence of AAS seeking support from physicians are the legal status of AAS, AAS users' engagement with health services and their perceptions of the services provided by physicians to people using AAS [8]. In countries where the possession of AAS without a medical prescription is illegal, it is reasonable to expect that some AAS users will refrain from disclosing the use of AAS to a physician [9]. The legal status of AAS can also influence the service provided to AAS users by physicians, as doctors are usually not allowed to prescribe AAS for the purposes of enhancement or to regulate hormonal levels after the use of AAS, a practice also known as post-cycle therapy (PCT) [10]. Whilst physicians are trained to treat the use of other illegal substances such as heroin and cocaine, many of them admit a lack of training and experience in recognising and treating adverse effects of AAS [11–14]. Another reason given by some AAS users to refrain from seeking medical support is the stigma and judgmental attitudes experienced in their contact with health professionals [13,15]. Due to these factors – and possibly others, such as the provision of health services in their locality, financial limitations, etc. – some AAS users rely on self-conducted research to manage their use and adverse effects of AAS and/or seek the support of informal sources such as friends and online forums [16,17]. Finally, some AAS users reported not seeking physicians simply because they did not feel the need to do so, due to an absence of adverse effects or to a perception that these effects can be managed without the help of a medical professional [13].

In addition to the legal status of AAS use and the access to health services in different countries, different help-seeking behaviours are seen across subpopulations of AAS users [18]. Younger AAS users, for instance, seem to be less likely to engage with health services [19]. Some strength athletes tend to rely on other athletes, who are perceived as knowledgeable about AAS [20–22], whilst AAS users who are clients of the needle and syringe exchange programme (NSP) seem to

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be more likely to interact with health services [23,24]. However, to this date, no systematic comparison of the prevalence of AAS users seeking the support of physicians has been produced. The objective of this study is to estimate the overall prevalence of AAS users seeking support from physicians by conducting a systematic review and meta-analysis of surveys and interviews with AAS users. The secondary objectives are to compare the prevalence of AAS users seeking support from physicians in different locations and among subpopulations of AAS users and discuss some of the factors that could have influenced their engagement with physicians' support.

Methods

Overview

This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [25], and its protocol is registered in PROSPERO (registration number CRD42020177919). As a review of the literature, this study is exempted from ethical clearance by King's College London research ethics office.

Search strategy and selection criteria

A search strategy was designed to retrieve studies describing surveys and interviews with people using AAS. Searches were performed on MEDLINE, PsycINFO, Web of Science and SciELO in January 2022 with no restrictions on the date, location and language of studies. The search algorithm adapted to each database can be found in the supplementary material (eTable 1). A supplementary search was performed on the reference lists of eligible studies. Two independent researchers (JA and AK) performed the screening, data extraction and assessed the risk of bias. Disagreements were resolved by discussion with the third researcher (PD).

Results of the searches were exported to Rayyan QCRI [26] for screening and removing of duplicates. Titles, abstracts and the full text of studies were screened for eligibility against our inclusion and exclusion criteria (Table 1). A spreadsheet was utilised to summarise descriptive data of selected studies, i.e. location of study, subpopulation of AAS users, the number of participants, number and gender of AAS users, and the number of AAS users who sought support from physicians.

Table 1: Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria

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Quantitative and qualitative studies with people using AAS for the purposes of image and performance enhancement.	Studies with patients using AAS for the treatment of medical conditions. Studies with prisoners, who have limited access to external health services. Studies with animal subjects and in vitro analysis of AAS' effects.
Studies informing the number of AAS users seeking physicians to receive information and prevent and treat AAS-related health conditions. Studies where the source of support or information can be understood as being a physician (e.g. doctor, medical doctor, health professional).	Studies not informing the number of AAS users in the sample who seek support from physicians. Studies where the contact with the physician cannot be understood as the participant's choice, such as involuntary admissions, post-mortem analysis and cohorts of patients selected for studies of specific health conditions.
Peer-reviewed studies. Studies published at any time. Studies published in any language, as long as it is possible to retrieve relevant data from the authors or articles.	'Grey literature' (non-peer-reviewed studies and reports). Case studies and interviews with a single AAS user.

Risk of bias assessment

The risk of bias was assessed in two stages. Initially, the quality and internal validity of studies were evaluated using the Mixed Methods Appraisal Tool (MMAT)[27]. The MMAT is comprised of five quality-assessment criteria: (i) Is the sampling strategy relevant to address the research question?; (ii) Is the sample representative of the target population?; (iii) Are the measurements appropriate?; (iv) Is the risk of nonresponse bias low?; and (v) Is the statistical analysis appropriate to answer the research question?. For the purposes of this review, studies with more than two negative or unknown responses to MMAT's assessment criteria were considered to have high risk of bias. In the second stage, studies were assessed for risk of publication bias. The risk of publication bias was assessed by visual inspection of asymetry in a funnel plot [28], Egger's test for asymmetry [29], and a rank correlation test [30].

Data synthesis

The data synthesis for meta-analysis was performed extracting the number of AAS users who reported seeking any kind of support from physicians in the selected studies. For the purpose of effect size calculations, the number of AAS users in each study was used as the number of participants of interest, and the number of those who informed seeking a physician was used as the number of cases in each study. When a study informed more than one number or percentage of AAS users seeking support from physicians, the higher value of male AAS users seeking medical support was considered, as only a few studies included female participants.

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A meta-analysis was performed to estimate the overall prevalence of AAS users seeking support from physicians. A random-effects model was chosen to better incorporate the dispersions of prevalence rates across studies and the different approaches to the research question [31]. Heterogeneity was measured using the I^2 index, which describes the percentage of variation of prevalence rates across a group of studies that is due to differences between studies (e.g. different sample sizes, populations or methods) [32,33]. A Baujat plot was produced to identify studies that could influence the overall result [34]. The secondary outcomes were measured by the prevalence rates of studies grouped by location and subpopulation of AAS users.

Univariate and multivariate meta-regressions were performed to measure the impact of study level moderators on the prevalence of AAS users seeking support from physicians. Based on the variables utilised by Sagoe et al.[35], four moderators were hypothesised to have an impact on the prevalence of AAS users seeking support from physicians (location of studies, subpopulation of AAS users, time of publication and study design). Additionally, two other moderators commonly used in prevalence studies [36,37] were included post hoc (sample size and risk of bias). The selection and coding of moderators followed consensus procedures. The time of publication was categorised as before and after the year 2000, as we hypothesised that the availability of the internet and on-line support communities of AAS users [16] – could impact the prevalence of AAS users seeking support from physicians. Risk of bias was categorised according to the number of negative or unknown responses to MMAT's assessment criteria [27]. For each moderator variable, the category with the highest number of studies was used as reference, and dummy variables were automatically generated. Statistically significant ($p < 0.05$) variables were entered into a multivariable model. The meta-analysis and meta-regression were conducted in R [38] using the metaphor package [39]. A full description of the codes and a dataset with the coded variables can be found in the supplementary material.

Patient and public involvement

It was not appropriate to involve patients or the public in the design, conduct, reporting or dissemination plans of this study.

Results

The searches identified 11,906 studies. After the removal of duplicates, 9,876 studies were screened by title and abstract. Among 285 full-text studies were assessed for eligibility, 31 were included in the review. A supplementary search on the reference list of included papers and previous reviews led to the inclusion of another five studies. A total of 36 studies were included in the review, as shown in the flowchart of Figure 1.

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Table 2 shows the summary characteristics of the studies included in this review. The studies were published between 1988 and 2021, with a total number of AAS users (nAAS) = 10,101, being 9,278 (91.85%) males, 179 (1.77%) females and 644 (6.38%) whose sex was not reported. Eight of the selected studies were located in the US (nAAS = 2,610; 25.84%), six in continental Europe (nAAS = 454; 4.49%), five in the UK (nAAS = 828; 8.20%), four in Australia (nAAS = 1,032; 10.22%), three in Brazil (nAAS = 260; 2.57%), five in Africa, Asia or the Middle East (nAAS = 1,110; 10.89%) and five studies were trans-regional (nAAS = 3,817; 37.79%). The selected studies presented a wide range of nAAS (median = 97.50; range = 4 to 2,385) and number of AAS users seeking support from physicians (nPhy; median = 31.50; range = 1 to 1,290). Regarding the subpopulation of AAS users, the most common were studies with nonspecific AAS users such as gym users and recreational athletes (21 studies, nAAS = 7,923; 78.44%), followed by studies with strength athletes such as bodybuilders, powerlifters or weightlifters (9 studies, nAAS = 801; 7.93%), studies with adolescents (4 studies, nAAS = 377; 3.73%) and studies with NSP clients (2 studies, nAAS = 1,000; 9.90%).

A large proportion of studies with adolescents (three out of four) and strength athletes (four out of nine) were located in the US, followed by a single study from different countries. The two studies with NSP clients were located either in the UK or Australia. All the five trans-regional studies were conducted with a nonspecific subpopulation of AAS users.

Table 2: Summary characteristics of selected studies

Authors, year	Location	Subpopulation	n	nAAS	nPhys n (%)
Yesalis et al., 1988 [40]	US	strength athletes	45	15 M	8 (53.33)
Johnson et al., 1989 [41]	US	adolescents	853	95 M	28 (29.47)
Kisling et al., 1989 [42]	Denmark	nonspecific	157	85 M	21 (24.71)
Lindstrom et al., 1990 [43]	Sweden	strength athletes	138	138 M	12 (8.70)
Terney et al., 1990 [44]	US	adolescents	2113	94 (67 M/27 F)	5 (5.32)
Tanner et al., 1995 [45]	US	adolescents	6930	184 (139 M/45 F)	33 (17.93)
Korkia et al., 1997 [46]	UK	nonspecific	1667	110 (97 M/13 F)	39 (35.45)
Bolding et al., 1999 [47]	UK	nonspecific	1004	81 M	25 (30.86)
Augé et al., 1999 [48]	US	strength athletes	17	17 (14 M/3 F)	8 (47.05)
Peters et al., 1999 [49]	Australia	nonspecific	100	100 (94 M/6 F)	42 (42.00)
Perry et al., 2005 [50]	US	strength athletes	207	207*	46 (22.22)
Parkinson et al., 2005 [51]	Trans-region	nonspecific	500	500 (494 M/6 F)	185 (37.00)
Pope et al., 2004 [7]	US	strength athletes	80	43 M	16 (37.21)
Striegel et al., 2006 [52]	Germany	nonspecific	621	84 (75 M/9 F)	47 (55.95)
Cohen et al., 2007 [53]	US	nonspecific	1955	1955 M	1290 (65.98)
Al-Falasi et al., 2008 [54]	UAE	nonspecific	154	34 M	4 (11.76)

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Larance et al., 2008 [55]	Australia	nonspecific	60	60 M	46 (76.66)
Posiadała et al., 2010 [56]	Poland	nonspecific	50	18 M	2 (11.11)
Gradidge et al., 2011 [57]	South Africa	adolescents	100	4 M	1 (25.00)
Ip et al., 2011[58]	Trans-region	nonspecific	1277	506 M	387 (76.48)
Santos et al., 2011 [59]	Brazil	strength athletes	123	41 M	4 (9.76)
Hope et al., 2013 [23]	UK	NSP clients	395	395 M	178 (45.06)
Raschka et al., 2013 [60]	Germany	nonspecific	484	79 (62 M/ 17 F)	30 (37.97)
Rowe et al., 2016 [61]	Australia	NSP clients	605	605 M	382 (63.14)
Westerman et al., 2016	Trans-region	nonspecific	231	231 M	153 (66.23)
Mooney et al., 2017 [62]	UK	nonspecific	377	26*	1 (3.85)
Zahnow et al., 2017 [19]	Trans-region	nonspecific	195	195*	68 (34.87)
Althobiti et al., 2018 [63]	Saudi Arabia	nonspecific	4860	476 M	181 (38.00)
Hill et al., 2019 [13]	UK	strength athletes	350	216*	91 (42.00)
Jacka et al., 2019 [64]	Australia	nonspecific	267	267 M	237 (88.76)
Macedo et al., 2019 [65]	Brazil	nonspecific	40	25 M	9 (36.00)
Pany et al., 2019 [66]	India	strength athletes	74	74 M	24 (32.43)
Pereira et al., 2019 [67]	Brazil	nonspecific	719	194 (149 M/45 F)	117 (60.31)
Uddin et al., 2019 [68]	Pakistan	nonspecific	841	512 M	9 (1.76)
Bonnecaze et al., 2020 [69]	Trans-region	nonspecific	2385	2385 M	1047 (43.90)
Jokipalo et al., 2021 [70]	Finland	strength athletes	50	50 (42 M/8 F)	15 (30.0)

AAS: androgenic-anabolic steroids. nAAS: number of AAS users in each study. nPhys: number of AAS users who informed seeking support from physicians. M = males. F = females. NSP: Needle and syringe exchange programme.
* Sex of AAS users not informed.

Risk of bias

According to the MMAT quality criteria, only seven (19.44%) studies did not present a risk of bias. The following number of studies had a negative or unknown response to MMAT assessment: Inappropriate or unclear sampling strategy (n = 0); sample representativeness low or unclear (n = 10; 27.78%); inappropriate or unclear measurements (n = 14; 38.89%); high or unclear risk of nonresponse bias (n = 19; 52.77%); and inappropriate or unclear statistical analysis (n = 2; 5.56%). The main cause of inappropriate or unclear measurements in some studies was the fact that the number of AAS seeking support from physicians was not clearly stated, requiring an extrapolation from the total number of participants. Ultimately, 12 studies (33.33%) were considered to have a high risk of bias (2 or more negative or unknown response to MMAT criteria). A full assessment of risk of bias can be found in the supplementary material (eTable 2). There was no evidence of publication bias, as indicated by visual inspection of the funnel plot (Figure 2), Egger's test for asymmetry (bias coefficient = 0.937, p = 0.349, 95% CI 0.49 to 0.45), and a rank correlation test (τ = -0.086, p = 0.4731).

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Meta-analysis of the overall prevalence of AAS users seeking support from physicians

The overall prevalence of AAS users seeking support from physicians obtained from 36 studies is 37.12% (95% CI 27.71 to 44.52). The smallest prevalence rate (1.76%; 95% CI 0.61 to 2.91; nAAS = 512) was observed in a study with gym users in Pakistan [68]. The highest prevalence rate (88.76%, 95% CI 77.46 to 100.06; nAAS = 267) was reported in a study with nonspecific AAS users in Australia [64]. Figure 3 shows a forest plot of prevalence rates, ordered by the effect sizes of all studies.

Meta-analysis of prevalence rates of studies grouped by location

When grouped by the location of studies, the highest prevalence of AAS users seeking support from physicians was seen among studies taking place in Australia (67.27%; 95% CI 47.29 to 87.25), followed by trans-regional studies (51.48%; 95% CI 35.26 to 67.71). The lowest prevalence of AAS users seeking support from physicians was seen in studies located in Africa, Asia or Middle East (21.02%; 95% CI 5.26 to 36.79). A forest plot of the prevalence rates of studies grouped by location is shown in Figure 4.

Meta-analysis of prevalence rates of studies grouped by subpopulation of AAS users

The highest prevalence of AAS users seeking support from physicians was seen among studies with NSP clients (54.13%; 95% CI 36.41 to 71.84), followed by studies with nonspecific AAS users (41.67%; 95% CI 31.23 to 52.12) and studies with strength athletes (27.83%; 95% CI 17.97 to 37.69). The lowest prevalence of AAS users seeking support from physicians was seen in studies with adolescents (17.27%; 95% CI 4.80 to 29.74). A forest plot of the prevalence rates of studies grouped by subpopulation of AAS users is shown in Figure 5.

Meta-regression exploring the variability in the prevalence of AAS users seeking support from physicians

The results of the meta-regression analyses are shown in Table 3. Univariable analyses showed that the prevalence of AAS users seeking support from physicians was significantly higher among studies located in Australia (β = 0.35, 95% CI 0.11 to 0.58, p = 0.005) and in studies utilising an online survey for data collection (β = 0.19, 95% CI 0.04 to 0.34, p = 0.014). These two variables were therefore eligible for inclusion in the multivariable regression analysis. An overall multivariable model with the variables was statistically significant (χ^2 (8) = 20.25, p = 0.009). Only studies located in Australia (β = 0.33, 95% CI 0.09 to 0.57, p = 0.007) remained a significant predictor of the prevalence of AAS users seeking support from physicians, suggesting that the prevalence of this behaviour is higher in Australia compared with other studies' locations.

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Table 3: Univariable and multivariable predictors of the prevalence of AAS users seeking support from physicians (N = 36)

		Univariable		Multivariable	
Variable	**N**	**Regression coefficient (95% CI)**	**SE**	**Regression coefficient (95% CI)**	**SE**
Location					
US	8	1	-	1	-
UK	5	-0.01 (-0.23 to 0.21)	0.11	-0.02 (-0.24 to 0.21)	0.12
Continental Europe	6	-0.05 (-0.26 to 0.16)	0.11	-0.02 (-0.24 to 0.19)	0.11
Australia	4	0.35 (0.11 to 0.58)**	0.12	0.33 (0.09 to 0.57)**	0.12
Brazil	3	0.02 (-0.24 to 0.29)	0.14	0.07 (-0.20 to 0.34)	0.14
Africa, Asia or Middle East	5	-0.12 (-0.34 to 0.11)	0.12	-0.07 (-0.30 to 0.17)	0.12
Trans-regional	5	0.19 (-0.03 to 0.40)	0.11	0.11 (-0.17 to 0.38)	0.14
Subpopulation					
Nonspecific AAS users	21	1	-		
Adolescents	4	-0.23 (-0.47 to 0.01)	0.12		
Strength athletes	9	-0.12 (-0.29 to 0.05)	0.09		
NSP clients	2	0.13 (-0.18 to 0.43)	0.15		
Sample size					
Small (<100)	18	1	-		
Medium (>100, < 1000)	16	0.12 (-0.03 to 0.27)	0.76		
Large (>1000)	2	0.25 (-0.06 to 0.55)	0.16		
Time of publication					
2005-2021	26	1	-		
1988-1999	10	-0.13 (-0.29 to 0.03)	0.08		
Study design					
Questionnaire	33	1	-	1	-
Interview	3	0.24 (-0.04 to 0.52)	0.14	0.14 (-0.14 to 0.42)	0.14
Online survey	9	0.19 (0.04 to 0.34)*	0.08	0.13 (-0.07 to 0.33)	0.10
Risk of bias					
Low (< 2)	24	1	-		
High (>= 2)	12	-0.41 (-0.20 to 0.12)	0.08		

*p < 0.05. **p < 0.01. N: number of studies. SE: standard error.

Discussion

We conducted a systematic review and meta-analysis and pooled data from 36 studies to estimate that the overall prevalence of AAS users seeking support from physicians is 37.12%. Higher prevalence rates of AAS users seeking support from physicians were seen among studies located in Australia (67.27%) and studies with NSP clients (54.13%). Lower prevalence rates were seen among studies located in Africa, Asia or the Middle East (21.02%) and studies with adolescents (17.27%). One of the general factors possibly influencing the prevalence of AAS users seeking support from physicians is the fact that men, who represent the majority of AAS

users [35], are generally less likely to seek medical support [71] – a tendency corroborated by the only two studies comparing the prevalence of male and female AAS users seeking support from physicians [19,48]. Other potential factors include the legal status of AAS use [9], the engagement of AAS users with health services [8], AAS users' perceptions of the service provided by physicians [7,24,72], and the stigma experienced by AAS users [13,15,73].

Our findings suggest that the legal status of AAS use is not always associated with AAS users' engagement with physicians' support. For example, studies located in Australia – where the possession of AAS is illegal [74] – showed the highest pooled prevalence of AAS users seeking support from physicians (51.13%). Besides, the estimated prevalence of AAS users seeking support from physicians in the US (32.91%) – where the possession of AAS is considered a federal crime [75]– was similar to the estimated prevalence in the UK (31.43%) and Brazil (35.23%), where the use and possession of AAS are not illegal [76,77]. Likewise, a low prevalence of AAS users seeking support from physicians was seen in studies from countries in Africa, Asia or the Middle East (21.02%), where anecdotal reports suggest loose enforcement of the prohibition of AAS use [78–81] or, in the case of India, where there are no laws regulating the use and commerce of AAS [82]. Two studies from German showed prevalence rates of AAS users seeking support from physicians of 37.97% [60] and 55.95% [52], while a single study from Poland reported a prevalence of 11.11% [56], despite the use of AAS not being illegal in both countries – unless, in the case of Germany, if AAS are used for the purpose of doping in sport competitions [83,84]. Nevertheless, the legal status of AAS use could have influenced the small prevalence of AAS users seeking support from physicians (8.70%) seen in a single study from Denmark, where the use of AAS is not only illegal but where gym users can be subject to urinalysis to screen for the use of AAS [85]. Therefore, it is reasonable to assume that the existence of laws regulating the use of AAS might have an impact on the engagement of some AAS users with physicians' support, but their relevance is possibly influenced by other variables, such as the actual enforcement of regulations and cultural factors involving individuals' help-seeking behaviours.

Other factors might have influenced the prevalence of AAS users seeking support from physicians across the locations analysed by this study. In Australia, for example, there seems to be an active effort to educate physicians about the management of the non-prescribed use of AAS [86] which could reflect a willingness to discuss AAS use with the medical community. Although data from Australia was based on a small number of studies, results of the multivariate meta-regression analysis showed that the prevalence rate of studies published in Australia was the only variable with a statistically significant impact on the overall prevalence of AAS users seeking support from physicians estimated by this study.

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The engagement of AAS users with health services could have influenced the comparatively high prevalence (54.13%) of AAS users seeking support from physicians seen among clients of the NSP – namely those assessing NSP services that provide information about injection practices and adverse effects of AAS [23,61]. Among NSP clients, those seeking support from physicians are more likely to have diagnostic screening for health conditions potentially associated with the use of AAS – despite some NSP clients considering physicians a less reliable source of information about AAS than NSP workers [61]. Despite the growing numbers of AAS users seeking the NSP, only a minority of primary NSP units offer specialised advice about AAS [87]. Besides, the majority of people using injectable drugs access the NSP via retail pharmacies, where the services are frequently limited to the exchange of injectable material [88].

The lower pooled prevalence of AAS users seeking support from physicians was seen among adolescents (17.27%). Although low rates of engagement with health services are seen among adolescents in general [89], some factors could have influenced an even lower prevalence of adolescent AAS users seeking support from physicians. Firstly, the prevalence of some health conditions potentially associated with the use of AAS – such as cardiovascular disease [90] – is lower among adolescents. Secondly, it is possible that the illegality of AAS use has deterred adolescents from seeking physicians more than other subpopulations of AAS users. As observed by Terney et al. [44], physicians were allowed to prescribe AAS for enhancement purposes in the US until 1988 and were frequently reported as a source of AAS to adolescents [44,45,91]. As near all of the selected studies with adolescents were located in the US, it is possible that the criminalization of AAS use has driven adolescent AAS users further away from seeking the support of physicians.

Among strength athletes who use AAS, we estimated a prevalence of seeking support from physicians of 27.83%. Strength athletes who use AAS have been described as having a perception that the use of AAS can be safely managed, namely with the support of other AAS users who share their objectives, training routines and lifestyles [20,22]. The self-research and trial-and-error experiences with AAS, combined with aesthetical and performance goals frequently considered exaggerated by people outside of the community of strength athletes [92] probably contribute to the perception that physicians are less knowledgeable about AAS than some members of this subpopulation. Furthermore, it is possible that some strength athletes are even more subject to stigma than other AAS users, due to their unusually muscular physiques or to a prejudice towards bodybuilding and other strength-related disciplines [93].

Regarding the kind of support sought by AAS users from physicians among the 36 selected studies, 16 (44.44%) reported that AAS users sought physicians as a source of information about the use and adverse effects of AAS. The remaining studies described AAS users' contact with

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physicians in many different ways, such as having close contact with physicians [52], seeking a doctor for interpretation of health checks [13], and disclosing the use of AAS to a physician [7,69]. We considered that further exploration of the types of support sought by AAS users would lie beyond the scope of this study, as they have been recently investigated by other reviews [17,94,95].

Limitations of this study

The prevalence of AAS users seeking support from physicians varied widely across the selected studies, and this variation was only minimally explained by the meta-regression and the comparisons between subgroups of studies. Furthermore, the selected studies investigated and described the help-seeking behaviours of AAS users in many different ways that were synthesised as a single variable for the purpose of comparison. The pooled prevalence in different locations was based on a limited sample of highly heterogenic studies. For instance, the estimated prevalence of AAS users seeking support from physicians from Australia was based on only four studies – three with a nonspecific population of AAS users and one with NSP clients – comprising 10.22% of the total number of AAS users from selected studies. This review did not distinguish data from male and female AAS users, as the sex of participants or differences in help-seeking behaviour between male and female participants were not reported by the majority of studies. The review only included studies from a few countries and, among those, many contributed with a single study unable to represent the local population of AAS users. As discussed in this review, the engagement of AAS users with physicians can be influenced by several factors, including attitudes that can vary widely between locations and subpopulations of AAS users. These limitations can compromise the generalisation of our results, and further studies are necessary to better understand the help-seeking behaviours of AAS users, namely among understudied locations and subpopulations of AAS users.

Implications for practice and policy

Available data suggest that factors such as the criminalisation of AAS use, the scarcity of physicians' knowledge about AAS and stigma against AAS users are barriers to the access of some AAS users to physicians. The results of this review can indicate locations and subpopulations of AAS users with higher engagement with physicians, so successful strategies can be replicated. Likewise, our results indicate the existence of under-studied and possibly under-supported populations of AAS users.

Conclusion

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This meta-analysis was the first to systematically investigate the prevalence of AAS users seeking support from physicians. Our findings suggest that the overall prevalence of AAS users seeking support from physicians is 37.12%, with considerable variation across locations and among subpopulations of AAS users. This study highlights the importance of understanding the help-seeking behaviours of AAS users and improving their access to physicians.

Acknowledgments

We thank the participants and authors of the selected studies, the developers of R and the mefator package, and the reviewers of this journal for their valuable comments.

Contributors

JA planned the study and conducted the meta-analysis. JA and AK conducted the selection of studies and the assessment of the risk of bias. AK and PD contributed to the interpretation of data. JA, AK and PD reviewed and approved the final draft.

Funding

The first author was funded by King's College London International Scholarship (RDPL3PRPCA - 2019-20).

Competing interests

None.

Ethics approval

As a review of the literature, this study was exempt from ethical approval by King's College London Research Ethics Office.

Provenance and peer reviewed

Not commissioned; externally peer reviewed.

Data availability statement

No additional data are available.

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Figure 1: Flowchart of the inclusion of studies in the review

Figure 2: Funnel plot of studies included in the analysis with pseudo 95% confidence intervals

Figure 3: Forest plot of the pooled prevalence of AAS users seeking support from physicians

Figure 4: Forest plot of prevalence rates of studies, grouped by location

Figure 5: Forest plot of prevalence rates of studies, grouped by subpopulation of AAS users

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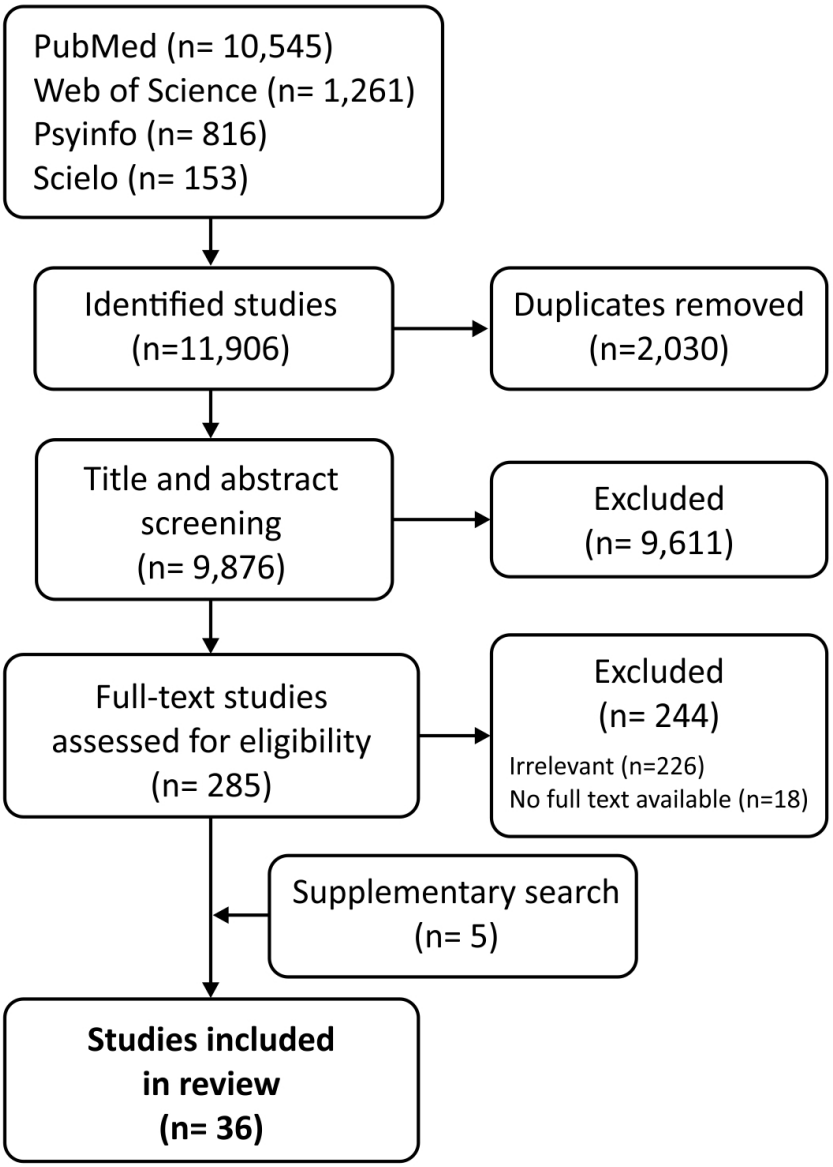


Figure 1: Flowchart of the inclusion of studies in the review

92x128mm (300 x 300 DPI)

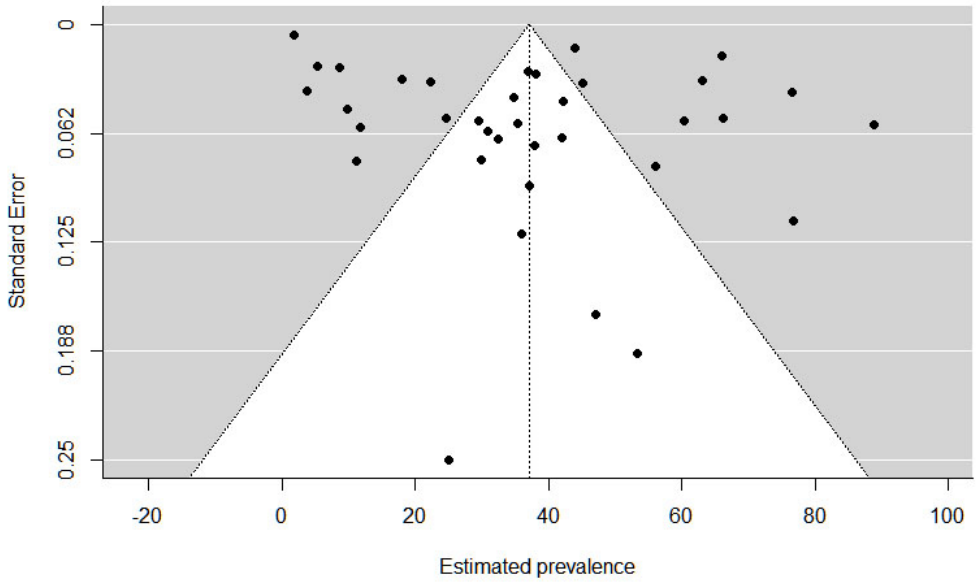


Figure 2: Funnel plot of studies included in the analysis with pseudo 95% confidence intervals

162x98mm (120 x 120 DPI)

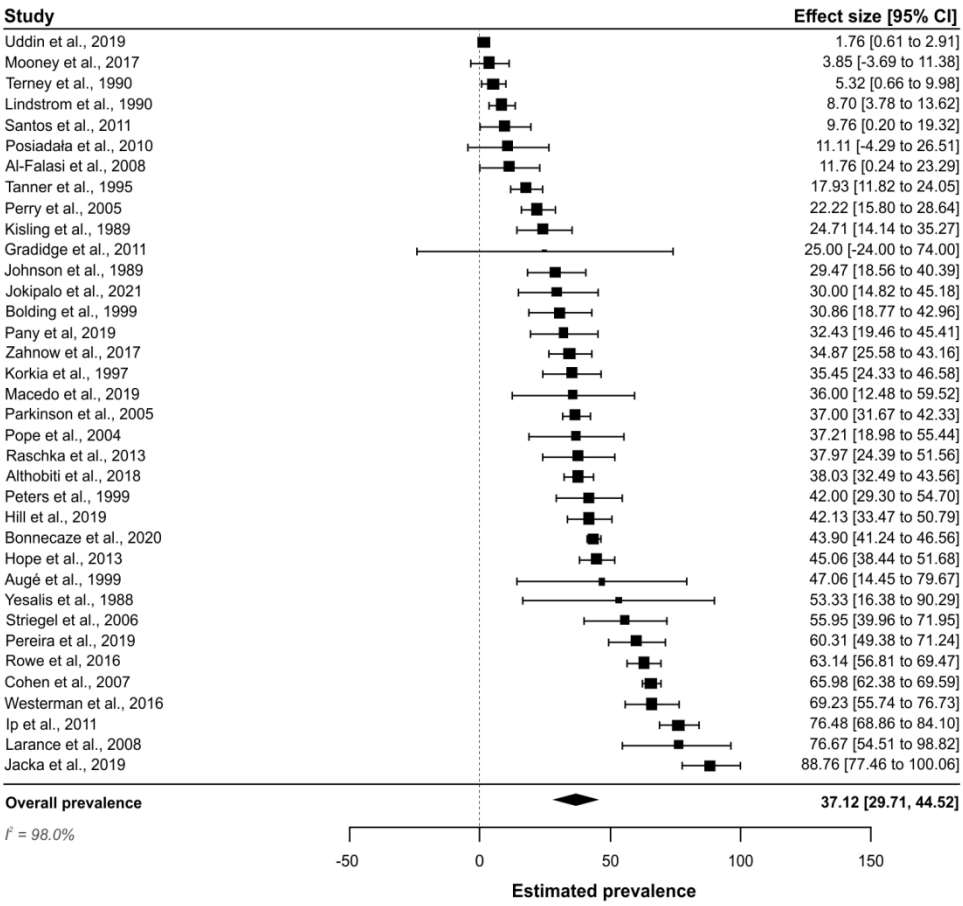


Figure 3: Forest plot of the pooled prevalence of AAS users seeking support from physicians

138x126mm (300 x 300 DPI)

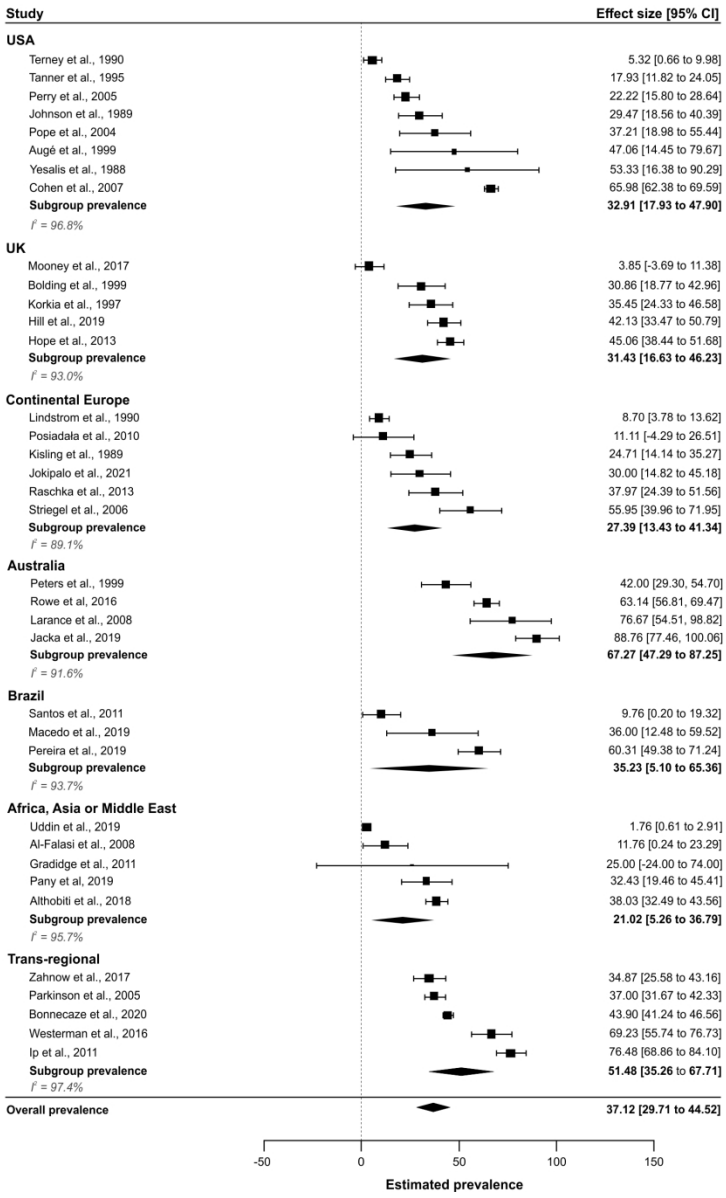


Figure 4: Forest plot of prevalence rates of studies, grouped by location

138x221mm (300 x 300 DPI)

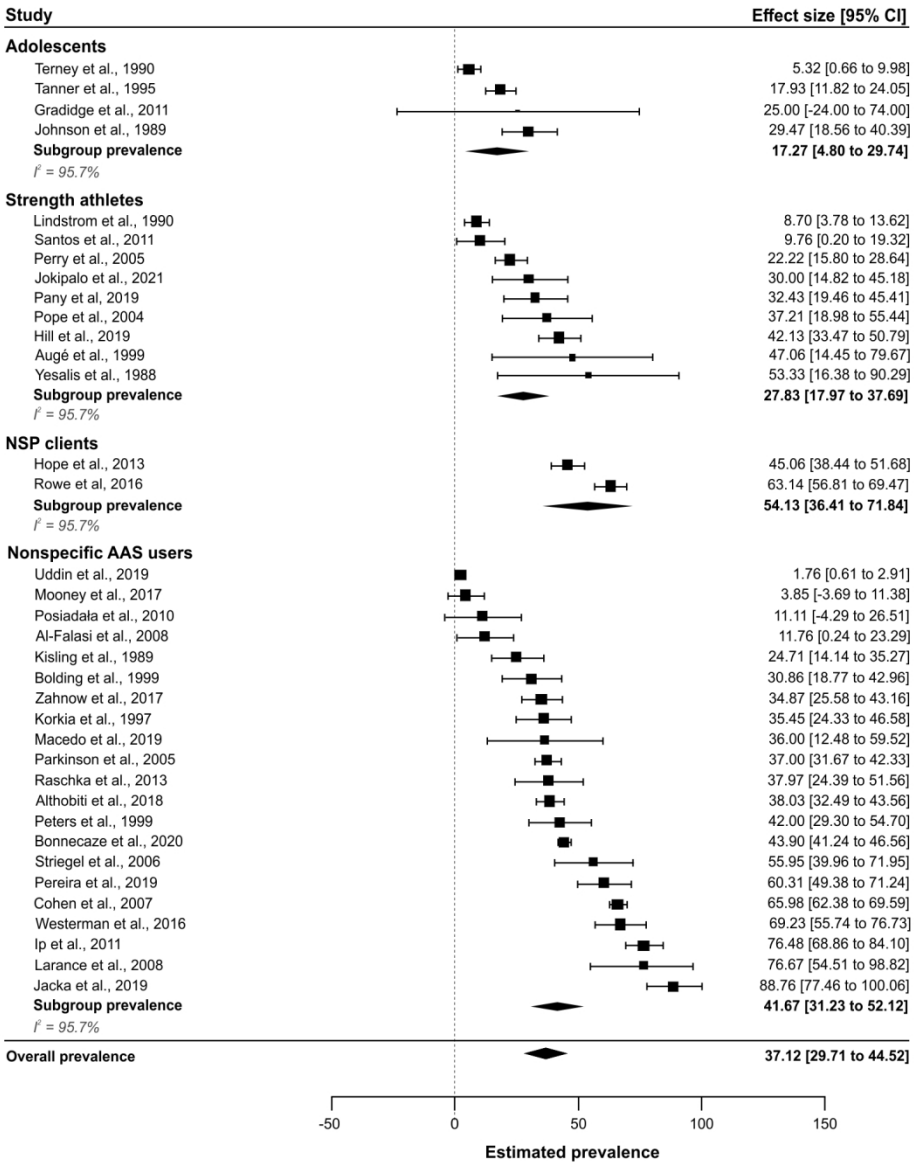


Figure 5: Forest plot of prevalence rates of studies, grouped by subpopulation of AAS users

138x174mm (300 x 300 DPI)

Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

Amaral JMX, Kimergård A, Deluca P

SUPPLEMENTARY MATERIAL

eTable 1: Search algorithms

PUBMED (results: 10,545)
((agents, anabolic[MeSH Terms]) OR (doping in sport[MeSH Terms]) OR (drugs, performance enhancing[MeSH Terms])) OR ((anabolic steroid*[Title/Abstract]) OR (anabolic androgenic[Title/Abstract]) OR (anabolic agent*[Title/Abstract]) OR (anabolic drug*[Title/Abstract]) OR (androgen* abuse[Title/Abstract]) OR (performance enhanc*[Title/Abstract]) OR (image enhanc*[Title/Abstract]) OR (synthetic androgen*[Title/Abstract]) OR (synthetic testosterone[Title/Abstract]) OR (testosterone ester*[Title/Abstract]) OR (designer steroid*[Title/Abstract]) OR (doping[Title/Abstract]) OR ("non prescript*" steroid*[Title/Abstract]) OR ("non medic*" steroid*[Title/Abstract]) OR ("non medic*" androgen*[Title/Abstract]) OR ("non prescript*" androgen*[Title/Abstract])) NOT ((review[Title/Abstract]) OR (animal*[Title/Abstract]) OR (bovine[Title/Abstract]) OR (in vitro[Title/Abstract]) OR (mice*[Title/Abstract]) OR (pig[Title/Abstract]) OR (pigs[Title/Abstract]) OR (rat[Title/Abstract]) OR (rats[Title/Abstract]) OR (mouse[Title/Abstract]) OR (spectrometry[Title/Abstract]) OR (nano*[Title/Abstract]) OR (conductor*[Title/Abstract]) OR (semiconductor[Title/Abstract]) OR (electro*[Title/Abstract]) OR (optic*[Title/Abstract]) OR (hydrogen[Title/Abstract]) OR (superconductor*[Title/Abstract]) OR (catalyst*[Title/Abstract]) OR (thermoelectric[Title/Abstract]) OR (ions[Title/Abstract])) AND (humans[Filter])
Web of Science (results: 1,261)
((TS=("doping in sport" OR "performance enhancing drug*"OR "anabolic steroid*" OR "anabolic androgenic*" OR "anabolic agent*" OR "anabolic drug*")) NOT (TS=(review OR animal* OR bovine OR “in vitro” OR mice* OR pig OR pigs OR rat OR rats OR mouse OR spectrometry OR nano* OR conductor* OR semiconductor OR electro* OR optic* OR hydrogen OR superconductor* OR catalyst* OR thermoelectric OR ions))) Filters: Articles, Sport Sciences, Medicine (all), Psychology (all), Toxicology
APA PsycNet (results: 816) A
((anabolic or doping or performance enhancing) not (review or animal or bovine or vitro or mice* or pig* or rat* or mouse or spectrometry or nano* or conductor* or semiconductor or electro* or optic* or hydrogen or superconductor* or catalyst* or thermoelectric or ions)).ti - limit to human
Scielo (results: 153)
(ti:((ti:((ti:(anabolic)) OR (ti:(doping)) OR (ti:(performance enhancing)))) AND NOT (ti:(review)))) AND NOT (ti:((ti:(animal)) AND NOT (ti:(bovine)) AND NOT (ti:(vitro)) AND NOT (ti:(mice*)) AND NOT (ti:(pig*)) AND NOT (ti:(rat*)) AND NOT (ti:(mouse)) AND NOT (ti:(spectrometry)) AND NOT (ti:(nano*)) AND NOT (ti:(conductor*)) AND NOT (ti:(semiconductor)) AND NOT (ti:(electro*)) AND NOT (ti:(optic*)) AND NOT (ti:(hydrogen)) AND NOT (ti:(superconductor*)) AND NOT (ti:(catalyst*)) AND NOT (ti:(thermoelectric)) AND NOT (ti:(ions))))

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eTable 2: Risk of bias

	Risk of bias: MMAT criteria*					
Authors, year	**1**	**2**	**3**	**4**	**5**	**Observations**
Yesalis et al., 1988	Y	N	?	Y	Y	Small sample of AAS users. Source of information described as 'Physician/pharmacist/medical literature'.
Johnson et al., 1989	Y	Y	Y	Y	Y	n/a
Kisling et al., 1989	Y	Y	?	Y	Y	Nonresponse rate not informed.
Lindstrom et al., 1990	Y	Y	Y	Y	N	The number of AAS users seeking a physician must be inferred from the text.
Terney et al., 1990	Y	Y	N	Y	Y	The number of AAS users seeking support from physicians was extrapolated from the total number of participants.
Peters et al., 1993	Y	Y	Y	?	Y	Nonresponse rate not informed.
Tanner et al., 1995	Y	Y	N	Y	Y	The number of AAS users seeking support from physicians was extrapolated from the total number of participants.
Korkia et al., 1997	Y	Y	Y	N	Y	Nonresponse rate = 41%.
Bolding et al., 1999	Y	Y	?	Y	Y	Support described as 'regular health checks to screen for steroid-related health problems'.
Auge et al., 1999	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Perry et al., 2005	Y	Y	N	Y	Y	The number of AAS users seeking a physician must be inferred from a graph and was extrapolated from the total number of participants.
Parkinson et al., 2005	Y	Y	Y	?	Y	Nonresponse rate not informed.
Pope et al., 2004	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Striegel et al., 2006	Y	Y	Y	?	Y	Nonresponse rate not informed.
Cohen et al., 2007	Y	Y	?	?	Y	Nonresponse rate not informed. Support described as 'willing to seek medical supervision'.
Al-Falasi et al., 2008	Y	N	N	N	Y	Nonresponse rate = 56%. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Larance et al., 2008	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Posiadala et al., 2010	Y	N	Y	N	?	Small sample of AAS users. Nonresponse rate = 28.57%.
Gradidge et al., 2011	Y	Y	N	Y	Y	Small sample of AAS users. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Ip et al., 2011	Y	Y	Y	Y	Y	n/a
Santos et al., 2011	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Hope et al., 2013	Y	Y	Y	Y	Y	n/a
Raschka et al., 2013	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Mooney et al., 2016	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Rowe et al., 2016	Y	Y	Y	Y	Y	n/a
Westerman et al., 2016	Y	Y	Y	Y	Y	n/a
Zahnow et al., 2017	Y	Y	Y	?	Y	Nonresponse rate not informed.
Althobiti et al., 2018	Y	Y	N	?	Y	Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Hill et al., 2019	Y	Y	Y	Y	Y	n/a
Jacka et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.

Macedo et al., 2019	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Pany et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.
Pereira et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.
Uddin et al., 2019	Y	Y	N	Y	Y	The number of AAS users seeking a physician must be inferred from the text.
Bonnecaze et al., 2020	Y	Y	Y	Y	Y	n/a
Jokipalo et al., 2021	Y	Y	Y	Y	Y	n/a

* MMAT's quality-assessment criteria:
1 - Is the sampling strategy relevant to address the research question?
2 - Is the sample representative of the target population?
3 - Are the measurements appropriate?
4 - Is the risk of nonresponse bias low?
5 - 'Is the statistical analysis appropriate to answer the research question?'
Y = yes. N = no. ? = not clear.

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R codes for the meta-analysis, forest plots and meta-regression

The table below contains the dataset of the file 'metaAAS.csv'. The column 'ti' contains the number of androgenic-anabolic steroids (AAS) users in each study, and the column 'xi', the number of AAS users who sought support from physicians. The subsequent columns contain the coded variables utilised in the forest plots and meta-regression.

study id	studies	ti	xi	locat	pop	size	time	bias	design
1	Yesalis et al.,1988	15	8	0	1	0	0	0	0
2	Johnson et al.,1989	95	28	0	0	0	0	1	0
3	Kisling et al.,1989	85	21	2	3	0	0	1	0
4	Lindstrom et al.,1990	138	12	2	1	1	0	1	0
5	Terney et al.,1990	94	5	0	0	0	0	1	0
6	Peters et al.,1993	100	42	3	3	0	0	1	0
7	Tanner et al.,1995	184	33	0	0	1	0	1	0
8	Korkia et al.,1997	110	39	1	3	1	0	1	0
9	Bolding et al.,1999	81	25	1	3	0	0	1	0
10	Auge et al.,1999	17	8	0	1	0	0	0	1
11	Perry et al.,2005	207	46	0	1	1	1	1	0
12	Parkinson et al.,2005	500	185	6	3	1	1	1	2
13	Pope et al.,2004	43	16	0	1	0	1	0	1
14	Striegel et al.,2006	84	47	2	3	0	1	1	0
15	Cohen et al.,2007	1955	1290	0	3	2	1	0	2
16	Al-Falasi et al.,2008	34	4	5	3	0	1	0	0
17	Larance et al.,2008	60	46	3	3	0	1	0	1
18	Posiadala et al.,2010	18	2	2	3	0	1	0	0
19	Gradidge et al.,2011	4	1	5	0	0	1	1	0
20	Ip et al.,2011	506	387	6	3	1	1	1	2
21	Santos et al.,2011	41	4	4	1	0	1	0	0
22	Hope et al.,2013	395	178	1	2	1	1	1	0
23	Raschka et al.,2013	79	30	2	3	0	1	0	0
24	Mooney et al.,2016	26	1	1	3	0	1	0	2
25	Rowe et al.,2016	605	382	3	2	1	1	1	0
26	Westerman et al.,2016	231	153	6	3	1	1	1	2
27	Zahnow et al.,2017	195	68	6	3	1	1	1	2
28	Althobiti et al.,2018	476	181	5	3	1	1	0	0
29	Hill et al.,2019	216	91	1	1	1	1	1	2
30	Jacka et al.,2019	267	237	3	3	1	1	1	2
31	Macedo et al.,2019	25	9	4	3	0	1	0	0
32	Pany et al.,2019	74	24	5	1	0	1	1	0
33	Pereira et al.,2019	194	117	4	3	1	1	1	0
34	Uddin et al.,2019	512	9	5	3	1	1	1	0
35	Bonnecaze et al.,2020	2385	1047	6	3	2	1	1	2
36	Jokipalo et al.,2021	50	15	2	1	0	1	1	2

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R codes

```
## Loading packages previously installed
library("robumeta")
library("metafor")
library("dplyr")
library("forestplot")
library("checkmate")
library("nloptr")
library("meta")
library("CompQuadForm")

## Meta-analysis of all studies
metaAAS <- read.csv("metaAAS.csv")
View(metaAAS)
metaAAS <- escalc(measure="IR", xi=xi, ti=ti, data=metaAAS, slab=paste (studies))
res <- rma(yi, vi, data=metaAAS)
res
confint(res)

## Baujat plot
b_res <- rma(xi, ti, data=metaAAS, slab=study.id)
baujat(b_res)
inf <- influence(res)
print(inf)
plot(inf)


### funnel plot
funnel(res,atransf=mytransf,xlab = "Estimated prevalence")
regtest(res)
ranktest(res)
warnings()


### Cross-tabulation of the number of studies – location vs subpopulation
addmargins(table(metaAAS$locat, metaAAS$pop))


### Setup functions for the forest plot
mlabfun <- function(text, res) {
 list(bquote(paste(.(text),
                   " (Q = ", .(formatC(res$QE, digits=2, format="f")),
                   ", df = ", .(res$k - res$p),
                   ", p ", .(metafor:::.pval(res$QEp, digits=2, showeq=TRUE, sep=" ")), "; ",
                   I^2, " = ", .(formatC(res$I2, digits=1, format="f")), "%, ",
                   tau^2, " = ", .(formatC(res$tau2, digits=2, format="f")), ")")))}
mytransf<-function(x)
 (x)*100


## Forest plot of all studies
forest (res, transf=mytransf, cex=.7,header=c("Authors and Year", "Effect size [95%
CI]"),xlab="Prevalence rate",
      mlab=mlabfun("RE Model for All Studies", res), order="obs", refline=NA)
```

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```
## Forest plot - subgroup LOCATION
forest (res, transf=mytransf, cex=.6, refline=NA,top=0,
      order=metaAAS$locat,
      rows=c(65:58,53:49,44:39,34:31,26:24,19:15,10:6),
      mlab=mlabfun("RE Model for All Studies", res),
      header=c("Authors and year", "Effect size [95% CI]"),
      xlab="Estimated prevalence")


### switch to bold italic font
par(font=4)

### add text for the subgroups
text(-130, c(54,45,35,27,20,11), pos=4,
   c("UK","Continental Europe","Australia","Brazil",
     "Africa,Asia and Middle East","Trans-regional"))

### fit random-effects model in the seven subgroups
res.0 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="0"), data=metaAAS)
res.1 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="1"), data=metaAAS)
res.2 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="2"), data=metaAAS)
res.3 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="3"), data=metaAAS)
res.4 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="4"), data=metaAAS)
res.5 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="5"), data=metaAAS)
res.6 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="6"), data=metaAAS)

### add summary polygons for the seven subgroups
addpoly(cex=.5,res.0, row=56.5, mlab=mlabfun("RE Model for Subgroup", res.0),transf=mytransf)
addpoly(cex=.5,res.1, row=47.5, mlab=mlabfun("RE Model for Subgroup", res.1),transf=mytransf)
addpoly(cex=.5,res.2, row=37.5, mlab=mlabfun("RE Model for Subgroup", res.2),transf=mytransf)
addpoly(cex=.5,res.3, row=29.5, mlab=mlabfun("RE Model for Subgroup", res.3),transf=mytransf)
addpoly(cex=.5,res.4, row=22.5, mlab=mlabfun("RE Model for Subgroup", res.4),transf=mytransf)
addpoly(cex=.5,res.5, row=13.5, mlab=mlabfun("RE Model for Subgroup", res.5),transf=mytransf)
addpoly(cex=.5,res.6, row=4.5, mlab=mlabfun("RE Model for Subgroup", res.6),transf=mytransf)


### fit meta-regression model to test for subgroup differences
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(locat), data=metaAAS)
resMeta
confint(resMeta)

### add text for the test of subgroup differences
text(-130, -3, pos=4, cex=0.6,
   bquote(paste("Test of moderator (location): ",Q[M], " = ",
            .(formatC(resMeta$QM, digits=2, format="f")),
            ", df = ", .(resMeta$p - 1),
                        ", p = ", .(formatC(resMeta$QMp, digits=2, format="f")))))
```

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```
##Forest plot -subgroup SUBPOPULATION
forest (res, transf=mytransf, cex=.6, refline=NA,top=0,
      order=metaAAS$pop,
      rows=c(53:50,45:37,32:31,26:6),
      mlab=mlabfun("RE Model for All Studies", res),
      header=c("Authors and year", "Effect size [95% CI]"),
      xlab="Estimated prevalence")

### fit random-effects model in the four subgroups
res.0 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="0"), data=metaAAS)
res.1 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="1"), data=metaAAS)
res.2 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="2"), data=metaAAS)
res.3 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="3"), data=metaAAS)

### add summary polygons for the four subgroups
addpoly(cex=.5,res.0, row=48, mlab=mlabfun("RE Model for Subgroup", res.0),transf=mytransf)
addpoly(cex=.5,res.1, row=35, mlab=mlabfun("RE Model for Subgroup", res.1),transf=mytransf)
addpoly(cex=.5,res.2, row=29, mlab=mlabfun("RE Model for Subgroup", res.2),transf=mytransf)
addpoly(cex=.5,res.3, row=4, mlab=mlabfun("RE Model for Subgroup", res.3),transf=mytransf)


### fit meta-regression model to test for subgroup differences
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(pop), data=metaAAS)
resMeta
confint(resMeta)

### add text for the test of subgroup differences
text(-130, -3, pos=4, cex=0.6,
   bquote(paste("Test of moderator (sub-population): ",Q[M], " = ",
            .(formatC(resMeta$QM, digits=2, format="f")),
            ", df = ", .(resMeta$p - 1),
            ", p = ", .(formatC(resMeta$QMp, digits=2, format="f")))))


### Univariate meta-regression model. Moderator: LOCATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(locat), ref="0"),  data=metaAAS)
resMeta
confint(resMeta,transf=mytransf)


### Univariate meta-regression model. Moderator: SUBPOPULATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(pop), ref="3"), data=metaAAS)
resMeta
confint(resMeta)

### Univariate meta-regression model. Moderator: SAMPLE SIZE
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(size), ref="0"), data=metaAAS)
resMeta
confint(resMeta)

### Univariate meta-regression model. Moderator: TIME OF PUBLICATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(time), ref="1"), data=metaAAS)
resMeta
confint(resMeta)
```

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Univariate meta-regression model. Moderator: STUDY DESIGN

```
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(design), ref="1"),
data=metaAAS)
resMeta
confint(resMeta)
```

Univariate meta-regression model. Moderator: RISK OF BIAS

```
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(bias), ref="1"), data=metaAAS)
resMeta
confint(resMeta)
```

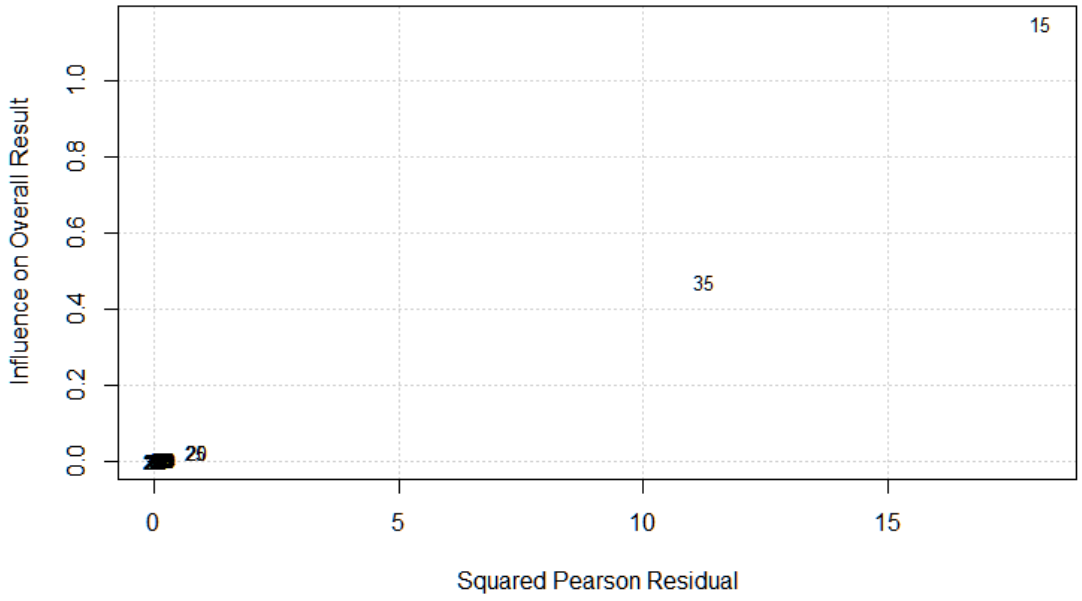
Multivariate meta-regression model. Moderators: LOCATION + STUDY DESIGN

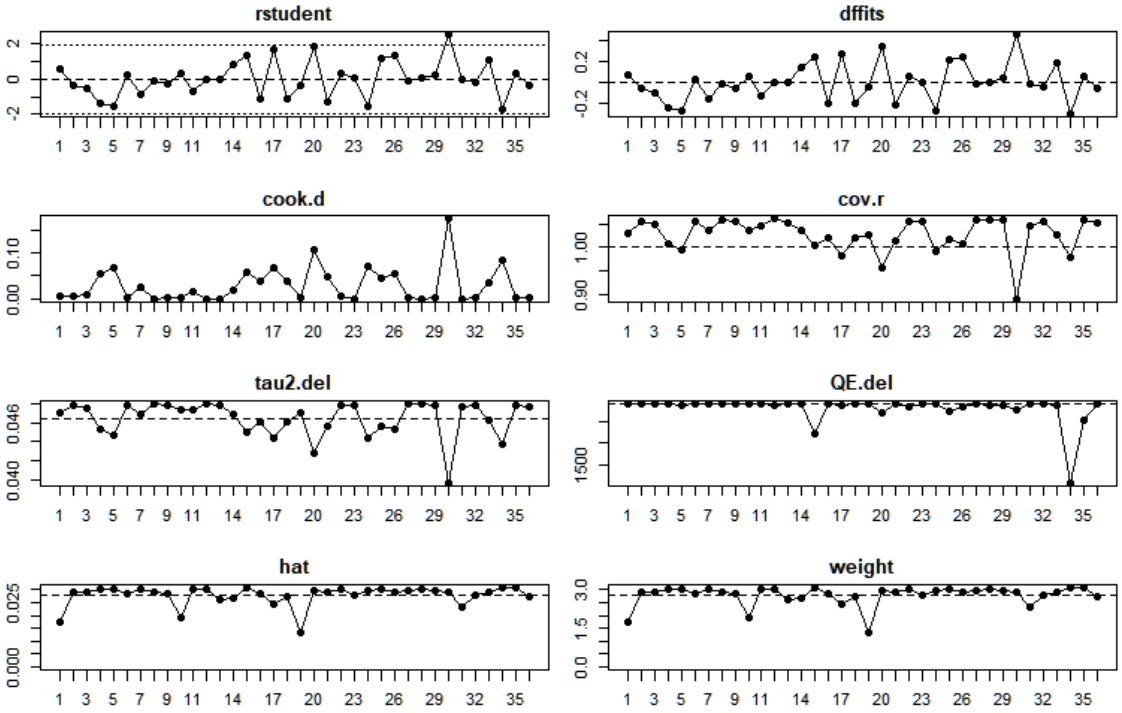
```
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(locat)+relevel(factor(design),
ref="0"), data=metaAAS)
resMeta
confint(resMeta)
```

Results of the Baujat plot

	rstudent	dffits	cook.d	cov.r	tau2.del	QE.del	hat	weight	dfbs	inf
Yesalis et al.,1988	0.5691	0.0751	0.0057	1.0296	0.0470	2901.5926	0.0174	1.7416	0.0749	
Johnson et al.,1989	-0.3444	-0.0615	0.0039	1.0562	0.0477	2901.1899	0.0288	2.8822	-0.0615	
Kisling et al.,1989	-0.5615	-0.0986	0.0099	1.0507	0.0475	2903.7126	0.0289	2.8937	-0.0986	
Lindstrom et al.,1990	-1.3464	-0.2365	0.0546	1.0080	0.0453	2888.7020	0.0303	3.0336	-0.2363	
Terney et al.,1990	-1.5188	-0.2662	0.0681	0.9939	0.0446	2872.4020	0.0304	3.0377	-0.2658	
Peters et al.,1993	0.2168	0.0346	0.0012	1.0567	0.0478	2891.9587	0.0282	2.8198	0.0346	
Tanner et al.,1995	-0.8914	-0.1576	0.0250	1.0380	0.0468	2904.9166	0.0301	3.0115	-0.1576	
Korkia et al.,1997	-0.0751	-0.0154	0.0002	1.0593	0.0479	2896.1842	0.0288	2.8752	-0.0154	
Bolding et al.,1999	-0.2797	-0.0501	0.0026	1.0566	0.0478	2901.0530	0.0284	2.8416	-0.0501	
Auge et al.,1999	0.3664	0.0502	0.0025	1.0367	0.0473	2902.0604	0.0193	1.9265	0.0501	
Perry et al.,2005	-0.6882	-0.1224	0.0152	1.0477	0.0473	2903.7785	0.0301	3.0053	-0.1225	
Parkinson et al.,2005	-0.0059	-0.0036	0.0000	1.0626	0.0480	2858.4148	0.0303	3.0265	-0.0036	
Pope et al.,2004	0.0035	-0.0017	0.0000	1.0536	0.0478	2900.9937	0.0259	2.5919	-0.0017	
Striegel et al.,2006	0.8253	0.1366	0.0188	1.0357	0.0468	2884.0498	0.0269	2.6891	0.1365	
Cohen et al.,2007	1.3765	0.2473	0.0594	1.0032	0.0451	2203.3151	0.0305	3.0526	0.2470	
Al-Falasi et al.,2008	-1.1563	-0.1977	0.0388	1.0208	0.0460	2903.5873	0.0286	2.8616	-0.1977	
Larance et al.,2008	1.6754	0.2655	0.0680	0.9820	0.0444	2878.6129	0.0241	2.4114	0.2664	
Posiadala et al.,2010	-1.1536	-0.1919	0.0366	1.0200	0.0461	2904.0446	0.0271	2.7140	-0.1920	
Gradidge et al.,2011	-0.3687	-0.0438	0.0019	1.0253	0.0470	2904.9093	0.0131	1.3108	-0.0436	
Ip et al.,2011	1.9001	0.3414	0.1077	0.9560	0.0428	2681.1281	0.0298	2.9778	0.3407	
Santos et al.,2011	-1.2671	-0.2186	0.0470	1.0136	0.0456	2901.6068	0.0292	2.9249	-0.2186	
Hope et al.,2013	0.3645	0.0618	0.0039	1.0575	0.0477	2842.9150	0.0300	3.0010	0.0619	
Raschka et al.,2013	0.0375	0.0040	0.0000	1.0575	0.0479	2897.1920	0.0279	2.7864	0.0040	
Mooney et al.,2016	-1.5763	-0.2731	0.0714	0.9891	0.0444	2889.9271	0.0298	2.9798	-0.2728	
Rowe et al.,2016	1.2229	0.2170	0.0463	1.0150	0.0457	2712.0790	0.0301	3.0072	0.2168	
Westerman et al.,2016	1.3474	0.2350	0.0539	1.0053	0.0452	2825.5656	0.0290	2.8960	0.2349	
Zahnow et al.,2017	-0.1027	-0.0204	0.0004	1.0610	0.0479	2890.1445	0.0296	2.9607	-0.0204	
Althobiti et al.,2018	0.0413	0.0047	0.0000	1.0624	0.0480	2856.9754	0.0302	3.0228	0.0047	
Hill et al.,2019	0.2275	0.0373	0.0014	1.0592	0.0479	2876.4907	0.0295	2.9508	0.0373	
Jacka et al.,2019	2.5277	0.4496	0.1749	0.8926	0.0397	2756.3671	0.0287	2.8694	0.4492	
Macedo et al.,2019	-0.0457	-0.0092	0.0001	1.0485	0.0476	2902.8855	0.0235	2.3471	-0.0091	
Pany et al.,2019	-0.2084	-0.0378	0.0015	1.0570	0.0478	2900.6346	0.0281	2.8096	-0.0378	
Pereira et al.,2019	1.0596	0.1829	0.0333	1.0252	0.0462	2848.8713	0.0288	2.8819	0.1829	
Uddin et al.,2019	-1.7165	-0.3019	0.0858	0.9758	0.0437	1099.7502	0.0307	3.0725	-0.3012	
Bonnecaze et al.,2020	0.3139	0.0533	0.0029	1.0598	0.0478	2520.6043	0.0306	3.0626	0.0534	
Jokipalo et al.,2021	-0.3118	-0.0543	0.0030	1.0537	0.0477	2902.8279	0.0272	2.7230	-0.0543	

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Influence on Overall Result
Squared Pearson Residual
15
35
29
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15


rstudent
dffits
cook.d
cov.r
tau2.del
QE.del
hat
weight

PRISMA 2020 Main Checklist

Topic	No.	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Section 1, Lines 126-128
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section 1, Lines 120-123
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section 1, Lines 123-125, 126-128
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 2, Line 157 (Table 1)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 2, Line 141
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 2, Line 144 (Figure 1)
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2, Lines 146-150
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 2, Lines 159-163
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 2, Lines 159-163
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 2, Lines 159-163

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Topic	No.	Item	Location where item is reported
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2, Lines 165-174
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Section 2, 176-200
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	Section 2, 198-200
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 2, 176-200
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 2, 176-200
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 2, 176-204
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Section 2, Lines 182-190
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Section 2, 201-204
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Section 2, Lines 201-204
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Section 2, 201-204
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 2, Line 154 (Figure 2)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	n/a
Study characteristics	17	Cite each included study and present its characteristics.	Section 3, Line 217 (Table 2)
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section 3, Lines 226-233
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Section 3, Line 217 (Table 2)
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section 3, Lines 226-233

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Topic	No.	Item	Location where item is reported
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Section 3, Lines 235-390
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Section 3, Lines 240-375
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Section 3, Lines 365-372
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Section 3, Lines 240-375
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Section 3, Lines 240-375
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section 4, Lines 377-419
	23b	Discuss any limitations of the evidence included in the review.	Section 4, Lines 420-438
	23c	Discuss any limitations of the review processes used.	Section 4, Lines 420-438
	23d	Discuss implications of the results for practice, policy, and future research.	Section 5, Lines 446-449
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Section 2, Lines 130-132
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Section 2, Lines 130-132
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	n/a
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Section 6, Lines 461-453
Competing interests	26	Declare any competing interests of review authors.	Section 6, Line 453
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Section 3, Lines 238-239

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PRIMSA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesize results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. MetaArXiv. 2020, September 14. DOI: 10.31222/osf.io/v7gm2. For more information, visit: www.prisma-statement.org

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